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31 May 1978
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24 OCTOBER 1975

MILITARY SPECIFICATION

NAVIGATION DISPLAY SET, AN/ASN-124

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1.0 SCOPE

1.1 Scope - The equipment covered by this specification shall utilize sensor inputs described herein and provide simultaneous inertial and dead reckoning navigation solutions in both a Geographical and a Tactical mode. The set shall also provide for the processing and display of a stabilized plot of selected markers.

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FSC 5826

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* 1.2 Classification - The equipment covered by this specification shall consist of the following items:

Assigned AN Nomenclature	Type Designation	Applicable Paragraph
Controller, Marker Entry	C-9850/ASN-124	3.5.1
Control, Keyer	C-9857/ASN-124	3.5.2
Indicator, Digital Display	ID-2035/ASN-124	3.5.3
Controller, Marker Plotter	C-9855/ASN-124	3.5.4
Converter-Interface	CV-3332/ASN-l24	3.5.5
Control, Tactical Display	C-9862/ASN-124	3.5.6
Computer, Navigation	CP-1224/ASN-124	3.5.7
Converter, Signal Data	CV-3289/ASN-124	3.5.8
Converter, Radar Buffer	CV-3290/ASN-124	3.5.9
Battery Unit	BA-597/ASN-124	3.5.10
Controller, Mode	C-9856/ASN-124	3.5.11
Control-Indicator (2)	C-9863/ASN-124	3.5.12
Interconnecting Box	J-3349/ASX-124	3.5.13
Mounting Base	MT-4950/ASN-124	3.5.14
Mounting Base	MT-4953/ASN-124	3.5.15

1.3 Associated Equipment - The equipment shall operate to the requirements of this specification with the associated equipment listed in Paragraph 6.8.

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2.0 APPLICABLE DOCUMENTS

* 2.1 General - The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-W-5088	Wiring, Aircraft, Selection and Installation of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-C-18191	Computer Group, Navigational, AN/ASA-13 and AN/ASA-13A
MIL-T-18303	Test Procedures, Preparation and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-P-18987	Plotting Board, Tactical Display, PT-396/AS
MIL-I-19165	Indicator Group, AN/APA-125
MIL-R-19877C	Receiver-Indicator, R-976B/ALD-2 (U)
MIL-R-21113	Radar Set, AN/APS-80(V)
MIL-I-21685	Indicator, Compass Repeater, Rotating Dial, Type MW-2
MIL-I-22075	Indicator, Bearing-Distance-Heading ID-663/U
MIL-M-22747	Monitor, Error Voltage MX-2230/ASG

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MIL-N-23407	Navigation Set, Radar AN/APN-153(V)
MIL-A-23670	Attitude Heading Reference System AN/ASN-50
MIL-H-27269	Horizontal Situation Indicator Group AF-A24J-1
MIL-M-38510	Microcircuits, General Specification for
MIL-P-55640	Printed Wiring Boards, Multilayer (Plated-through Hole)
MIL-D-81358	Display Group, Tactical Data AN/ASA-66
MIL-A-81403	Altimeter-Encoder, AAU-21/A
MIL-C-81486 (AS)	Computer-Recorder Group, Sonar AN/AQA-7(V) (U)
MIL-N-81497A,	Navigation Set, Inertial, AN/ASN-84
MIL-N-81675(AS)	Navigation Set, Omega AN/ARN-99(V)1
MIL-A-81852(AS)	Altimeter-Encoder AAU-32/A
MIL-R-81859	Sonobuoy Receiver Set, AN/ARR-72 (U)
MIL-D-85009(AS)	Display Group, Tactical Data AN/ASA-66B

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirement for
AR-104	Reliability Development Test for Avionics Equipment, Requirements for
AS-l283(AV) AS-l284(AV) AS-l285(AV) AS-l286(AV)	Alphanumerics, Digital Sweep Frequency Generator, and Cass Modifications to the Computer-Recorder Group, Sonar, AN/AQA-7(V)(U) Bearing Bias Correction, Automatic Threshold and Doppler Bearing Tracker Modifications to the Computer-Recorder Group, Sonar AN/AQA-7(V)
EI-632	Avionics Installation Instructions for Navigation-Display Set, AN/ASN-124

STANDARDS

MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment

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MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and Systems of Units, Electromagnetic Interference Technology
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-749	Preparation and Submission of Data for Approval of Nonstandard Parts
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794D	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-883	Test Methods and Procedures for Microelectronics
MS17322	Meter, Time Totalizing, Miniature, Digital, 115 Volt, 400 Hertz
MS25212	Control Panel, Console Type, Aircraft Equipment, Basic Dimensions
MS27651	Meter, Time Totalizing, Hermetically Sealed, 115V, 400 Hz

Federal Standards

FED-STD-595	Colors

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3.0 REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronics

(a) The Microelectronics shall meet MIL-M-38510 class "B" or equivalent. Where MIL-M-38510 slash sheets do not exist, the contractor specifications shall define the equivalency. The Contractor shall provide more than one source for microelectronics when available.

(b) All hybrid microelectronics shall be built to the approved contractor specifications. The intent of MIL-M-38510 shall be followed by the Contractor to the maximum extent practical.

(c) The requirements of AR-5 shall be met where dual sources are available. When dual sources are available, interchangeability shall be certified by the Contractor.

(2) Semiconductor Devices

JAN TX types shall be used where required to achieve specified reliability of the equipment.

(3) Discrete Components

Where standard MIL types satisfy all of the design requirements, they shall be used. Established Reliability (ER) type passive parts shall be used where required to achieve specified reliability of the equipment. Non MIL types may be used with the approval of the procuring activity.

(4) Non-Repairable Assemblies

(a) Non-repairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the sub-assembly and the amount of circuitry to be included therein shall be approved by the procuring activity.

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(b) Any screening of non-repairable subassemblies shall be done on complete assemblies so as to include screening of the assembly operations.

(c) When previously produced models of this equipment did not use non-repairable subassemblies, the design shall not be changed to employ non-repairable subassemblies without the approval of the procuring activity.

(d) Non-repairable assemblies shall be considered non-standard parts.

* 3.2.1 Non-standard Parts and Materials Approval - Non-standard parts and materials which have been approved for use by the procuring activity in other military equipment shall be considered approved for use in this equipment. When non-standard parts or materials not previously approved are used, approval for use shall be obtained. Approval for the use of non-standard parts and materials (including electron tubes, transistors, and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-STD-749 (Part 1 and 2 data). Microelectronic devices shall be approved in general in accordance with AR-5 and this specification. The vendor shall certify that all non-standard microelectronic devices shall have been subjected to MIL-STD-883, Group B and C testing, per generic families as a mini-mum.

3.2.2 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10.

* 3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship, except as otherwise specified herein. Copper cladding on internal layers of multilayer printed wiring boards shall be one ounce per square foot, minimum.

Electronic assemblies previously produced to an earlier issue of MIL-E-5400 need not be re-designed.

The requirement of MIL-P-55640 for continuous solder plug filling of the interracial plated-through holes of printed wiring boards shall not apply.

The following exceptions to the requirements of MIL-STD-275 for printed wiring boards shall apply:

References are to Paragraph Numbers of MIL-STD-275.

(1) 4.2 It shall be permissible to use separate hole chart drawings and artwork drawings of different sizes so as to take advantage of automated and electronic data processing techniques. The separate drawings shall be referenced on the printed wiring board detail drawing.

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(2)	5.1.5.3	Where high density packaging techniques are used, 0.0065 inch minimum spacing may be used between the outer periphery of a plated through hole terminal area and an adjacent conductor. In such a case the voltage between conductors shall be limited to 20 volts DC or AC peak.
(3)	5,1.6, 5.2.1, 5.3.1	Jumper wires may be added to existing terminal areas to supplement the original etched pattern, where authorized by approved design changes. When heat sink core construction is used, component leads may be soldered to a flat surface pattern on the component side of the board, using approved contractor processes.
(4)	5.1.7.2, 5.5.4	The requirement for solder plating and tin-lead coating shall not apply to areas covered by a heat sink. On high density boards, the solder plug shall wet a minimum of 0.001 inch outward from the hole. Continuous solder plug filling of the interracial plated-through holes shall not be required.
(5)	5.2.2	The size of terminal areas may be reduced where standard military specification components have terminal spacing which is incompatible with the requirements of this paragraph. Where high density packaging techniques are used, the surrounding area shall be at least 0.012 inch greater than the diameter of the hole.
(6)	5.3.2, 5.5.3	Hole-to-lead size difference shall not exceed 0.034 inch.
(7)	5.7.1, 5.2.1.1	The part attachment method of Paragraph 5.2.1.3 may be used.
(8)	5.7.1.2	Component leads may not necessarily be clinched in alignment with the conductor path. The leads may extend beyond the edge of the terminal area and may not necessarily make contact with the conductor pattern before soldering.
(9)	5.7.1.3	Where feasible, straight-through leads shall be swaged. Clearance for component leads which extend beyond the foil surface more than 0.060 inch shall be a design consideration and shall be specified on the drawing.
(10)	5.7.1.4	The outline of the lead shall be visible in the solder.

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(11) 5.7.2.5 Non-axial lead components may be mounted in contact with the printed wiring board, provided manufacturing processes insure the elimination of any foreign particles or fluids between the board and component.

(12) 5.8.2 Conformal coating applied in accordance with processes meeting procuring activity approval.

* 3.3.1 Total Weight - The total weight of the equipment excluding cables and mounting, shall be a minimum consistent with good design and shall not exceed 258 pounds in all possible configurations.

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a re-order from a supplier who has produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement intervals shall be specified by the contractor.

3.3.2.3 Reliability in Mean-Time-Between-Failures (MTBF) - The equipment shall have 100 hours mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following units shall each contain a time totalizing meter in accordance with MS17322 or MS327651.

Unit
Computer, Navigation
Battery Unit
Converter, Signal Data
Converter-Interface
Control, Tactical Display
Control, Keyer
Indicator, Digital Display

3.3.3 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

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3.3.4 Control Panel - The configuration of all panels must be submitted to the procuring activity for approval. All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, MIL-STD-462, and MIL-STD-463.

3.3.7 Provisions for Maintainability - The Maintainability Program, built-in test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10, except as modified herein. The Maintenance Concept to be implemented in the Maintainability Program shall be as follows:

(1) Organizational Maintenance

The equipment shall provide fault isolation to the WRA, utilizing BITE and supplementary procedures employing standard support equipment such as voltmeters and oscilloscopes. Under no circumstances will “carry-on” organizational test equipment be considered a viable supplement to BITE. Repair will be by replacement with a serviceable WRA, and the repair verified. Faulty WRA’s will be sent to the intermediate level for repair. The BITE shall include the capability for an Operational Readiness Test.

(2) Intermediate Level

Repair will consist of fault isolation to the SRA utilizing common and special support equipment, and replacement with a serviceable item. The repair will be verified and the faulty items will be returned to the depot for repair.

(3) Depot Level

This level includes all maintenance actions beyond the capability of the intermediate level, including repair of WRA’s and SRA’s by replacement of the faulty component(s).

3.3.7.1 Compatibility with VAST - VAST compatibility is not required.

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3.3.7.2 Operational Maintainability Requirements - The equipment covered by this specification shall be designed to achieve the following Maintainability parameters. These corrective maintenance times do not include aircraft access times and assume fault isolation with use of BITE, operational controls, and standard support equipment.

3.3.7.2.1 Mean Corrective Time (M_{ct}) Organizational Level - 0.25 hours.

3.3.7.2.2 Mean Corrective Time, 95th percentile ($M_{max_{ct}}$) Organizational Level - 0.5 hours.

3.3.7.2.3 Mean Time to Repair (MTTR), WRA, Intermediate Level - 1.0 hours.

3.3.7.2.4 Mean Time to Repair (MTTR), SRA - 1.5 hours.

3.3.7.3 AR-10A Compatibility Requirements - The requirements of AR-10A shall apply except as modified herein:

References are to Paragraph Numbers of AR-10A:

(1) 3.1.4.1 Newly-designed equipment shall utilize applicable military standard connectors with Requirement 10 of MIL-STD-454 as the preferred type.

(2) 3.3 The detection and location functions shall include those equipment features which are exclusively test functions, as well as those which, by their normal operation, provide detection and location capability.

(3) 3.3.2 Newly-designed equipment shall include features which will enable failures to be located on the ground and in flight. This built-in failure location shall not require the use of any PGSE test devices for the performance of organizational maintenance.

(4) 3.3.2.2 The intent of this paragraph of AR-10A shall be met by the existing features of the annunciators provided on the Mode Controller and the Digital Display Indicator, the visual presentations of the two CRT's, and Go/No-Go Indicators.

(5) 3.3.2.3 The intent of this paragraph of AR-10A shall be met by the annunciation features provided on the Mode Controller, the Digital Display Indicator, and the visual presentations of the two CRT Displays.

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(6) General:

(a) The maintainability indices of Paragraphs 3.4.2 and 3.4.3 of AR10A shall be considered in the design and the following shall be the minimum requirements:

LRA ratio	.714
NA ratio of BITE detected failures	.87 Does not include ambiguities with associated equipment.
FI ratio	As specified in AR10A. Loading of constants or scale factors into the Navigation Computer and other adjustments required by associated equipment shall not be included.
QRA ratio	.8
SNA ratio	.85 for 1 SRA .9 for 2 SRA's 1.0 for 3 SRA's
SFI ratio	.95

(b) The requirements for test points specified in Section 3.5 of AR-10A shall guide the design of new items only as necessary to meet the requirements of Section 3.3.7.2 of this specification.

(c) Previously produced Items shall not be required to meet any requirement specified in AR-1OA.

3.3.7.4 Packaging Requirements - The equipment shall be packaged in modular form so that, to the maximum extent practical, all subassemblies shall be of the plug-in type and should be easily removable.

3.3.7.5 Protective Devices - Overload protection devices shall be used. The NC and SDC shall be protected by single phase circuit breakers. Circuit breaker reset mechanisms and fuses, when used, shall be accessible from the exterior of the equipment.

3.3.7.6 Maintenance Module Accessibility - Maintenance module arrangement will be such that access to each module does not require the removal of adjacent modules or parts other than access panels.

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3.3.7.7 Keying - Modules and connectors, except those required for BNC connectors, shall be keyed to prevent the insertion of a connector or module into an improper location within or on the. equipment.

3.3.7.8 Part Replacement - Part replacement on the modules shall be possible by skilled technicians working in a designated service area such as an intermediate maintenance activity. Use of potting compounds, excessive moisture/fungus proofing compounds, or any other substance to mechanically support piece parts is forbidden.

Use of potting compounds for moisture/fungus proofing purposes is allowed, provided the following minimum requirements are met:

(1) A 30-mil thickness is not exceeded.
(2) Reference designators and/or part identifications remain clearly legible.
(3) Excessive use of hard curing compounds (such as some epoxies) to hold parts in place is prohibited.
(4) Part replacement is not significantly impeded.

3.3.7.9 Non-Repairable Assemblies - All designs for non-repairable assemblies shall have procuring activity approval, prior to release for fabrication.

3.3.7.10 Adjustments - Procuring activity approval is required for design and use of all periodic adjustment, alignment, or calibration functions except for those specified herein. There shall be no adjustments at organizational level except for operator controls as specified herein.

3.3.7.11 Access Panels - Access panels shall be retained by captive fasteners,

3.3.7.12 Service Access - Devices requiring removal periodically, for inspection and/or servicing, shall be retained by quick-disconnect fasteners; i.e., hand-operated, Dzus type, etc.

3.3.7.13 Mounting Devices - Connectors, receptacles, circuit breakers or any other electronic parts, shall not be a part of any mounting device associated with the equipment.

3.3.7.14 Handles - A handle or handles shall be provided if required to afford one man optimum handling. Handle location shall be appropriate for ease of handling, installation, and removal.

3.3.8 Nomenclature and Nameplates - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

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3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C ± 10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 VAC single phase, 400 ± 20 Hz and 27.5 ± 0.5 VDC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1A equipment, except as modified herein:

* 3.3.10.1 Vibration - The equipment shall operate to the requirements of this specification when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400, except as modified below:

(1) Curve I shall be limited to 5g.
(2) The Tactical Display Control shall meet Curve I or 5 g's, whichever is less, for X and Y axis; and Curve I and III from 5 to 160 Hz, 2 g's from 160 to 400 Hz, 5 g's (Curve 111) from 400 to 500 Hz for Z axis. Cross-over points 160 Hz and 400 Hz (Z-axis Tactical Display Control Unit) are not considered resonances.

3.3.10.2 Temperature - The minimum temperature for satisfactory performance to the requirements of this specification shall be -30°C. The battery contained in the Battery Unit shall not be subjected to temperature less than -40°C or greater than +55°C. Battery protection for power interruptions shall not be required at temperatures below -15°C, or above +45°C.

3.3.11 Warm-Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 3 minutes under standard conditions, and 5 minutes at -30°C.

Alignment time for the Inertial Measurement Set is not included in warm-up time.

3.3.12 Input Electrical Power

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704, except as modified herein. The power required shall not exceed the specified amounts.

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(1) AC Power (Single Phase), 115V, Category B, 15 VA.

* (2) AC Power (Three Phase), 115/200V, Category B, 1295 VA.

(3) DC Power, 28V, Category B, 14 watts

3.3.12.1.1 Transient Limits - The Integrated Tactical Display System (ITADS) shall give satisfactory performance when the Operating Power of 3.3.12.1 is within the bounds of the transient AC limits 2 and 3 of Figure 3 of MIL-STD-704, except that the upper operating limit shall be 130 volts and the lower operating limit shall be 100 volts. The ITADS shall give satisfactory performance when the Operating Power of 3.3.12.1 is within the transient frequency limits 2 and 3 of Figure 5 of MIL-STD-704. The ITADS shall give satisfactory performance when the Operating Power of 3.3.12.1 is within the transient DC limits 2 and 3 of Figure 9 of MIL-STD-704 except the lower limit shall be +25 volts.

3.3.12.1.2 Degraded Performance - Degraded performance will be permitted from the ITADS for voltage transients not exceeding 0.5 second during normal electric system operation. Operation of the ITADS shall return to normal with no resulting damage to the equipment. Degraded operation of the ITADS may occur when the input power exceeds the limits of Paragraph 3.3.12.1 but remains within limits 1 and 4 of Figures 3, 5, and 9 of MIL-STD-704. However, no damage shall be suffered by the ITADS and normal operation of the ITADS shall automatically resume upon return of the input power voltage to levels within the operating limits as defined above.

3.3.12.2 Synchro Power - Input power for synchro excitation shall be supplied as follows:

AC Power (Single Phase), 26V, 400 Hz, 6 VA.

3.3.12.3 Lighting Power - Input power for lighting shall not exceed 6 amps at 0 to 26 VAC.

3.3.12.4 Other Power - Input power from the PP-4964/ASN-84 power supply shall be supplied as follows:

(1) +5.25 ± .25 VDC, 8 Watts.

(2) +15 VDC (C), 0.5 Watts.

(3) -15 VDC (C), 0.3 Watts.

(4) +26 VDC, 1 Watt.

(5) -26 VDC, 1 Watt.

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3.4 Performance - Unless otherwise specified, values set forth to establish requirements of satisfactory performance apply to performance under both standard and service conditions. When reduced performance under service conditions is acceptable, tolerances or variances from performance under service conditions will be specified.

3.4.1 Operation

3.4.1.1 System Definition - The Set shall consist of those items of Paragraph 1.2. The system shall operate in conjunction with the associated P-3/B aircraft equipment listed in Paragraph 6.8. A functional block diagram of the system and the associated equipment is shown in Figure 1 including the items indicated in dashed outline. These items are included to show aircraft system integration, although the items are not directly interfaced with the system defined by this specification. The system shall provide for distribution of analog outputs to associated equipment as shown in Figure 2 when interconnected as shown.

3.4.1.1.1 Hardware - The Signal Data Converter (SDC) converts input analog signals from external equipments into digital data for processing by the Navigation Computer (NC). The SDC also provides conversion from digital to resolver and synchro outputs, level shifting of input pulse trains and discretes, light and logic power to external equipments, conversion of input signals from the OR-90/ARN-99(V)1 Omega Receiver Converter from parallel to serial digital data, and provides the interface with the Gyroscope Assembly accelerometer outputs and gyroscope torquing inputs. The SDC also provides a resolver to synchro buffer that provides the interface between the CN-1231/ASN-84 Gyroscope Assembly and the MX-6985/ASN-50 Compass Adapter Compensator. The SDC contains interfaces to interconnect the AN/AQA-7(V) Sonar Computer-Recorder Group (Difar), the C-7617/ARR-72 Dual Channel Control Indicator, the ITADS, the Marker Plotter Controller, and the CN-1231/ASN-84 Gyroscope Assembly with the NC.

The Navigation Computer contains the memory and logic controls required for performing arithmetic operations, contains serial digital input and output channels and input and output discretes under program control. It also interfaces directly with the Keyer Control.

The Converter-Interface interfaces with the IP886B/ASA-66 Tactical Data Display. The front panel contains DIFAR data controls and displays and a panel which. is illuminated to display the status of tactical markers. The chassis contains the electronics to interface with the computer subsystem and the AN/APS-80 Search Radar.

The Tactical Display Control contains the Trackball controls, a Marker Control keyset, and additional controls and indicators related to the display of tactical data.

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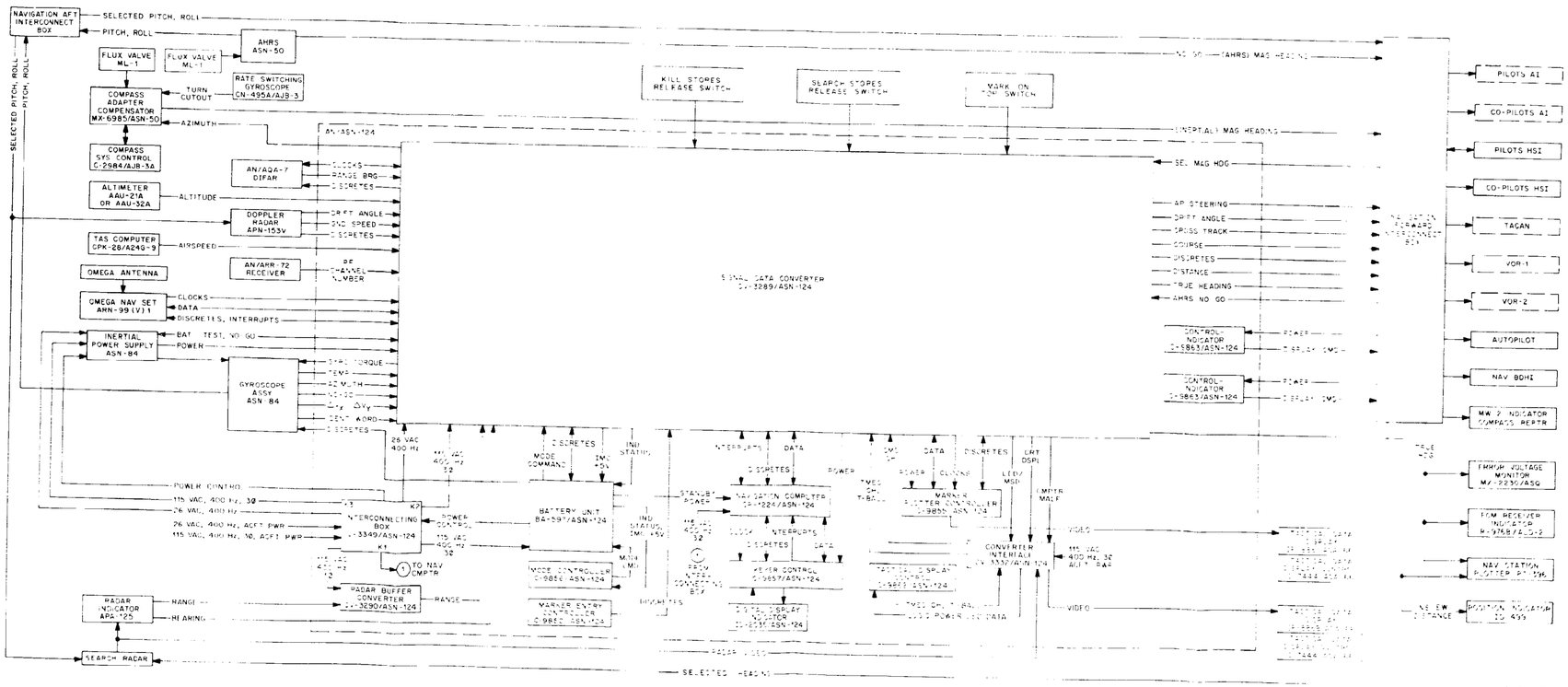


FIGURE 1. Functional block diagram, Navigation-Display Set, AN/ASN-124.

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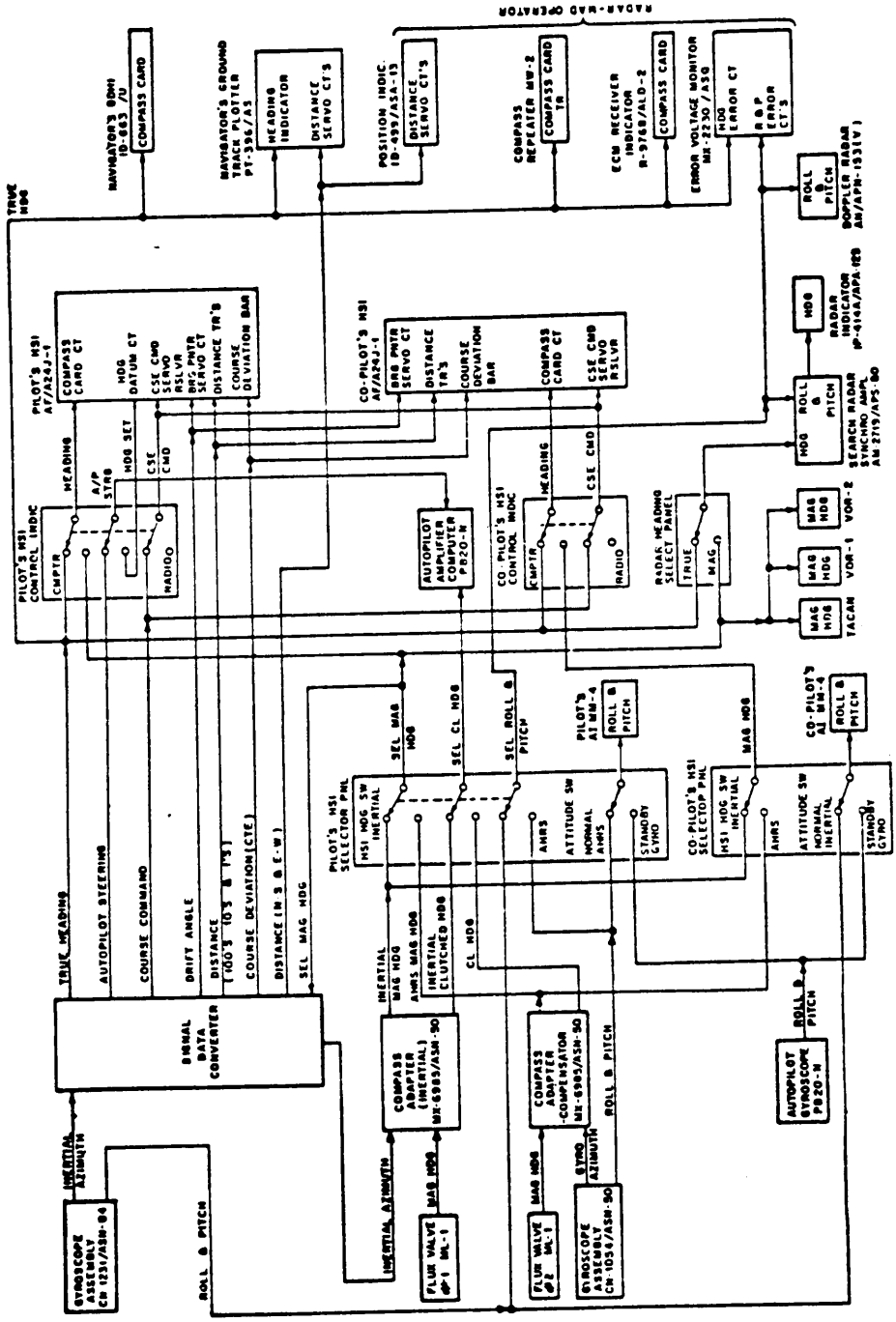


FIGURE 2. Analog output signal distribution diagram.

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The Digital Display Indicator contains the digital displays and the associated display electronics, indicator lights for monitoring various system functions, a BIT switch, and a dimming control. The digital displays consist of an alpha display row for operator cueing and two alpha-numeric display rows for parameters selected via the Keyer Control.

The Keyer Control contains the necessary switches and controls required to insert and control the navigation functions, and the associated electronics necessary to interface with the Digital Display Indicator and the Navigation Computer.

The Remote Marker Entry Controller is utilized for the entry of MAD and RADAR markers. It interfaces with the SDC.

The Marker Plotter Controller provides the Pilot's interface between the NC and the Pilot's IP-886A/ASA-66. It contains electronics to generate the symbols required by the IP-886A/ASA-66 and provides keys for initiation of correction functions, and for selection of the scale and center of the displayed information. It interfaces with the SDC and the IP-886A/ASA-66.

The Battery Unit provides battery power to the NC during power interruptions and provides sequencing and interlock functions for control of the ASN-84 Inertial Measurement System. It contains a nickel-cadmium battery, battery charge circuit, environmental controls for the battery, control circuits for the CN-1231/ASN-84 Gyroscope Assembly and power control circuits for the NC and SDC.

The Mode Controller provides the selection of the Inertial Measurement System operating modes and provides legends for operational status and recommended actions. It interfaces with the Battery Unit, the SDC, and PP-4964/ASN-84 Power Supply.

The Control-Indicator provides a switch for selecting either the radio navigation aids or the Navigation Display Set for controlling the data displayed on the pilot's and the copilot's HSI's. A true heading indicator light is provided as positive indication that true heading information is being displayed. Two Control-Indicators are provided, one each for the pilot and the co-pilot.

The Radar Buffer Converter processes outputs from the IP-414A/APA-125 Radar Indicator for transmission to the SDC.

The Interconnecting Box provides the system interconnect facility for necessary signal and power distribution. The box is designed to enhance system electromagnetic compatibility and to simplify cabling interconnections.

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3.4.1.1.2 Software - The system shall be programmed to perform the following basic functions.

(1) Inertial Measurement Set Management

(2) Navigation

(3) Control Panel Interface

(4) Display Interface

(5) System Self Test and Fault Isolation

3.4.1.2 Operating Modes - The system shall provide the capability to navigate in a Geographical (GEO) or in a Tactical (TAC) mode of operation. Within either GEO or TAC, the operating navigation solutions shall be Inertial (IN), Auxiliary Inertial (XN), Dead Reckoning (DR) and Omega, with the following capabilities:

(1) IN - Inertial velocities and barometric altitude. This is the pure inertial solution. It shall be used for platform management and shall not be susceptible to position update. It shall be possible to update any other solution using IN. It shall not be used for geographic or tactical navigation.

(2) XN - Inertial velocities and barometric altitude. This shall be a separate, auxiliary, inertial solution and shall be selectable for geographic or tactical navigation. It shall be possible to position update XN.

(3) DR - Doppler or true airspeed velocities, inertial or selected magnetic heading, and barometric altitude. DR shall be selectable for geographic or tactical navigation. It shall be possible to position update DR. Within the DR mode, the system shall automatically utilize the best heading and best velocity sensor available according to the following priority:

	Best Sensors
Geographic Navigation	Doppler - Inertial
Tactical Navigation	Navigator Specified

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(4) Omega - shall provide a solution of receiver position only, in Omega lines of position and in latitude/longitude. The Omega subsystem shall provide the following features:

(a) Automatic Signal Acquisition
(b) Automatic Synchronization
(c) Automatic Phase Correction
(d) Automatic or Manual Station Selections
(e) Automatic Antenna Lobe Selection

3.4.2 Accuracy

3.4.2.1 Input Conditions - The specification of system accuracy is based on the system inputs within the following limits:

3.4.2.1.1 System Inputs

True Heading (Inertial)
0.2 degrees initial heading error
0.02 degrees/hour heading drift rate

Magnetic Heading
0.5 degrees heading accuracy excluding uncompensated flux valve errors.

True Airspeed
±5 knots below 200 knots
±3 knots above 200 knots

Doppler Ground Speed
±0.35 knot
±0.23% of true value

Doppler Drift Angle
±0.23 degrees

Magnetic Variation
Known to ±0.5 degrees

Altitude
±30 feet ±0.5% of aircraft altitude

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* 3.4.2.1.2 Flight Conditions

(1) Aircraft "on top" Accuracy

(a) Tac Nav - The error in flying over a surface reference and marking an "on top" position shall not exceed 100 feet (1 σ). This surface reference may be a fixed point on the earth or a floating buoy.

(b) GEO-Nav - The error in flying over a known geographic point and marking an "on top" position shall not exceed 500 feet (1 σ). The location of this geographic reference shall be known to within 0.1 minutes in latitude and longitude.

(2) Surface Motion Accuracy

The surface motion due to wind and sea current shall be known to within ±15% of actual. The actual surface movement (current or tide) shall not exceed 5 knots during the Doppler Navigation Test.

(3) Doppler Random Bias Errors

The combined ground speed and drift angle errors of the APN-153 Doppler shall not contribute to more than 9 yards/minute (1 σ) overall velocity error.

(4) TAC NAV Test Conditions

(a) Airplane altitude 500 feet or less to minimize visual "on top" error.

(b) "On tops" made over a fixed ground reference to eliminate non-constant sonobuoy drift errors caused by variable ocean currents.

(c) "On-Top" revisit-time not less than 8 minutes during the bias velocity and the residual drift computation.

(d) The TAC NAV Tests shall be conducted under conditions where the wind is steady and not greater than 20 knots.

NOTE: A (1 σ) limit refers to a normal (Gaussian) distribution.

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(5) Omega Navigation Test Conditions

(a) Rate aiding information will be derived from the inertial system.

(b) Automatic station selection shall be used.

(c) Any Omega transmitter operating normally may be used during the test period. Normal operation shall be defined as a station that is transmitting, synchronized, and radiating a nominal 10 kw of power.

(d) Phaselock loop settling time is a function of signal to noise ratio. Therefore, adequate flight time will be allowed before measurements are made to allow all phaselock loops to achieve steady state under existing signal to noise ratio conditions.

* 3.4.2.2 NAV/ASW System Accuracy

The accuracy of NAV/ASW System shall be as follows when the conditions are in accordance with 3.4.2.1:

(1) GEONAV Mode

(a) IN Navigation: 1.5 nm/hr Circular Error Probable (CEP) for not greater than 10 hours and 2.75 nm/hr 90th percentile position error for not greater than 10 hours.

(b) DI Navigation: 0.5 NM ± 1.64% of distance traveled CEP when compensated for surface movement and system drift.

(c) DA Navigation: 3.5 NM per hour CEP.

(d) Omega Navigation: The radial position errors shall not exceed 2.0 nm (RMS).

(2) TACNAV Mode

(a) DA Navigation: 25 yards per minute averaged for no less than 30 min.

(b) AI Navigation: 35 yards per minute averaged for no less than 30 min.

(c) AA Navigation: 40 yards per minute averaged for no less than 30 min.

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3.5 Detail Requirement

* 3.5.1 Controller, Marker Entry (C-9850/ASN-124) - The Controller shall meet the following requirements:

3.5.1.1 Function - The Controller is used by the radar operator for the entry of MAD and radar contact coordinates.

3.5.1.2 Form Factor - The Controller, exclusive of front panel protrusions, attachment points, and connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
1.875	5.750	3.500 (behind mounting surface)

3.5.1.3 Weight - The weight of the Controller shall not exceed 1.0 pound.

3.5.1.4 Controls - The Controller shall provide the following switch functions. Figure 3 depicts front panel configuration.

3.5.1.4.1 MAD Contact Entry - The MAD pushbutton shall be a momentary action switch.

3.5.1.4.2 Radar Contact Entry - The Radar pushbuttons shall be 2 momentary action switches.

3.5.1.4.3 Pushbutton Illumination - The MAD and Radar pushbuttons shall be illuminated green. The pushbutton legends shall be translucent against a black background.

3.5.1.5 Controller Illumination - The Controller shall incorporate a legend panel with white letters on gray background, and white illumination for night viewing (See Figure 3). The panel and the pushbuttons shall utilize 0-26 VAC aircraft lighting power.

* 3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	SJTP02RE10-35PA(023)	Outputs to Computer
J2	SJTP02RE10-98PA(023)	Lighting Power

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3.5.1.7 Lighting Power - The equipment shall utilize aircraft lighting power in accordance with Paragraph 3.3.12 as follows:

0 - 26 VAC, 0.68 Amp, Maximum

3.5.1.8 Mounting - The Controller shall meet the requirements of MIL-C-6781 for Type 1 control panels.

3.5.1.9 Color - Surface shall be painted lusterless gray, No. 36231, in accordance with FED-STD-595.

3.5.2 Keyer Control (C-9857/ASN-124) - The Keyer control shall meet the following requirements:

3.5.2.1 Function - The Keyer Control shall interface directly with the NC. Display data shall be conveyed to the Digital Display Indicator for visual indication of navigation status. NC data shall be input and output in both serial and discrete forms. The serial data shall be synchronous with the NC supplied 1 MHz clock, gated into 16 pulses per word on dedicated lines.

3.5.2.2 Form Factor - The Keyer Control, exclusive of front panel protrusions, attachment points, and connector projections, shall be formed within the following maximum dimensions (inches). Connectors shall be mounted on the recessed bottom-rear of the unit.

Height	Width	Depth
7.880	5.750	6.500 (behind mounting surface)

3.5.2.3 Weight - The weight of the Keyer Control shall not exceed 8.5 pounds.

3.5.2.4 Contents - The Keyer Control shall contain logic circuitry functions interfacing both the NC and the Digital Display Indicator.

3.5.2.4.1 Logic Signal Circuitry - Logic signal circuitry shall be, or similar to, differential line receivers type µA9615 and line drivers type µA9614. Full compatibility shall exist. Circuit input impedance shall be a nominal 130-ohm dc balanced line load with full driver capability at this level. Logic level definition shall be as presented in Figure 4.

3.5.2.4.2 Interface with NC - Refer to data function of Section 3.5.2.1. All communications shall be via a “priority interrupt command word“ method. A command word of 16 bits shall precede all incoming and outgoing data, and shall contain the instructions specifying what data are incoming to the Keyer Control or outgoing to the NC in subsequent series of clock words.

3.5.2.4.2.1 Command Word - The equipment shall be such that several instructions specified in one command word shall be processed by the Keyer Control in a specific priority sequence. A reset pulse shall always precede a command word and shall terminate any processing in progress, enabling the following new command word.

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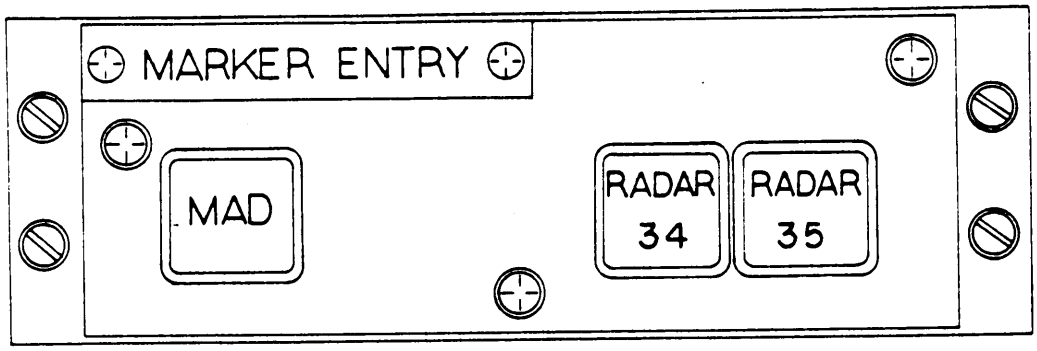


FIGURE 3. Marker Entry Controller.

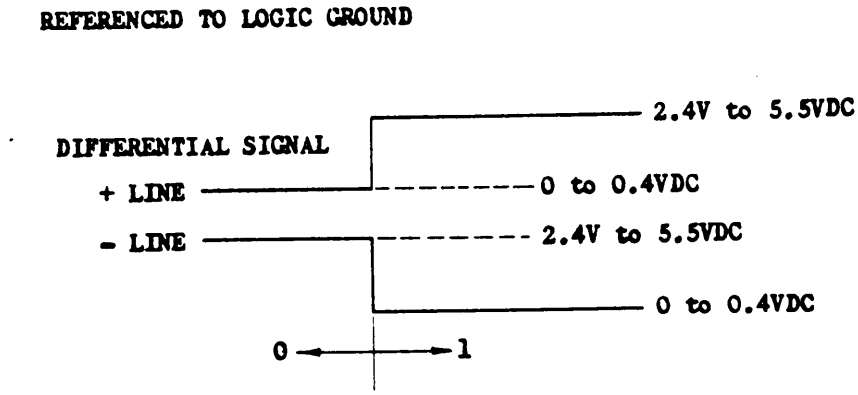


FIGURE 4. Logic level definition.

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3.5.2.4.2.2 Access Signals - Panel switch positions shall be interrogated at a cyclic rate. Immediate access signals to the NC formed by actuating momentary action push-button indicators (PBI) (Paragraph 3.5.2.5.3), shall be handled on an interrupt basis. The interrupt request to the NC shall be on a dedicated line, and shall simultaneously formulate an interrupt word in storage in the Keyer Control. When the NC is ready to accept this interrupt word, a reset pulse is transmitted followed by a command word requesting the Keyer Control to output the stored interrupt word.

3.5.2.4.2.3 Serial Data Format - Serial data format between the Keyer Control and NC shall be as presented in Figure 5.

3.5.2.4.2.4 Signal Inputs - Command reset input shall be a differential logical "1" pulse of 250 ± 50 ns duration. BIT status lights input shall be separate dedicated lines:

Logical "1" = Light illuminated.
Logical "0" = Light extinguished.

(1) MALF
(2) CMPTR An input to light "CMPTR" shall also activate "MALF".

3.5.2.4.2.5 Command Word Serial Input - Command Word serial input from NC to the Keyer Control shall be of serial format; characteristics of Paragraph 3.5.2.4.1, waveform as shown in Figure 5, and structure as tabulated.

MSB						COMMAND WORD						LSB
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

BIT	Function [1]	No. Words	Reference Paragraph	Table
0	INTERRUPT OUTGOING	1	3.5.2.4.3.3	
1	KEYER CONTROL OUTGOING	1	3.5.2.4.3.4	
2	SPARE			
3	SPARE			
4	SPARE			
5	SPARE			
6	SPARE			
7	SPARE			
8	SPARE			
9	DIGITAL DISPLAY I INCOMING	3	3.5.2.4.2.6	1
10	DIGITAL DISPLAY II INCOMING	2	3.5.2.4.2.7	1
11	DIGITAL DISPLAY III INCOMING	2	3.5.2.4.2.8	2
12	DIGITAL DISPLAY IV INCOMING	2	3.5.2.4.2.9	2
13	KEYER CONTROL DISPLAY INCOMING	2	3.5.2.4.2.10	1 and 2
14	SPARE			
15	KC LT INCOMING	1	3.5.2.4.2.11	

A logical one shall indicate the related word sequence has been commanded. Priority and sequence of processing shall progress from bit 0 through bit 15 with bit 0 processing first.

[1] Spare bits in the Command Word must be zero. When used in a word bit identification, a spare bit is presently unused and future use is possible. Hardware modification may be required for such future use.

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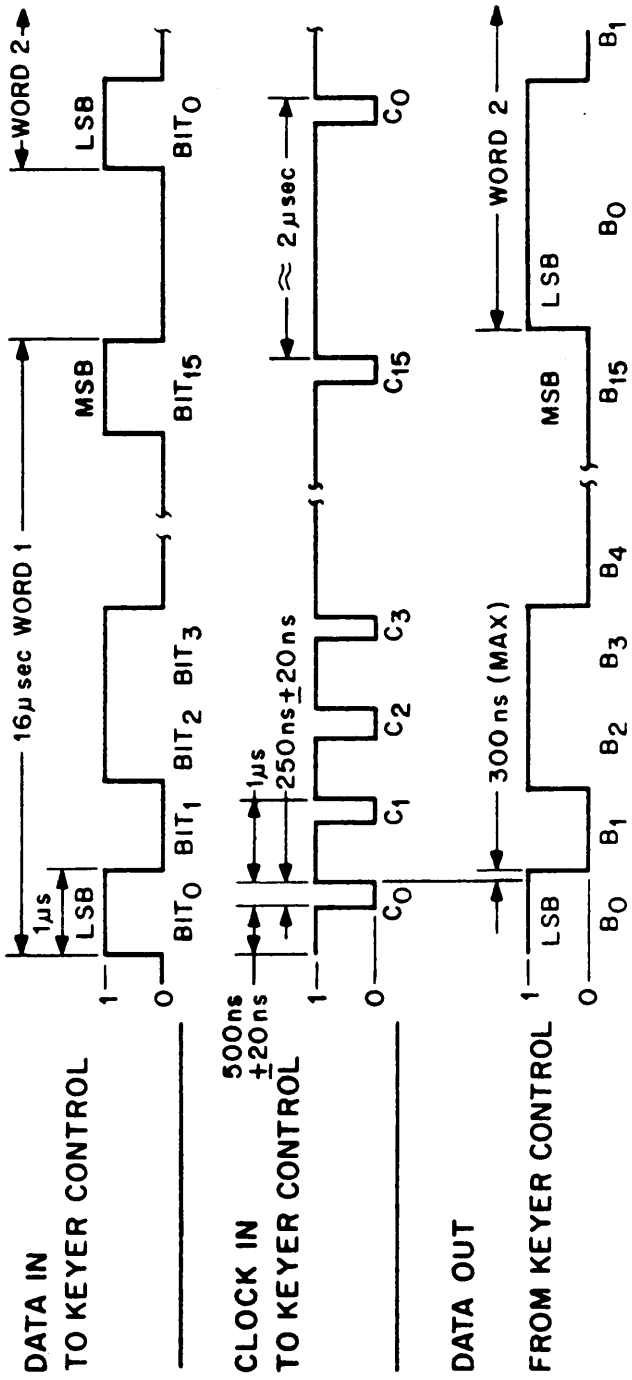
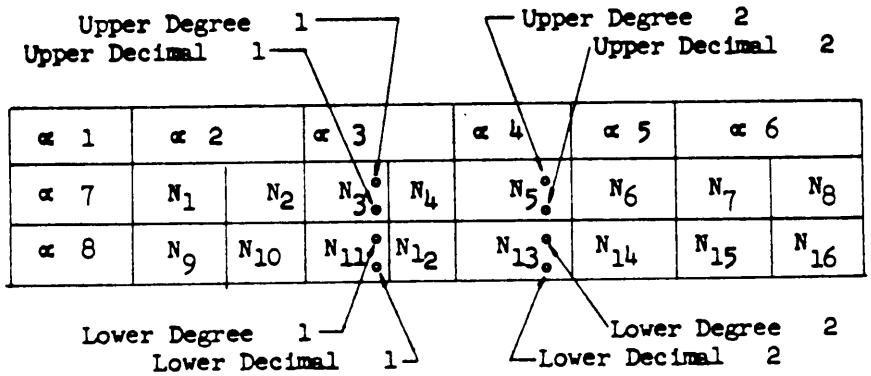


FIGURE 5. Serial data format, Keyer Control to NC.

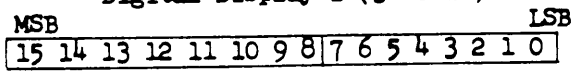
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3.5.2.4.2.6 Digital Display I Serial Input - Digital Display I Serial input from NC to Keyer Control skill be as tabulated. Alpha code shall apply as listed in Table I.



Display Positions of Alphas & Numerics

Digital Display I (3 words)

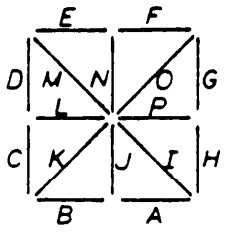


Typ each word, Word 1 sent out first, Bit 0 sent out first

BIT	FUNCTION, WORD 1		FUNCTION, WORD 2		FUNCTION, WORD 3	
0	ALPHA 2	BIT 0	ALPHA 4	BIT 0	ALPHA 6	BIT 0
1	ALPHA 2	BIT 1	ALPHA 4	BIT 1	ALPHA 6	BIT 1
2	ALPHA 2	BIT 2	ALPHA 4	BIT 2	ALPHA 6	BIT 2
3	ALPHA 2	BIT 3	ALPHA 4	BIT 3	ALPHA 6	BIT 3
4	ALPHA 2	BIT 4	ALPHA 4	BIT 4	ALPHA 6	BIT 4
5	ALPHA 2	BIT 5	ALPHA 4	BIT 5	ALPHA 6	BIT 5
6	ALPHA 2	BIT 6	ALPHA 4	BIT 6	ALPHA 6	BIT 6
7	ALPHA 2	BIT 7	ALPHA 4	BIT 7	ALPHA 6	BIT 7
8	ALPHA 1	BIT 0	ALPHA 3	BIT 0	ALPHA 5	BIT 0
9	ALPHA 1	BIT 1	ALPHA 3	BIT 1	ALPHA 5	BIT 1
10	ALPHA 1	BIT 2	ALPHA 3	BIT 2	ALPHA 5	BIT 2
11	ALPHA 1	BIT 3	ALPHA 3	BIT 3	ALPHA 5	BIT 3
12	ALPHA 1	BIT 4	ALPHA 3	BIT 4	ALPHA 5	BIT 4
13	ALPHA 1	BIT 5	ALPHA 3	BIT 5	ALPHA 5	BIT 5
14	ALPHA 1	BIT 6	ALPHA 3	BIT 6	ALPHA 5	BIT 6
15	ALPHA 1	BIT 7	ALPHA 3	BIT 7	ALPHA 5	BIT 7

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TABLE I. Alphanumeric bit code (ASCII).



BITS 6 and 7 are "DON'T CARE" states

OUTPUT SYMBOL	Address Lines						Output Segment													
	A_5	A_4	A_3	A_2	A_1	A_0	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
A	0	0	0	0	0	1	0	0	1	1	1	1	1	1	0	0	0	1	0	0	0	1
B	0	0	0	0	1	0	1	1	0	0	1	1	1	1	0	1	0	0	0	1	0	1
C	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0
D	0	0	0	1	0	0	1	1	0	0	1	1	1	1	0	1	0	0	0	1	0	0
E	0	0	0	1	0	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0
F	0	0	0	1	1	0	0	0	1	1	1	1	0	0	0	0	0	1	0	0	0	0
G	0	0	0	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	1
H	0	0	1	0	0	0	0	0	1	1	0	0	1	1	0	0	0	1	0	0	0	1
I	0	0	1	0	0	1	1	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0
J	0	0	1	0	1	0	0	1	1	0	1	1	0	0	0	1	0	0	0	1	0	0
K	0	0	1	0	1	1	0	0	1	1	0	0	0	0	1	0	0	1	0	0	1	0
L	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
M	0	0	1	1	0	1	0	0	1	1	0	0	1	1	0	0	0	0	1	0	1	0
N	0	0	1	1	1	0	0	0	1	1	0	0	1	1	1	0	0	0	1	0	0	0
O	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	1	0
P	0	1	0	0	0	0	0	0	1	1	1	1	1	0	0	0	0	1	0	0	0	1
Q	0	1	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0
R	0	1	0	0	1	0	0	0	1	1	1	1	1	0	1	0	0	1	0	0	0	1
S	0	1	0	0	1	1	1	1	0	0	1	1	0	1	0	0	0	0	1	0	0	1
T	0	1	0	1	0	0	0	0	0	0	1	1	0	0	0	1	0	0	0	1	0	0
U	0	1	0	1	0	1	1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	0
V	0	1	0	1	1	0	0	0	1	1	0	0	0	0	0	0	1	0	0	0	1	0
W	0	1	0	1	1	1	0	0	1	1	0	0	1	1	1	0	1	0	0	0	0	0
X	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	1	0
Y	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0
Z	0	1	1	0	1	0	1	1	0	0	1	1	0	0	0	0	1	0	0	0	1	0

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TABLE I. Alphanumeric bit code (ASCII) - Continued

OUTPUT SYMBOL	ADDRESS LINES						OUTPUT SEGMENT														
	A_5	A_4	A_3	A_2	A_1	A_0	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
[	0	1	1	0	1	1	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0
\	0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0
]	0	1	1	1	0	1	1	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0
^	0	1	1	1	1	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0
-	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sp	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
!	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0
'	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0
#	1	0	0	0	1	1	1	1	1	1	1	1	1	1	0	1	0	1	0	1	0	1
$	1	0	0	1	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1	0	1	1
%	1	0	0	1	0	1	1	1	0	0	1	1	0	0	1	0	1	0	1	0	1	0
OMIT	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
'	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
OMIT	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OMIT	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
*	1	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
+	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	1
,	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
-	1	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1
.	1	0	1	1	1	0	1	1	0	0	0	0	0	0	1	0	1	0	0	0	0	0
/	1	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0
0	1	1	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0
2	1	1	0	0	1	0	1	1	1	0	1	1	1	0	0	0	0	1	0	0	0	1
3	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	1
4	1	1	0	1	0	0	0	0	0	1	0	0	1	1	0	0	0	1	0	0	0	1
5	1	1	0	1	0	1	1	1	0	1	1	1	0	1	0	0	0	1	0	0	0	1
6	1	1	0	1	1	0	1	1	1	1	1	1	0	1	0	0	0	1	0	0	0	1
7	1	1	0	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0
8	1	1	1	0	0	0	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	1
9	1	1	1	0	0	1	0	0	0	1	1	1	1	1	0	0	0	1	0	0	0	1
OMIT	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OMIT	1	1	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0
=	1	1	1	1	0	1	1	1	0	0	0	0	0	0	0	0	0	1	0	0	0	1
>	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0
?	1	1	1	1	1	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1
BLANK	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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3.5.2.4.2.7 Digital Display II Serial Input - Digital Display II serial input from NC to Keyer Control shall be as tabulated. Alpha code shall apply as listed in Table I.

DIGITAL DISPLAY II (2 words)

MSB															LSB
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Typ Each Word, Word 1 Sent First, Bit 0 sent out first.

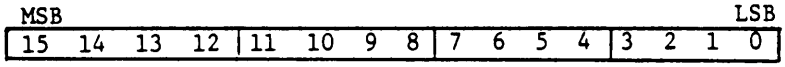
BIT	FUNCTION, WORD 1		FUNCTION, WORD 2
0	ALPHA 8	BIT 0	Lower Degree No. 2
1	ALPHA 8	BIT 1	Lower Decimal No. 2
2	ALPHA 8	BIT 2	Lower Degree No. 1
3	ALPHA 8	BIT 3	Lower Decimal No. 1
4	ALPHA 8	BIT 4	Upper Degree No. 2
5	ALPHA 8	BIT 5	Upper Decimal No. 2
6	ALPHA 8	BIT 6	Upper Degree No. 1
7	ALPHA 8	BIT 7	Upper Decimal No. 1
8	ALPHA 7	BIT 0	Spare
9	ALPHA 7	BIT 1	Spare
10	ALPHA 7	BIT 2	Spare
11	ALPHA 7	BIT 3	Spare
12	ALPHA 7	BIT 4	Spare
13	ALPHA 7	BIT 5	MODE LT
14	ALPHA 7	BIT 6	SYNC LT
15	ALPHA 7	BIT 7	Ω UPDATE LT

	LOGIC 1	LOGIC 0
UP & LWR DECIMALS	ON	OFF
UP & LWR DEGREES	ON	OFF
SYNC LT	ON	OFF
MODE LT	ON	OFF
Ω UPDATE LT	ON	OFF

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3.5.2.4.2.8 Digital Display III Serial Input - Digital Display III serial input from NC to Keyer Control shall be as tabulated. Numeric code shall apply as listed in Table II.

DIGITAL DISPLAY III (2 words)



Typ each word, Word 1 sent out first, Bit 0 sent out first.

BIT	FUNCTION, WORD 1		FUNCTION, WORD 2	
0	NUM 4	BIT 0	NUM 8	BIT 0
1	NUM 4	BIT 1	NUM 8	BIT 1
2	NUM 4	BIT 2	NUM 8	BIT 2
3	NUM 4	BIT 3	NUM 8	BIT 3
4	NUM 3	BIT 0	NUM 7	BIT 0
5	NUM 3	BIT 1	NUM 7	BIT 1
6	NUM 3	BIT 2	NUM 7	BIT 2
7	NUM 3	BIT 3	NUM 7	BIT 3
8	NUM 2	BIT 0	NUM 6	BIT 0
9	NUM 2	BIT 1	NUM 6	BIT 1
10	NUM 2	BIT 2	NUM 6	BIT 2
11	NUM 2	BIT 3	NUM 6	BIT 3
12	NUM 1	BIT 0	NUM 5	BIT 0
13	NUM 1	BIT 1	NUM 5	BIT 1
14	NUM 1	BIT 2	NUM 5	BIT 2
15	NUM 1	BIT 3	NUM 5	BIT 3

TABLE II. Typical numeric code.

NUMERIC	LSB BIT 0	BIT 1	BIT 2	MSB BIT 3
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	1	1	0	0
4	0	0	1	0
5	1	0	1	0
6	0	1	1	0
7	1	1	1	0
8	0	0	0	1
9	1	0	0	1
BLANK	1	1	1	1
ALL OTHER CODES FORBIDDEN				

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3.5.2.4.2.9 Digital Display IV Serial Input - Digital Display IV serial input from NC to Keyer Control shall be as tabulated. Numeric code shall apply as listed in Table II.

MSB				DIGITAL DISPLAY IV (2 words)											LSB
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Typ each word, Word 1 sent out first, Bit 0 sent out first

BIT	FUNCTION, WORD 1		FUNCTION, WORD 2	
0	NUM 12	BIT 0	NUM 16	BIT 0
1	NUM 12	BIT 1	NUM 16	BIT 1
2	NUM 12	BIT 2	NUM 16	BIT 2
3	NUM 12	BIT 3	NUM 16	BIT 3
4	NUM 11	BIT 0	NUM 15	BIT 0
5	NUM 11	BIT 1	NUM 15	BIT 1
6	NUM 11	BIT 2	NUM 15	BIT 2
7	NUM 11	BIT 3	NUM 15	BIT 3
8	NUM 10	BIT 0	NUM 14	BIT 0
9	NUM 10	BIT 1	NUM 14	BIT 1
10	NUM 10	BIT 2	NUM 14	BIT 2
11	NUM 10	BIT 3	NUM 14	BIT 3
12	NUM 9	BIT 0	NUM 13	BIT 0
13	NUM 9	BIT 1	NUM 13	BIT 1
14	NUM 9	BIT 2	NUM 13	BIT 2
15	NUM 9	BIT 3	NUM 13	BIT 3

3.5.2.4.2.10 Keyer Control Serial Input - Keyer Control Serial Input shall be as tabulated. Alpha code and numeric code shall apply as respectively listed in Tables I and II. Keyer Control dispolays identification shall be as depicted.

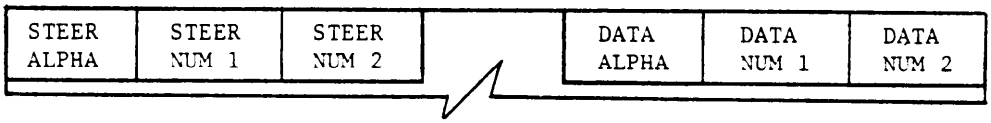
MSB				KEYER CONTROL DISPLAY (2 words)											LSB
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Typ each word, Word 1 sent first, Bit 0 sent out first

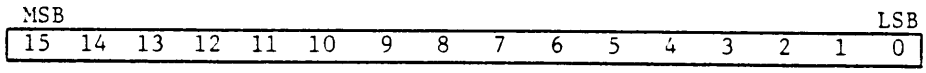
BIT	FUNCTION, WORD 1		FUNCTION, WORD 2	
0	STEER NUM 2	BIT 0	DATA NUM 2	BIT 0
1	STEER NUM 2	BIT 1	DATA NUM 2	BIT 1
2	STEER NUM 2	BIT 2	DATA NUM 2	BIT 2
3	STEER NUM 2	BIT 3	DATA NUM 2	BIT 3
4	STEER NUM 1	BIT 0	DATA NUM 1	BIT 0
5	STEER NUM 1	BIT 1	DATA NUM 1	BIT 1
6	STEER NUM 1	BIT 2	DATA NUM 1	BIT 2
7	STEER NUM 1	BIT 3	DATA NUM 1	BIT 3
8	STEER ALPHA	BIT 0	DATA ALPHA	BIT 0
9	STEER ALPHA	BIT 1	DATA ALPHA	BIT 1
10	STEER ALPHA	BIT 2	DATA ALPHA	BIT 2
11	STEER ALPHA	BIT 3	DATA ALPHA	BIT 3
12	STEER ALPHA	BIT 4	DATA ALPHA	BIT 4
13	STEER ALPHA	BIT 5	DATA ALPHA	BIT 5
14	STEER ALPHA	BIT 6	DATA ALPHA	BIT 6
15	STEER ALPHA	BIT 7	DATA ALPHA	BIT 7

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KEYER CONTROL DISPLAYS IDENTIFICATION



3.5.2.4.2.11 KC Light Incoming



BIT	FUNCTION
0	SPARE
1	SPARE
2	SPARE
3	SPARE
4	SPARE
5	POS UPDATE - GREEN LT
6	POS UPDATE - AMBER LT
7	SPARE
8	
9	
10	
11	
12	
13	
14	
15	SPARE

For All Lights: Logic 1 = Light ON
Logic 0 = Light OFF

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3.5.2.4.3 Signal Outputs to the NC - Signal outputs to the NC shall be functional interrupt pulses, discrete outputs, and serial data

3.5.2.4.3.1 Interrupt Pulse - The Interrupt pulse shall be a differential logical "1" pulse of 500 ns ± 40 percent duration. Other pulse characteristics shall be as detailed in Paragraph 3.5.2.4.1. An interrupt pulse shall be output within 1 ms whenever any switch function listed within the interrupt word of Paragraph 3.5.2.4.3.3 is actuated.

3.5.2.4.3.2 Discrete Outputs - Discrete outputs shall be steady state logic signals output on individually dedicated lines for the following functions:

BULK ERASE, BIT } Logical "1" = Switch depressed; Logical "0" = Switch released

Bulk Erase signal shall only be enabled if the ENTER PBI is also depressed. BIT signal shall output when the BIT input signal is received as a BIT CMD Logical "1" (Paragraph 3.5.2.4.4.7).

3.5.2.4.3.3 Interrupt Word Serial Output - Interrupt Word serial output to the NC shall be as tabulated. Keyboard bit code shall apply as listed in Table III. Logical "1" shall indicate PBI actuation, and all word bits shall be logical "0" after the word dump.

INTERRUPT WORD

MSB															LSB
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

BIT	FUNCTION
0	SPARE
1	SPARE
2	SPARE
3	SPARE
4	DRIFT ACPT
5	POS UPDATE SW
6	SPARE
7	SPARE
8	SPARE
9	ENTER SW
10	KEYBOARD BIT 0
11	KEYBOARD BIT 1
12	KEYBOARD BIT 2
13	KEYBOARD BIT 3
14	KEYBOARD BIT 4
15	KEYBOARD BIT 5

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TABLE III. Alphanumeric keyboard bit code.

KEY		IDENT [2]	BIT 0	BIT 1	BIT 2	BIT 3	BIT 4	BIT 5
PRI	SEC		LSB					MSB
A	1		1	0	0	0	0	0
B	4		0	1	0	0	0	0
C	7		1	1	0	0	0	0
D	-		0	0	1	0	0	0
E	2		1	0	1	0	0	0
F	5		0	1	1	0	0	0
G	8		1	1	1	0	0	0
H	0		0	0	0	1	0	0
I			1	0	0	1	0	0
J	3		0	1	0	1	0	0
K	6		1	1	0	1	0	0
L	9		0	0	1	1	0	0
M	+		1	0	1	1	0	0
N	MODE		0	1	1	1	0	0
O	GT GS		1	1	1	1	0	0
P			0	0	0	0	1	0
Q			1	0	0	0	1	0
R			0	1	0	0	1	0
S	POS		1	1	0	0	1	0
T	BRG RNG		0	0	1	0	1	0
U	TIME ETE		1	0	1	0	1	0
V	DRIFT		0	1	1	0	1	0
W	ALT		1	1	1	0	1	0
X	TH TAS		0	0	0	1	1	0
Y	MH VAR		1	0	0	1	1	0
Z	WD WV		0	1	0	1	1	0
CLEAR △			1	1	0	1	1	1
CLEAR ▽			0	0	1	1	1	1
SKIP			1	0	1	1	1	1
DATA EXIT			0	1	1	1	1	1

Bit code shall be all zeros when PBI's are not depressed.

[2] "Primary" and "Secondary" do not refer to priority of use but only alternate uses according to the specific parameter in use.

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3.5.2.4.3.4 Outgoing Word - Outgoing word to the NC shall be as tabulated.

MSB				OUTGOING WORD										LSB	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

BIT	FUNCTION
0	SPARE
1	SPARE
2	SPARE
3	SPARE
4	STEER SWITCH
5	STEER SWITCH
6	BLANK KEY
7	SPARE
8	DATA SWITCH
9	DATA SWITCH
10	DATA SWITCH
11	SPARE
12	SPARE
13	DISPLAY HOLD
14	TEST BIT - LOGIC ONE
15	TEST BIT - LOGIC ZERO

STEER SWITCH CODE		
POSITION	WORD BIT 4	WORD BIT 5
DISENG	0	0
NAV MAN	1	0
NAV AUTO	0	1
TACCO	1	1

DATA SWITCH CODE			
POSITION	WORD BIT 8	WORD BIT 9	WORD BIT 10
FLT	0	0	0
UPDATE	1	0	0
WAY PT	0	1	0
MARKER	1	1	0
MODE	0	0	1
SENSOR	1	0	1

PUSH BUTTONS	WORD BIT	ACTUATED (DOWN)	NOT ACTUATED (UP)
DISPLAY HOLD	13	1	0
BLANK KEY	6	1	0

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3.5.2.4.4 Interface with Digital Display Indicator - Refer to data function of Paragraph 3.5.2.1 and requirements listed in Paragraph 3.5.3. The data serial train for all sections of the Digital Display Indicator shall be output on a common line. Section selection shall be accomplished by outputting a gated clock on one of four clock outputs. Individual discrete outputs shall energize the two BIT status lights. Dimming and test signals shall be input to the Keyer Control to simultaneously provide these functions to the display.

3.5.2.4.4.1 Serial Data Outputs - Output characteristics and logic level definition shall be as defined in Paragraph 3.5.2.4.1 and Figure 4. Waveform shall be as depicted in Figure 6. Word structure, bit Functional assignments, and alphanumeric codes shall be as defined in Paragraphs 3.5.2.4.2.6 through 3.5.2.4.2.9.

3.5.2.4.4.2 Gated Clock Outputs - Four gated clock outputs, one for each section of the Digital Display Indicator, shall be provided. A clock signal shall appear individually on one line synchronously with data outputs. Characteristics and waveform shall be as defined in Paragraph 3.5.2.4.1 and Figure 6.

3.5.2.4.4.3 Reset Output - A differential logical "1" pulse shall be put out upon receipt of each command word. Waveform shall be as depicted in Figure 7.

3.5.2.4.4.4 Discrete Outputs - Dedicated steady state logic level outputs from the Keyer Control to the Digital Display Indicator shall be provided for the following BIT status indicators:

MALF, CMPTR } Logical "1" = Light illuminated
Logical "0" = Light extinguished

3.5.2.4.4.5 Signal Input - Signal inputs to the Keyer Control from the Digital Display Indicator shall be unbalanced single-ended logic signals on dedicated lines. Logic levels shall be referenced to logic grounds as listed:

Logical "1" = 2.4 to 5.5 VDC
Logical "0" = 0 to 0.4 VDC

3.5.2.4.4.6 Display Test Signal Input - A steady state logic level shall be input simultaneously with the lamp test of the Digital Display Indicator as listed.

Logical "1" = Lamps extinguished
Logical "0" = Lamps illuminated

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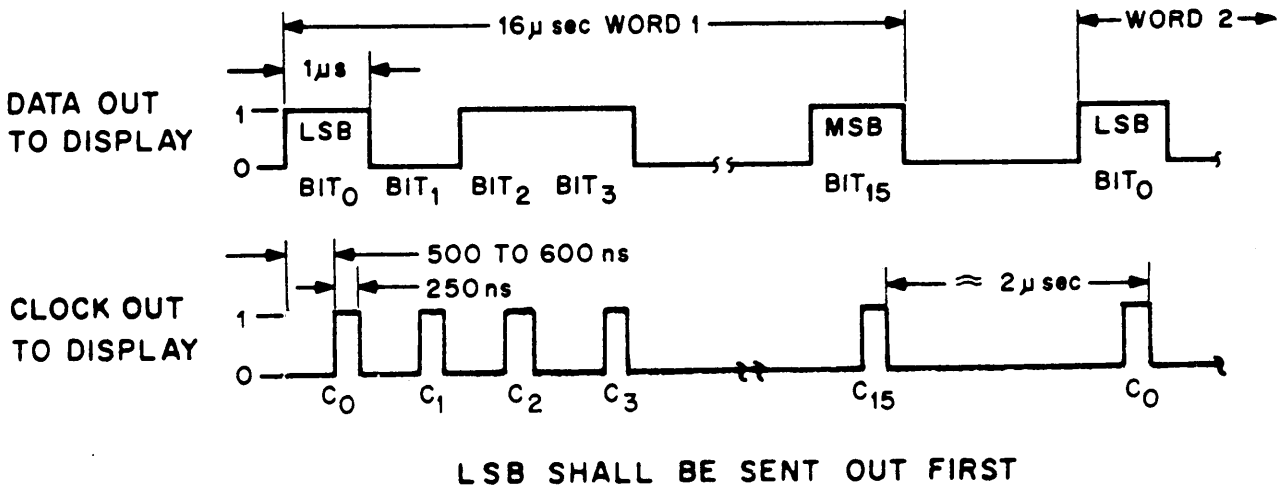


FIGURE 6. Serial data waveform.

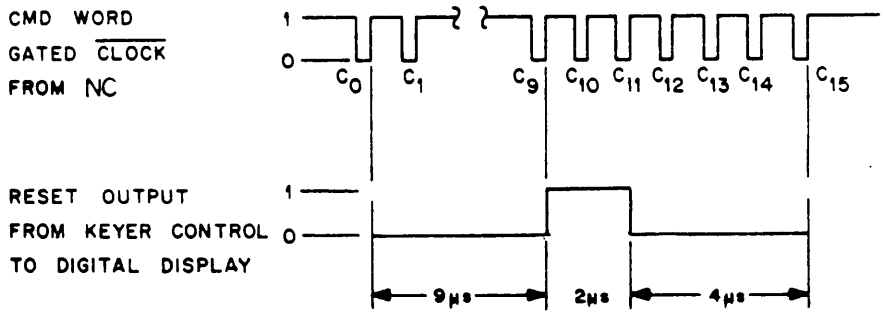


FIGURE 7. Reset output.

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3.5.2.4.4.7 BIT Signal Input - A steady state logic level shall be an input from the Digital Display Indicator. When the BIT CMD is received, the Keyer Control shall output the discrete BIT (logical "1") per Paragraph 3.5.2.4.3.2.

3.5.2.5 Controls - The Keyer Control shall contain the designated controls and display windows with positional placements and labeling depicted in Figure 8.

Labeling for rotary switches shall be transillumi-nated for low-ambient viewing. Knobs shall be compatible gray per FED-STD-595. All lighting elements shall be front replaceable.

The numeric, flight data access control, and miscel-laneous control PBI's (24) shall be of black background; remaining PBI's (12) shall be of gray background. Labeling of all alphanumerics, flight data access control, and program control PBI's (32) shall be white. Illumination of all alphanumeric, flight data access control, and program control PBI's shall be white per MIL-P-7788 derived from 5 VAC excitation. controlled by an external 26 VAC dimming control. Legends of the four miscellaneous control PBI's shall be translucent for green or amber illumination as specified below.

3.5.2.5.1 STEER - The STEER control shall be a single-deck four position switch. The switch control shall enable assigned bits into the SC on the Outgoing Word as defined in Paragraph 3.5.2.4.3.4.

3.5.2.5.2 DATA - The DATA control shall be a single-deck six position rotary switch. The switch control shall be used in conjunction with the associated Digital Display window and keyboard to select, monitor, insert, and change navigation data list parameters. Position definitions with control functions shall enable assigned bits into the NC on the Outgoing Word as defined in Paragraph 3.5.2.4.3.4.

3.5.2.5.3 Keyboard - The keyboard control shall consist of an array of momentary action and four pushbutton indicator switches (PBI) arranged and labeled as depicted in Figure 8. PBI actuation of the keyboard shall enable insertion of flight data into the navigation system and select parameters for display on the Digital Display Indicator (Paragraph 3.5.3). PBI control functions shall be in four groupings as defined.

*3.5.2.5.3.1 Flight Data Access Control - Flight data shall be immediately accessible in preemption over sequential access lists. Access control shall consist of the 10 PSI's listed. PBI actuation shall initiate respective bit insertion of the Interrupt Word into the NC as defined in Paragraph 3.5.2.4.3.3.

MODE[N]	POS[S]	TIME[U] ETE	ALT[W]	MH[J] VAR
GT[O] GS	BRG[T] RNG	DRIFT[V] BIAS	TH[X] TAS	WD[Z] WV

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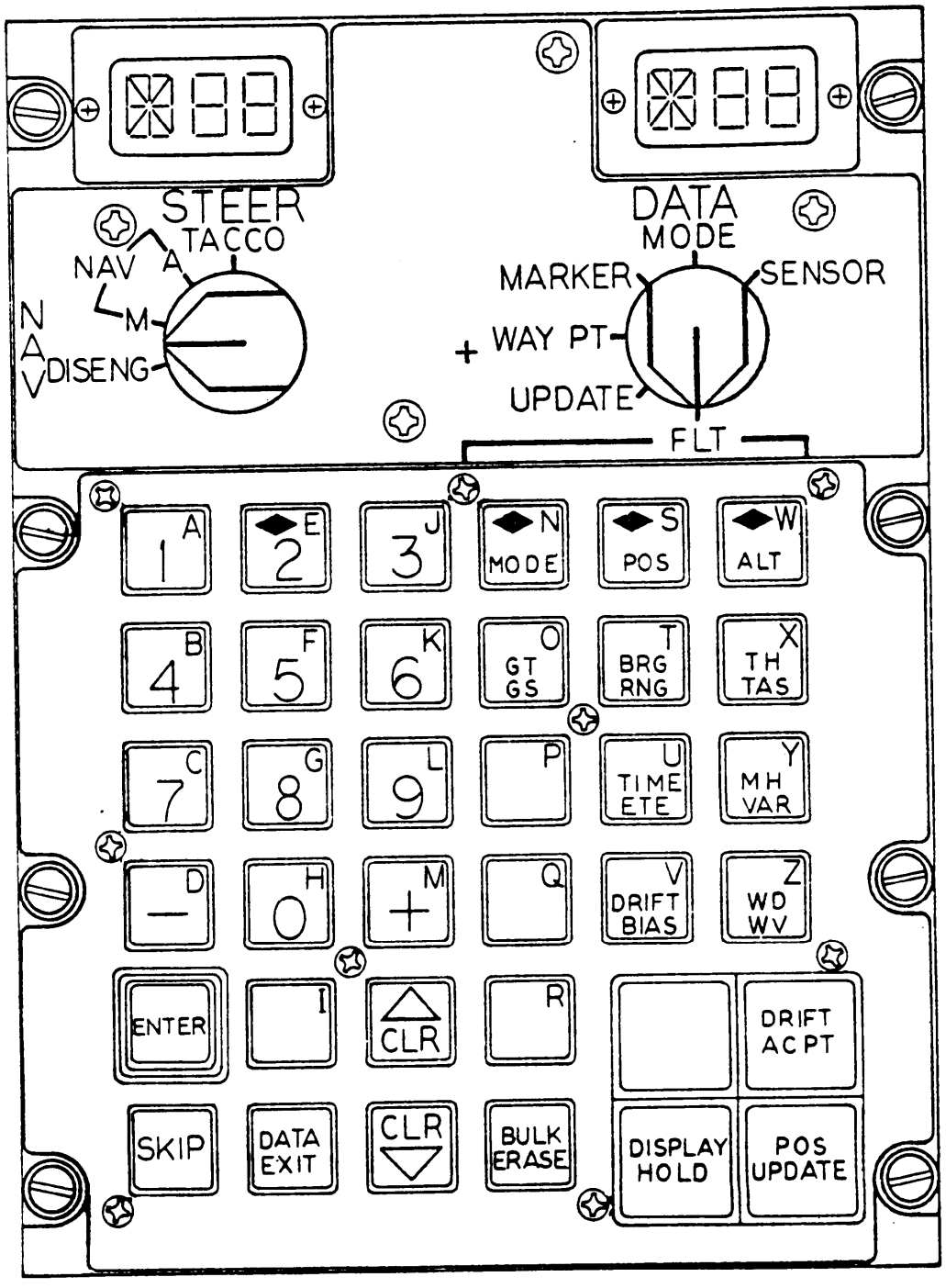


FIGURE 8. Keyer Control front panel.

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3.5.2.5.3.2 Program Control - Program control shall consist of 6 momentary PBI's. Actuation of the program control PBI's shall enable assigned bits into the NC on the Interrupt Word as defined in Paragraph 3.5.2.4.3.3.

CONTROL PBI		
Δ CLR	SKIP	ENTER
CLR ∇	DATA EXIT	BULK ERASE

ENTER enables data displayed on the alphanumeric display into the NC memory. BULK ERASE in conjunction with ENTER erases all data from NC memory.

3.5.2.5.3.3 Miscellaneous Control - Four PBI's shall be provided in the lower right portion of the keyboard panel. Switch actuation shall enable bits into the NC as defined in Paragraph 3.5.2.4.3.4. 'The blank switch shall be installed and wired for future use (TBD). Switch functions shall be as follows:

(Blank)	Alternate action
DRIFT ACPT	Momentary action; illuminated green
DISPLAY HOLD	Alternate action; normally illuminated green, amber when depressed, remains amber until again depressed, returns to green.
POS UPDATE -	Momentary action; illuminates green or amber under software control

3.5.2.5.3.4 Alpha and Numeric Control - Sixteen alpha and numeric PBI's shall be provided to enable assigned bits into the NC on the Interrupt Word as defined in Paragraph 3.5.2.4.3.3.

1^{A}	2^{E}	3^{J}	I
4^{B}	5^{F}	6^{K}	P
7^{C}	8^{G}	9^{L}	Q
$-^{D}$	0^{H}	$+^{M}$	R

3.5.2.6 Display Windows - The Display Windows shall each provide an alpha and two numerics controlled by the serial data word initially generated as defined in Paragraph 3.5.2.4.2.10.

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3.5.2.7 Panel Illumination - The Control shall incorporate an integrally illuminated information panel per MIL-P-7788, Class 2W, Type V. (See Figure 8.) Excitation shall be 5 VAC derived from the dimmable 26 VAC from the external dim control.

*3.5.2.8 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function
J1	SJTP02RE18-35S(023)	Digital Displays
J2	SJTP02RE14-35P(023)	Data and Discretes
J4	MS27508E24F24PB	Logic and Lighting Power

3.5.2.9 System Power - The equipment shall utilize system power as follows:

(1) 115 VAC, 400 Hz, 3 0, 60 VA, from the SDC.

(2) 0 - 26 VAC, 400 Hz, 1 0, 0.8 Amp, from the SDC.

(3) 0 - 4.5 VDC regulated, from the Digital Display Indicator, 6.5 watts.

(4) 5 VDC regulated, 24 watts from the SDC.

3.5.2.10 Mounting - Chassis and mounting requirements of YIIL-C-6781 and MS25212 shall apply.

3.5.2.11 Color - Surface shall be painted lusterless gray, No. 36231 in accordance with FED-STD-595.

3.5.3 Digital Display Indicator (DDI) ID-2035/ASN-124 - The Digital Display Indicator shall meet the following requirements:

3.5.3.1 Function - The Digital Display Indicator shall interface directly with the Keyer Control as defined in Paragraph 3.5.2. The Digital Display Indicator shall function as the primary navigational alphanumeric display from the NC to the operator. The displays shall be capable of being updated independently and in any sequence.

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3.5.3.2 Form Factor - The Digital Display Indicator, exclusive of front panel protrusions, attachment points, and connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
3.000	5.750	5.75 (behind mounting surface)

3.5.3.3 Weight - The weight of the Digital Display Indicator shall not exceed 3.0 pounds.

3.5.3.4 Contents - The Digital Display Indicator shall contain logic circuitry functions interfacing the Keyer Control.

3.5.3.4.1 Data Serial Train - The data serial train for all four sections of the Digital Display Indicator shall be entered on a common line. Selection shall be by presenting a gated clock on only one of the four clock inputs. Data for alphas shall be coded American Standard Code for Information Interchange (ASCII), and the code for numerics shall be 4-bit binary-coded decimal (BCD). Individual bits shall control decimals, degrees, and non-BIT status lights.

3.5.3.4.2 Logic Signal Circuitry - Logic signal circuitry shall be of the differential line receiver type, similar to µA9615, and shall be compatible with signals supplied by a differential line driver similar to µA9614. Logic level definition shall be as presented in Figure 4.

3.5.3.4.3 Gated Clock Inputs - Four dedicated clock inputs (one for each display section) shall be provided. The positive - going clock, on positive line, shall appear on one input at one time during updating of the section. Clock gating shall be in word lengths of 16 clock-cycles with word numbers as listed:

Section I - 3 words
Section II, III, and IV - 2 words

Waveforrn and synchronization with data input shall be as depicted in Figure 9.

3.5.3.4.4 Reset Input - The reset input shall be a differential logical “1” going pulse of 2 µsec duration preceding any display update. The reset pulse shall be sent out by the Keyer Control whenever a command word is received from the NC. Based on this logic, a reset pulse can be received without a subsequent series of gated-clock signals, however a gated-clock signal shall always be preceded by a reset pulse (Figure 9).

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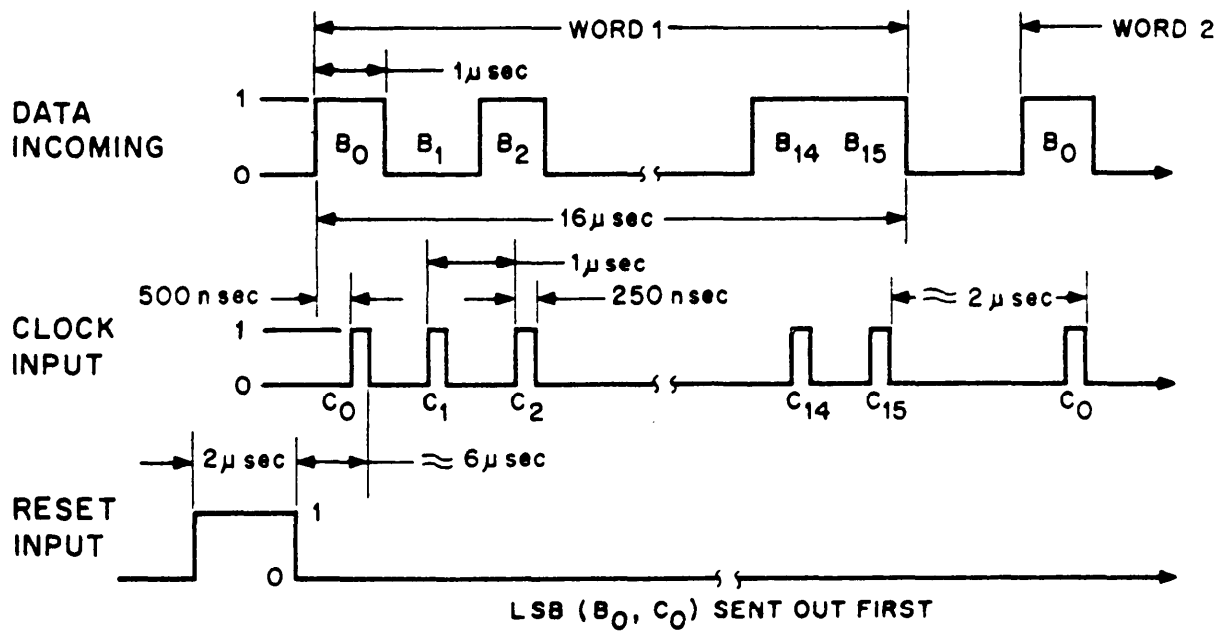


FIGURE 9. Gated clock format.

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3.5.3.4.5 Serial Data Input - Serial data input shall be of waveform and synchronization as depicted in Figure 9. Input shall be on a common line for all display sections. Steering (directing) and clocking shall be according to Paragraph 3.5.3.4.3.

The Serial Data Input Word Structure, and Digital Displays II, III, and IV shall be as listed. Bit "0" shall be sent out first.

DATA	SECTION	TABLE
Serial Data Input Word Structure	3.5.2.4.2.6	1
Digital Display II	3.5.2.4.2.7	1
Digital Display III	3.5.2.4.2.8	2
Digital Display IV	3.5.2.4.2.9	2

3.5.3.4.6 Discrete Inputs - Individually dedicated logic level discrete inputs shall be provided for MALF and CMPTR BIT status indicators.

(1) MALF Command - An input command to energize the MALF lamp shall only illuminate the lamp of interest.

(2) CMPTR Command - An input command to energize the CMPTR lamp shall illuminate both the MALF and CMPTR lamps. The CMPTR input shall be biased to the logic 1 state. Logic levels shall be as listed:

Logical "1" = Light(s) illuminated
Logical "0" = Light(s) extinguished

3.5.3.4.7 Logic Signal Outputs - Signal outputs shall be unbalanced single-ended logic signals on dedicated output lines. Logic levels shall be referenced to logic ground as listed.

Logical "1" = 2.4 to 5.5 VDC
Logical "0" = 0 to 0.4 VDC

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3.5.3.4.7.1 Display Test Signal Output - A steady state logic level shall be a simultaneous output to the Keyer Control with the lamp test of the display as listed:

Logical "1" = Lamps extinguished
Logical "0" = Lamps illuminated

3.5.3.4.7.2 BIT Signal Output - A steady state logic level shall be output to indicate actuation of the BIT pushbutton switch as listed:

Logical "1" = Switch depressed
Logical "0" = Switch released

3.5.3.5 Displays and Controls - The Digital Display Indicator shall contain designated controls and displays with positional placements and labeling depicted in Figure 10. Labeling shall be transilluminated for low-ambient viewing. Display illumination shall be controlled by the rotatable DIM knob. The knob shall be finished compatible gray per FED-STD-595. All lighting elements shall be front replaceable. Panel illumination shall be white per MIL-P-7788 derived from 5 VAC excitation controlled by a dimmable 26 VAC cockpit dim control. The three Advisory and two BIT Status Indicators shall be identified by illumination of an amber light mounted immediately below each listed indicator.

3.5.3.5.1 Data Display - The data display shall consist of three rows of alphanumeric characters. The top row shall contain six alpha-numeric characters, primarily of interactive routine. The middle and bottom rows shall each contain one alpha and eight numeric characters. The two rows of data shall allow simultaneous display of logically grouped parametric pairs. Data insertion shall first be presented at extreme left as a parameter identifier with inserted numeric values first appearing at extreme right, shifting left across display in sequence of entry. The displays shall be segmented incadescent type, with an external red filter.

3.5.3.5.2 Test Control - Test shall be effected by actuating the two PBI's indicated.

LMP - All filaments of the Data Display Indicator lamps and the five Advisory Status Indicator lamps shall illuminate simultaneously with steady state logic level signal output per Paragraph 3.5.3.4.7.1.

BIT - Steady state logic level shall be output per Paragraph 3.5.3.4.7.2 and 3.5.2.4.4.7.

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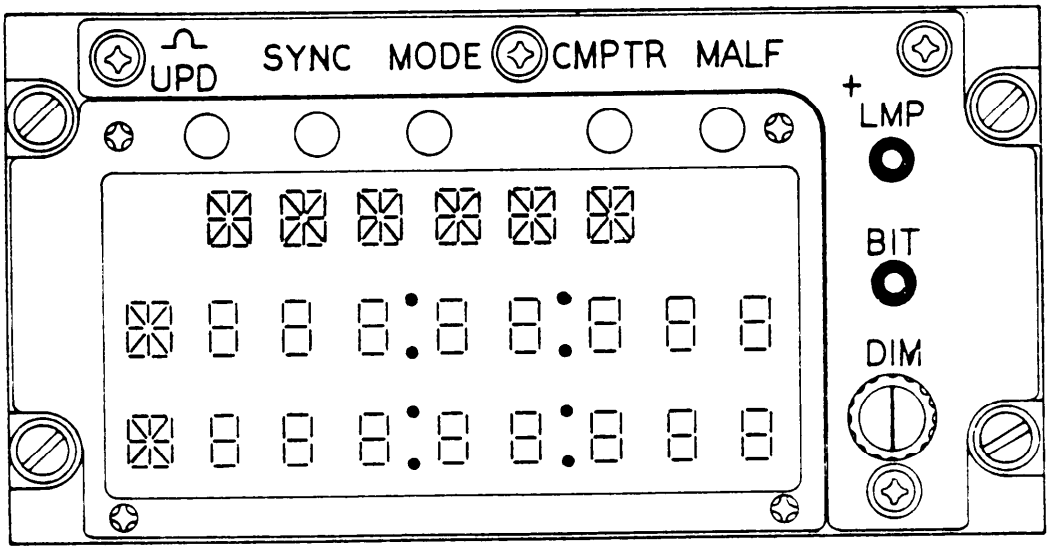


FIGURE 10. Digital Display Indicator front panel.

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*3.5.3.6 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function
J1	SJTP02RE18-35PA(023)	Digital Displays Logic Power

3.5.3.7 System Power - The equipment shall utilize system power from the Keyer Control as follows:

(1) 115 VAC, 400 Hz, 1 0, Category B, 0.004 Amps

(2) 0 - 5 VAC, 400 Hz, 1 0, 0.6 Amps

(3) 6 VDC, 6.5 Amps.

(4) 5 VDC, regulated, 2.6 Amps.

3.5.3.8 Mounting - Chassis and mounting requirements of MIL-C-6781 and MS25212 shall apply.

3.5.3.9 Color - Surface shall be painted lusterless gray, No. 36231 in accordance with FED-STD-595.

3.5.4 Control, Marker Plotter C-9855/ASN-124 - The Control shall meet the following requirements:

3.5.4.1 Function - The Control is used by the pilot to control the entry and display of data on the Tactical Data Display IP-886/MA-66 (or equivalent). It includes the symbol generator circuits which drive that Display.

3.5.4.2 Form Factor - The Marker Plotter Control, exclusive of front panel protrusions, attachment points, and connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
5.250	5.750	9.000 (behind mounting surface)

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3.5.4.3 Program Control - The weight of the Control shall not exceed 8 pounds.

3.5.4.4 Contents - The Control shall contain the following major functions:

3.5.4.4.1 Command Word - The Control shall provide three Command Words, 1 through 3, as presented in Figure 11. The number of word pairs (NOWP) shall indicate the symbol data that is to be transmitted. Whenever the Command Words are transmitted, the Scale Factor data must also be transmitted. NOWP shall be in binary-2 code from 2 through 32 as shown in Figure 11.

3.5.4.4.2 Symbol Generator

3.5.4.4.2.1 Symbology - The Control shall provide electronic signals to the Tactical Data Display IP-886A/ASA-66 (or equivalent) to generate the following symbology:

3.5.4.4.2.1.1 Scale Factor - This shall be a four digit numeric as dictated by Input Data Words 4-7. The numerals shall be 0.125″ high with a width to height ratio of 3:4. Spacing between numerals shall be approximately 1/2 the numeral width. The numerals shall be located in the 6 o'clock position near the perimeter of the display (See Figure 12).

3.5.4.4.2.1.2 Aircraft - This symbol shall be a circle with a line extending from its center. The center of the circle represents aircraft position; the line direction represents aircraft ground track. The circle shall be 0.25″ diameter. The line shall be 0.50″ long. The center of the circle shall be positioned as dictated by Input Data Words 8 and 9. The line shall extend from the center at an angle as dictated by Input Data Words 10 and 11. (See Figures 13 and 14.)

3.5.4.4.2.1.3 Fly to Point - This symbol shall be an "X". The "X" shall consist of two 0.35″ lines intersecting each other at 90 degrees and intersecting the vertical or Y axis of the display at 45 degrees. The intersection of the lines shall be as dictated by Input Data Words 12 and 13. (See Figure 15.)

3.5.4.4.2.1.4 Markers - A marker shall be shown by a dot followed by two numerals. The dot shall indicate the marker location and be positioned as dictated by Input Data Words 16-61 in successive word pairs, with one pair for each marker. The numerics shall be 0.125″ high with a width to height ratio of 3:4. Spacing between the dot and numerics and between the two numerals shall be approximately 1/2 the numeral width. The dot shall be located at the mid-point of the numeral height. (See Figure 17.)

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Bit No.	Word 1	Word 2	Word 3
LSB 1	0	0	NOWP : 2^5
2	↓	↓	0
3	↓	↓	0
4	↓	↓	0
5	↓	↓	0
6	↓	0	1
7	↓	SCALE: 2^0	0
8	↓	SCALE: 2^1	0
9	↓	SCALE: 2^2	0
10	↓	SCALE: 2^3	1
11	↓	SCALE: 2^4	0
12	↓	NOWP: 2^0	0
13	↓	NOWP: 2^1	0
14	↓	NOWP: 2^2	0
15	↓	NOWP: 2^3	0
MSB 16	0	NOWP: 2^4	0

SCALE	2^0	2^1	2^2	2^3	2^4
2	0	1	0	0	0
3	1	1	0	0	0
4	0	0	1	0	0
5	1	0	1	0	0
. . .					
30	0	1	1	1	1

NOWP	2^0	2^1	2^2	2^3	2^4	2^5
2	0	1	0	0	0	0
3	1	1	0	0	0	0
4	0	0	1	0	0	0
5	1	0	1	0	0	0
. . .						
30	0	1	1	1	1	0
31	1	1	1	1	1	0
32	0	0	0	0	0	1

FIGURE 11. Command words and NOWP binary-2 code.

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Bit No.	Word 4	Word 5	Word 6	Word 7
LSB 1 2 3 4 5 6	Thousand Digit Code	Hundreds Digit Code	Tens Digit Code	Unit Digit Code
7 8 9 10 11 12 13 14 15 MSB 16	X Position	Y Position	X Position	Y Position

Digit	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Ø	1	1	0	0	0	0
1	1	1	0	0	0	1
2	1	1	0	0	1	0
3	1	1	0	0	1	1
4	1	1	0	1	0	0
5	1	1	0	1	0	1
6	1	1	0	1	1	0
7	1	1	0	1	1	1
8	1	1	1	0	0	0
9	1	1	1	0	0	1
*	1	0	1	0	1	0
N	0	0	1	1	1	0
T	0	1	0	1	0	0

The X, Y-position data for Wd 4 & Wd 5 refers to .XXXX position as shown:

The X, Y-position data for Wd 6 & Wd 7 refers to XX.XX position as shown:

FIGURE 12. Scale factor data words 4 through 7.

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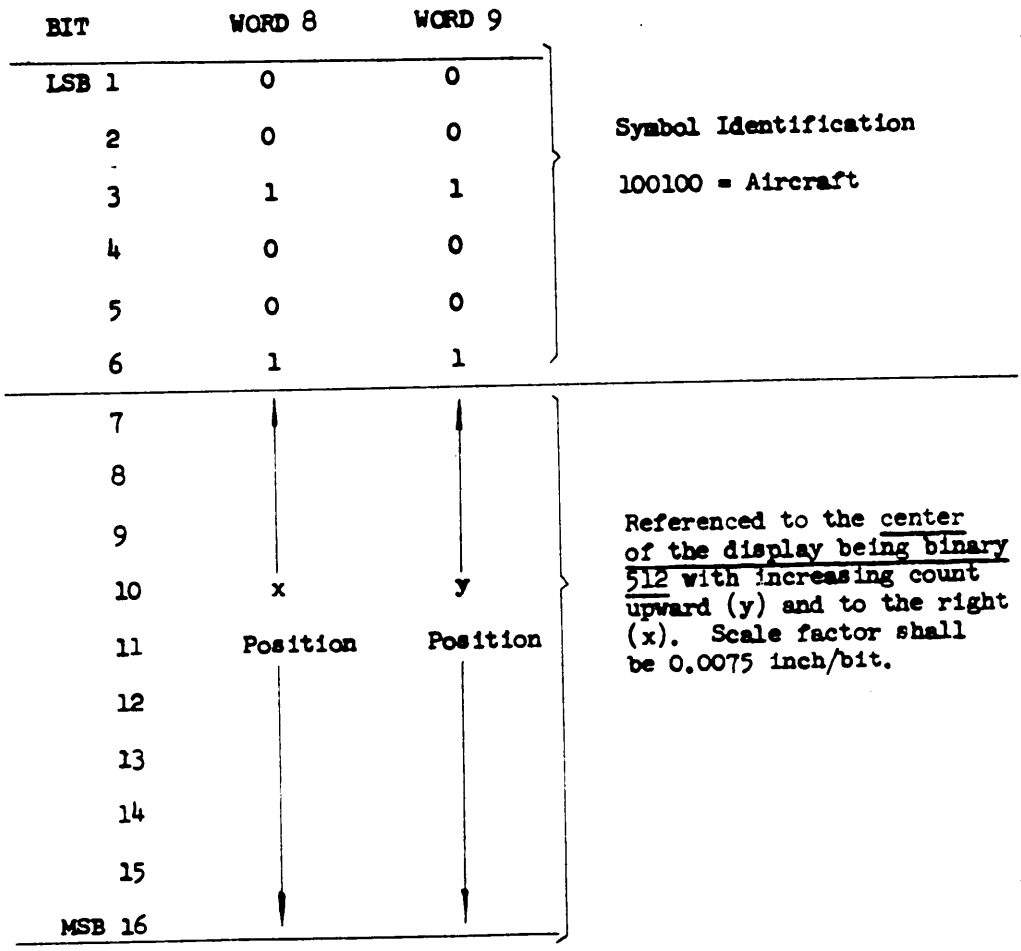


FIGURE 13. Aircraft symbol position data words 8 and 9.

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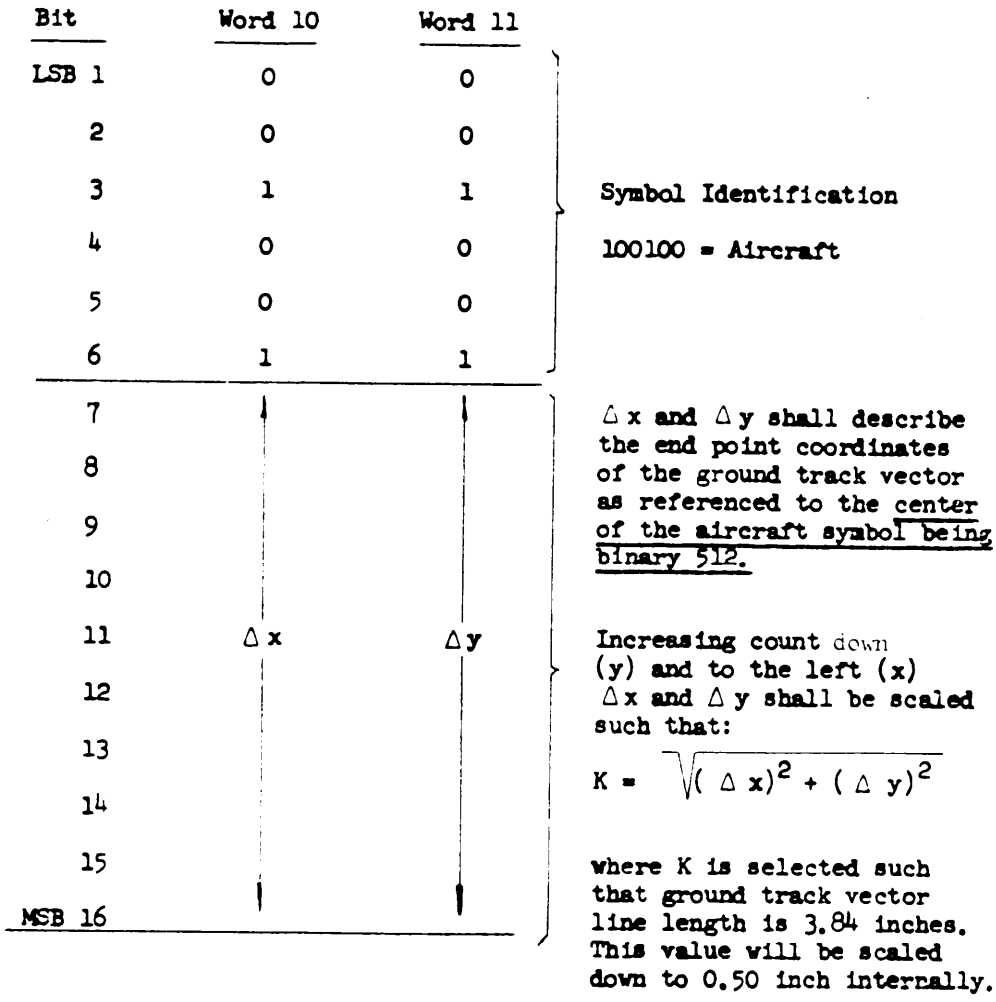


FIGURE 14. Aircraft symbol vector termination data words 10 and 11.

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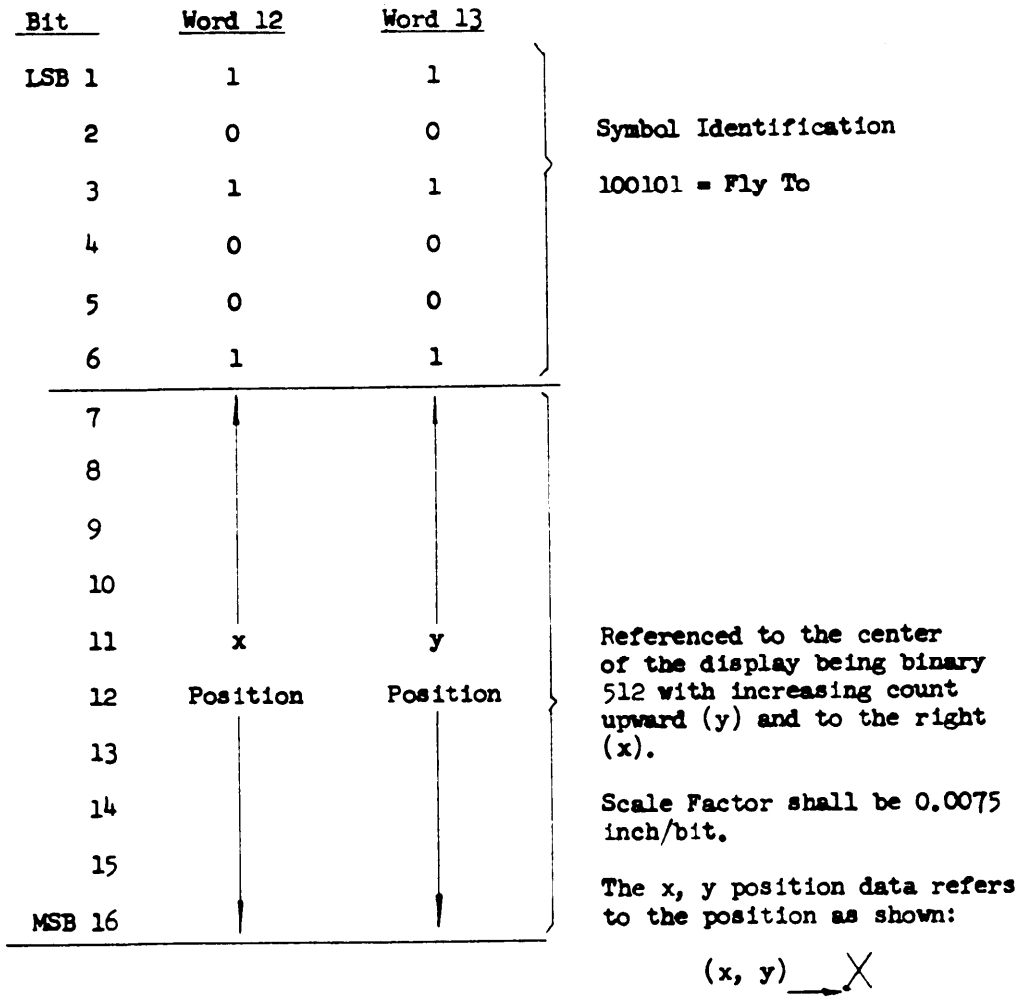


FIGURE 15. Fly to position data words 12 and 13.

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Bit	Word 14	Word 15
LSB 1	0	0
2	1	0
3	0	0
4	1	0
5	0	0
6	1	1
7		
8		
9		
10		
11	x	y
12	Position	Position
13		
14		
15		
MSB 16		

Symbol Identification for Priority Fly To (asterisk).

The Priority Fly To marker shall count as one of the 15 markers of Para. 3.5.4.4.2.3 and Figure 17.

Referenced to the center of the display being binary 512 with increasing count upward (y) and to the right (x).

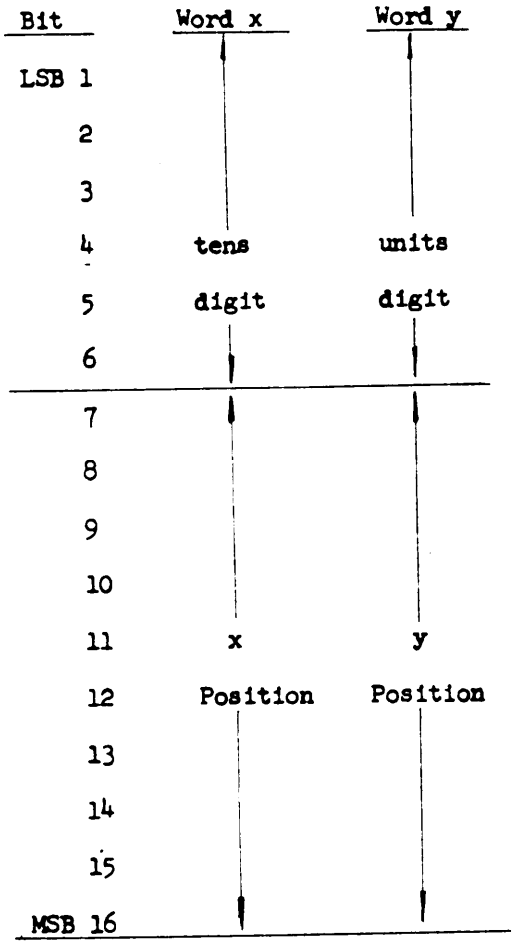
Scale Factor shall be 0.0075 inch/bit.

The x, y position data refers to the position as shown:

(x, y)—·

FIGURE 16. Priority fly to position data words 14 and 15.

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Marker two digit number identifier. (LSB first), referenced to the display center being binary 512 with increasing count upward (y) and to the right (x).

Scale factor shall be 0.0075 inch/bit.

The x, y position data refer to position as shown:

——— •XX

Nominal word allocation shall be:

1 Command - 3 words

1 Scale Factor - 4 words

1 acft. symbol - 4 words

1 fly-to symbol - 2 words

15 markers - 30 words

FIGURE 17. Marker symbol position data words X and Y.

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3.5.4.4.2.1.5 Malfunction - The malfunction symbol shall be a large "X". The "X" shall consist of two 2.0" lines intersecting each other at 90 degrees. The symbol shall be positioned in the center of the CRT intersecting the vertical or Y axis of the display at 45 degrees. Display of the Malfunction symbol shall result in removal of all other symbols.

3.5.4.4.2.1.6 Priority Fly To - This symbol shall be an *. The * shall be 0.125 in height and with a width to height ratio of 3:4. See Figure 16.

3.5.4.4.2.2 Scaling

3.5.4.4.2.2.1 Scale Factor - The scale factor shall be 0.0075 inch/bit and 0.543 inch/volt, nominal. These values are adjustable at the Display Unit ± 10%.

3.5.4.4.2.2.2 Sense - The center of display shall be defined as binary 512 in x = 0 VDC and binary 512 in y = 0 VDC. Increasing count and voltage shall be defined as upward or to the right from display center, except for aircraft vector per Figure 14.

3.5.4.4.2.3 Display Parameters - The Control shall provide the capability to display 15 markers (Figure 17) simultaneously without perceptible flicker.

3.5.4.4.2.3.1 Brightness - The Control shall provide signals to the CRT Display Unit (ASA-66A or equivalent - P43 phosphor) such that a brightness level of 500 foot-lamberts minimum can be achieved by any symbol.

3.5.4.4.2.3.2 Uniformity - The Control shall provide signals to the CRT Display Unit such that the display illumination level shall be uniform within 25% from symbol to symbol.

3.5.4,4.2.3.3 Update Rate - The Control shall update the CRT Display Unit 40 times/second minimum.

3.5.4.4.3 Input/Output Signal Characteristics - General timing relationships and signal characteristics shall be as shown in Figure 18. The Control shall be serviced by the NC on a periodic non-interrupt basis.

3.5.4.4.3.1 Input Signal Characteristics - The following signals shall come from the NC via the SDC.

3.5.4.4.3.1.1 Serial Digital Input - The input data, clock, data enable, output data enable, and output data clock signals shall be balanced lines with high-noise immunity circuits (Type µA9614/9615, or equivalent).

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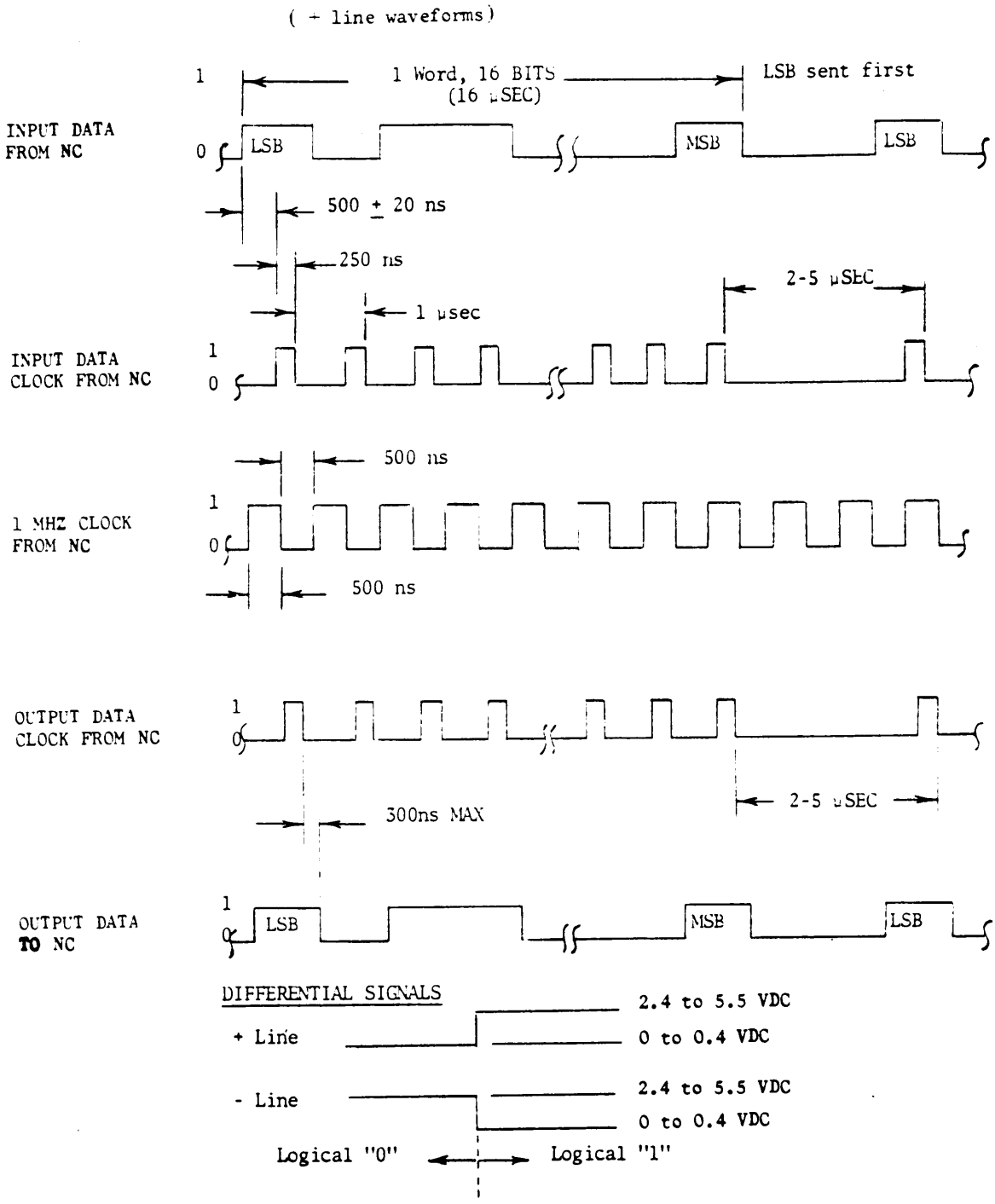


FIGURE 18. Input - output signal characteristics.

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*3.5.4.4.3.1.1.1 Input Data - The data shall consist of a serial train of a variable number of words, each containing 16 bits transmitted LSB first. The bit rate shall be 1 MHz. A spacing of 2 to 5 µsec shall exist between words. A spacing of at least 60 milliseconds shall exist between the end of a data train and the beginning of the next data train.

These data shall describe the following symbol types and be sent in the following order:

(1) Command Words (3 words)
(2) Scale Factor Symbol (4 words)
(3) Priority Fly To Symbol (2 words)
(4) Aircraft Symbol (4 words)
(5) Fly to Symbol (2 words)
(6) Marker Symbols (2 words each)

Data shall be organized in word pairs, such that symbol identification is established by bits 1 - 6 from word pairs, and x and y position data are established by bits 7 through 16 from word pairs. All position data are natural binary with LSB first. Word formats shall conform to Figures 11 through 17.

*3.5.4.4.3.1.1.2 Input Data Clock - A signal data input clock shall be provided consisting of sets of pulses with 16 pulses per set. Clock sets shall be separated by a 2 to 5 µsec pause. The clock rate shall be 1 MHz and shall be synchronous with the data. (See Figure 18.)

3.5.4.4.3.1.1.3 Input Data Enable - A discrete signal shall be provided which precedes the first clock pulse by 5 ± 2 µsec and remains until 5 ± 2 µsec after the last clock pulse.

3.5.4.4.3.1.1.4 1 MHz Clock - A continuous 1.0 MHz clock shall be provided. See Figure 18 for waveform definition. Computer Malfunction shall be indicated by a continuous logic “1” instead of the 1.0 MHz clock. When the continous logic “1” is received the malfunciton “X” symbol will be displayed on the CRT.

3.5.4.4.3.1.1.5 Output Data Enable - An Output Data Enable discrete shall be provided to request that the Output Words (2) be transmitted from the Control to the NC. The discrete signal shall precede the first clock pulse by 5 ± 2 µsec and remains until 5 ± 2 µs after the last clock pulse.

3.5.4.4.3.1.1.6 Output Data Clock - A signal Output Data Clock shall be received from the NC to facilitate transmission of the Output This clock signal shall consist of 2 sets of 1 MHz pulses, each set containing 16 pulses with 2 to 5 microseconds between sets.

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3.5.4.4.3.1.2 Discretes - The following discrete signals shall be received from the Navigation Computer on high noise immunity, balanced line circuits:

(1) Fly To - Illuminate FLY TO indicator light
(2) Enter - Illuminate the ENTER key
(3) Pattern Correct - Illuminate the PATT key

Each discrete shall be considered active when the line receiver is driven to the logic "1" state. See Figure 18 for differential signal logic definition.

3.5.4.4.3.2 Output Signal Characteristics

3.5.4.4.3.2.1 Deflection - The x deflection signal shall drive the horizontal deflection amplifier of the CRT display subassembly.

The y deflection signal shall drive the vertical deflection amplifier of the CRT display subassembly.

Both x and y Deflection Signals shall have the following characteristics:

(1) Scaling: From 0 to ± 6 to ± 8 volts deflects the beam one radius (± 3.8")

Nominal scaling - 1.84 volts/inch

(2) Sense: Positive signals deflect the beam upward and to the right.

(3) Impedance: The deflection signals shall be capable of driving 90 ohms and 100 pf thru a maximum cable length of 12 feet.

3.5.4.4.3.2. Blanking - The blanking signal shall provide Z axis modulation to the CRT display subassembly. It shall have the following electrical characteristics:

(1) Levels: The unblanked (white) signal shall be +5 ± 1 volts. The blanked (black) signal shall be 0 ± 1 volt.

(2) Impedance: The blanking signal shall be capable of driving 90 ohms and 100 pf thru a maximum cable length of 12 feet.

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3.5.4.4.3.2.3 Video - The video signal shall provide automatic control to maintain uniform brightness of the symbols being presented on the CRT.

(1) Levels: The output video level shall be between + 1.0 volt to 2.5 volts.

(2) Impedance: The video signal shall be capable of driving 90 ohms and 100 pf thru a maximum cable length of 12 feet.

3.5.4.4.3.2.4 Serial Digital Output - The following channel shall be established to transmit data from the Control to the Navigation Computer. The output data signals shall use high noise immunity balanced line circuits.

3.5.4.4.3.2.4.1 Output Data - The data shall consist of a serial train of two words, each containing 16 bits. The bit rate shall be 1 MHz. A spacing of 2 to 5 µsec shall exist between words.

The word format shall be as shown in Figure 19,

3.5.4.4.4 Built-In-Test (BIT) - The unit shall provide for two levels of test.

3.5.4.4.4.1 Hardware BIT - Circuits within the Control shall provide continuous monitoring as follows:

(1) Detection of failure to receive any logical "1's" on the Input Data Word line for a period of 0.5 to 3.0 seconds.

(2) Detection of failure to receive Input Data Clock for a period exceeding 0.5 to 3.0 seconds.

(3) Detection of failure to receive Input Data Ready for a period of 0.5 to 3.0 seconds.

(4) 1 MHz clock, ref. Para. 3.5.4.4.3.1.1.4.

Detection of any one of the above abnormalities shall result in display of the Malfunction Symbol. (Ref 3.5.4.4.2.1.5.)

3.5.4.4.4.2 Software BIT - Subsystem loop test capability shall be provided by a discrete output to the Computer operated by the Test switch.

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BIT	WORD 1		WORD 2	
LSB 1	TEST		INSR MKR	
2	AUTO SCALE		MKR CORR	
3	SPARE = 1		ACFT CORR	
4	SPARE = 0		GEO CORR	
5	DECR SCALE		SPARE = 1	
6	INCR SCALE		SPARE = 0	
7	RECTR ACFT		PATT CORR	LSB
8	RECTR FLY TO		ENTER	transmitted
9	1		0	first
10	2	Units BCD	0	
11	4		0	
12	8	MKR THUMBWHEEL SELECTOR	0	
13	1		0	
14	2	TENS BCD	0	
15	4		0	
MSB 16	8		0	

FIGURE 19. Output data word format.

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3.5.4.5 Operator Controls and Displays - See Figure 20

3.5.4.5.1 Marker Identification - The Control shall provide marker selection capability via two digit thumbwheel selector.

(1) Tens digit - 0 thru 6 and stop
(2) Units digit - 0 thru 9 and continuous to 0

3.5.4.5.2 Plotter Control - The Control shall provide momentary action, green lighted push-buttons switches for selection of:

3.5.4.5.2.1 Recenter on Aircraft - ACFT

3.5.4.5.2.2 Recenter on Fly to Point - FLY TO

3.5.4.5.2.3 Scale Factor Increase - INCR

3.5.4.5.2.4 Scale Factor Decrease - DECR

3.5.4.5.2.4 Automatic Downscaling - AUTO. The key shall be alternate action; illuminated green when the function is disengaged, and amber when engaged.

3.5.4.5.3 Test - The Control shall provide a test key-TEST. The key shall be alternate action; illuminated green when disengaged, and amber when engaged. Activation of the test function shall cause test of all lamps on the unit, and generation of the Test discrete to the NC.

3.5.4.5.4 On-Top - The Control shall provide the following pushbutton keys to designate the type of “ON TOP” correction:

(1) Geographic - GEO
(2) Aircraft - ACFT
(3) Marker - MKR

Each key shall be a momentary actuation type with electronic memory interlock. The key’s inactive color shall be illuminated green. When operated, the key changes to amber and remains amber until one of the following keys is operated:

(1) ENTER
(2) INSR MKR
(3) GEO or ACFT or MKR

If ENTER is depressed, all keys in the row revert to green. If any of the other keys, in the row, are depressed, the last key depressed will remain amber. All other keys will be green.

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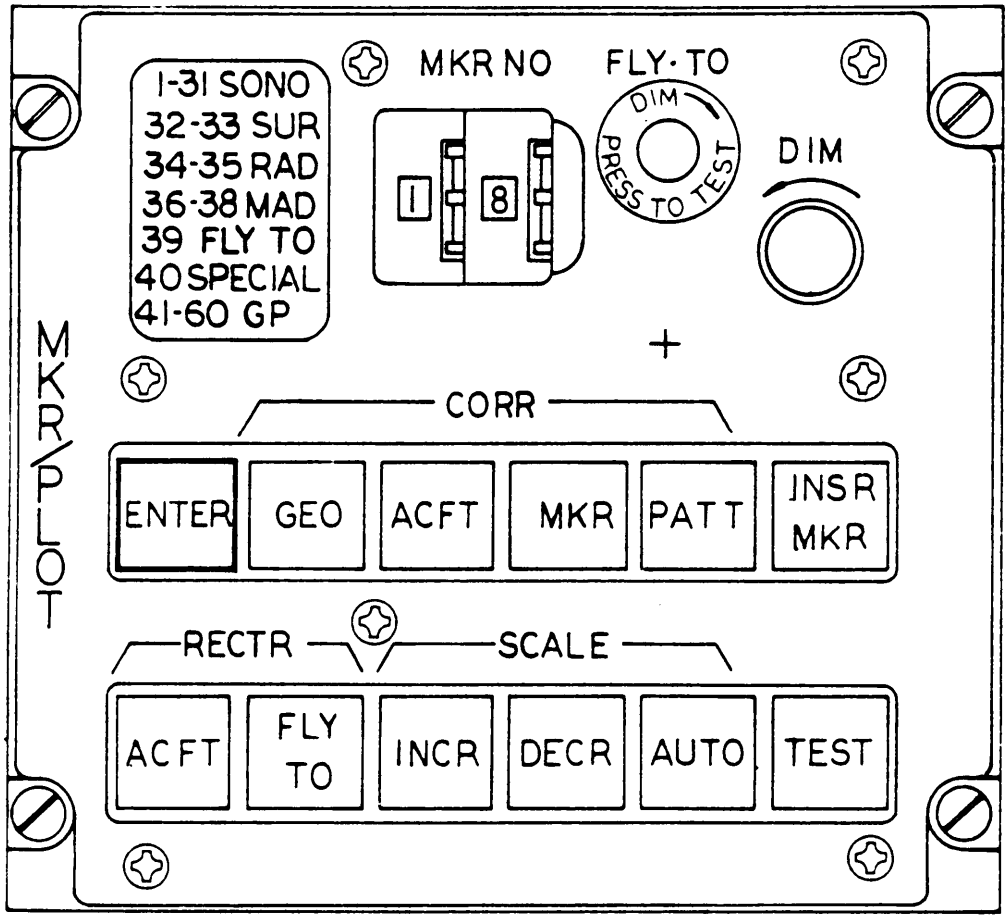


FIGURE 20. Marker Plotter Controller panel layout.

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3.5.4.5.5 Pattern Correct - The Control shall provide a pushbutton key to designate execution of a pattern correct - PATT. The key shall be a momentary actuation type, and shall be illuminated green when inactive. Illumination of the PATT key shall be under software control and shall be amber when active.

3.5.4.5.6 Insert Marker - The Control shall provide a push-button key to designate the insertion of a marker - INSR MKR. The key shall be momentary actuation with electronic memory interlock as in Paragraph 3.5.4.5.4. The key illumination color shall be greeen when inactive and amber when active.

3.5.4.5.7 Entry - The Control shall provide a pushbutton key to enter correction and marker insertion data - (ENTER). The key shall be a guarded momentary actuation type. The key shall be illuminated green when inactive, and amber when active. Key illumination shall be under software control.

3.5.4.5.8 Fly-to-Light - The Control shall provide a light to announce the "fly-to" function when illuminated. The light shall be off when the function is inactive, green when active, and labeled - FLY TO. The light shall be readily visible from as wide a viewing angle as practical. The light shall be under software control.

3.5*4.5.9 Dimming Control - The Control shall provide a single dimming control for lighting power for the "FLY TO" light and all push-button key illumination. Clockwise rotation shall result in increased brightness.

3.5.4.5.10 Marker Table - The Control shall be labeled with a marker number table on the face of the lighting panel as follows:

1 - 31 SONO
32 - 33 SUR
34 - 35 RAD
36 - 38 MAD
39 FLY TO
40 SPECIAL
41 - 60 GP

3.5.4.6 Control Illumination

3.5.4.6.1 Panel Lighting - Panel lighting shall conform to MIL-P-7788D, Class 1-R, Type V. Excitation shall be 26 to 5 VAC externally controlled.

3.5.4.6.2 Push-Button Lighting - Push-button nomenclature shall be white translucent when unlit, green for standby lighted, and amber for active per Paragraph 3.5.4.5. Background shall be black.

Push-button and "Fly To" lamp dimming shall be controlled by a self-contained dimmer available to the operator. Excitation shall be 26 VAC, 400 Hz, aircraft lighting power.

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3.5.4.7 Input Power

3.5.4.7.1 Aircraft Power - The equipment shall utilize aircraft power in accordance with Paragraph 3.3.12 as follows:

(1) 0-26 VAC, 400 Hz, 10, 0.5 Amp.
(2) 26 VAC, 400 Hz, 10, 1.8 Amps

3.5.4.7.2 System Power - The equipment shall utilize system power from the SDC as follows:

(1) +13.5 VDC, nominal 550 mAmp, regulated, ± 10%
(2) -13.5 VDC, nominal 550 mAmp, regulated, ± 10%
(3) 5.2 VDC, nominal 2.5 Amp, regulated ± 5%

*3.5.4.8 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	M81511/51F-E-03P2	Logic Power
J2	SJTP02RE10-5P(023)	Lighting Power
J3	DAGE 4679-1	Video Output
J4	DAGE 4679-1	Horiz. Deflection
J5	DAGE 4679-1	Vert. Deflection
J6	SJTP02RE14-35PC(023)	Data and Discretes
J7	DAGE 4679-1	Blanking
J8	SJTP02RE18-35P(023)	Test

3.5.4.9 Mounting - The Control shall meet the requirements of MIL-C-6781 for Type I control panels.

3.5.4.10 Color - The surface shall be painted lusterless black, No. 37038, in accordance with FED-STD-595.

3.5.5 Converter-Interface CV-3332/ASN-124 - The Converter shall meet the following requirements.

3.5.5.1 Function - The Converter shall provide the TACCO with readouts of RF channels and the bearings or ranges from Traces U and L, Grams I, II, III, and IV of both Sensor Stations. It shall provide interface with an IP-886B/ASA-66 Display. It shall provide controls to update the bearing or range data displayed on the IP-886B/ASA-66 Tactical Display. The front panel of the unit shall conform to Figure 21. Interface signal flow is shown in the block diagram of Figure 22. The Converter-Interface and the Tactical Display Control, together, make up the Integrated Tactical Display System (ITADS).

3.5.5.2 Form Factor - The Converter, exclusive of front panel protrusions, attachment points, fan and connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
16.4	9.000	18.000 (behind front panel)

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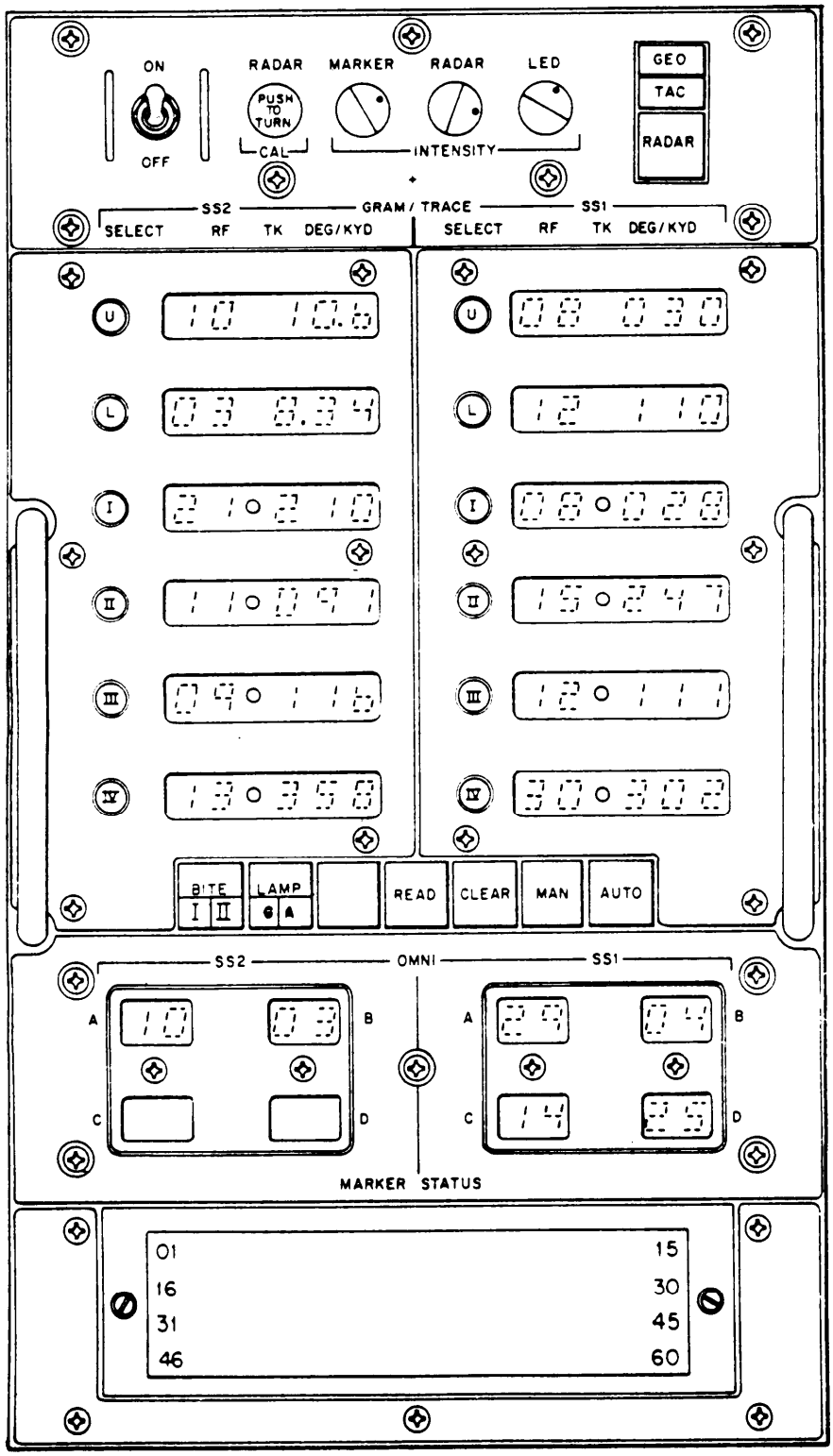


FIGURE 21. Converter-Interface.

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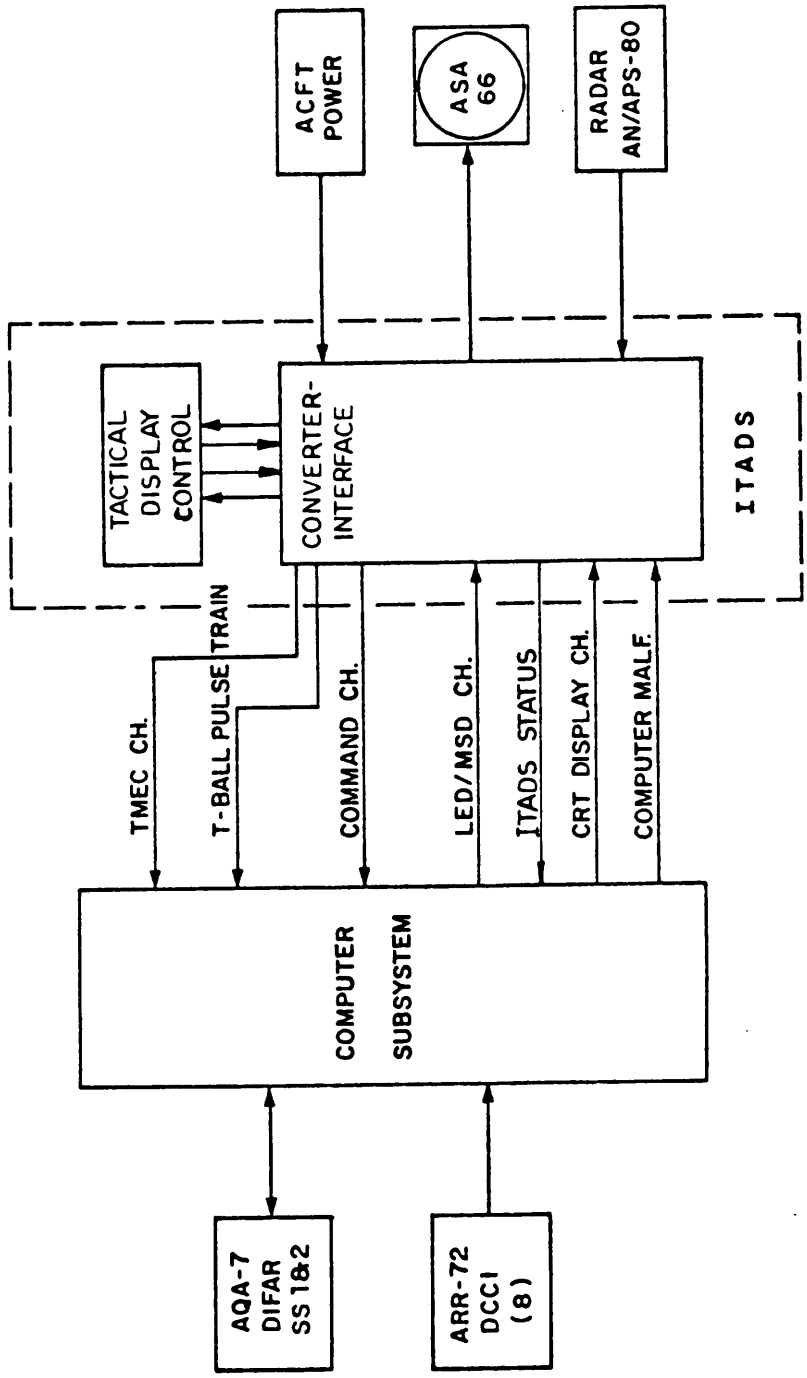


FIGURE 22. ITADS functional block diagram.

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3.5.5.3 Weight - The weight of the Converter shall not exceed 64 pounds.

3.5.5.4 Contents - The Converter shall contain the following major functions.

3.5.5.4.1 Radar Interface - The Converter shall accept radar sweep input data from the AN/APS-80 Radar Set. The Converter shall process these signals so as to provide outputs to the IP-886B/AsA-66 for PPI radar display.

3.5.5.4.2 Tactical Display Control (TDC) Interface - The Converter shall provide logic power and LED Display Data to the TDC. (Paragraph 3.5.6).

3.5.5.4.3 IP-886B/ASA-66 Interface - The Converter shall provide video, deflection, and unblinking signals to the IP-886B/ASA-66 for display of tactical data. Interface requirements shall be as defined in MIL-B-81358 for the IP-886B/ASA-66.

3.5.5.4.4 Computer Subsystem Channel Interfaces - The Converter interface with the Computer Subsystem (consisting of two system units: the NC and the SDC) shall provide serial data, clock signals, and discretes necessary to communicate TACCO decisions to the NC and receive display data in return. The interface shall be on a non-interrupt basis under NC control. All data shall be transmitted LSB first.

3.5.5.4.4.1 Command Channel - The Command Channel shall interface from the Converter to the SDC. Timing shall be in accordance with Figure 23 and the Command word format shall be in accordance with Figure 24.

The Command Word discrete shall be set true each time a function code is formulated and shall remain true until clocking is initiated. It shall be set false again within 10 µsec after the first clock is received.

The SS-1 and SS-2 select portions of the word shall represent the state of the Gram/Trace Select PBI's at the time a function code is formulated. An active pushbutton state shall set the corresponding select bit to true.

The design shall be such that a new command word is not formulated until the old (previous) word is read into the computer.

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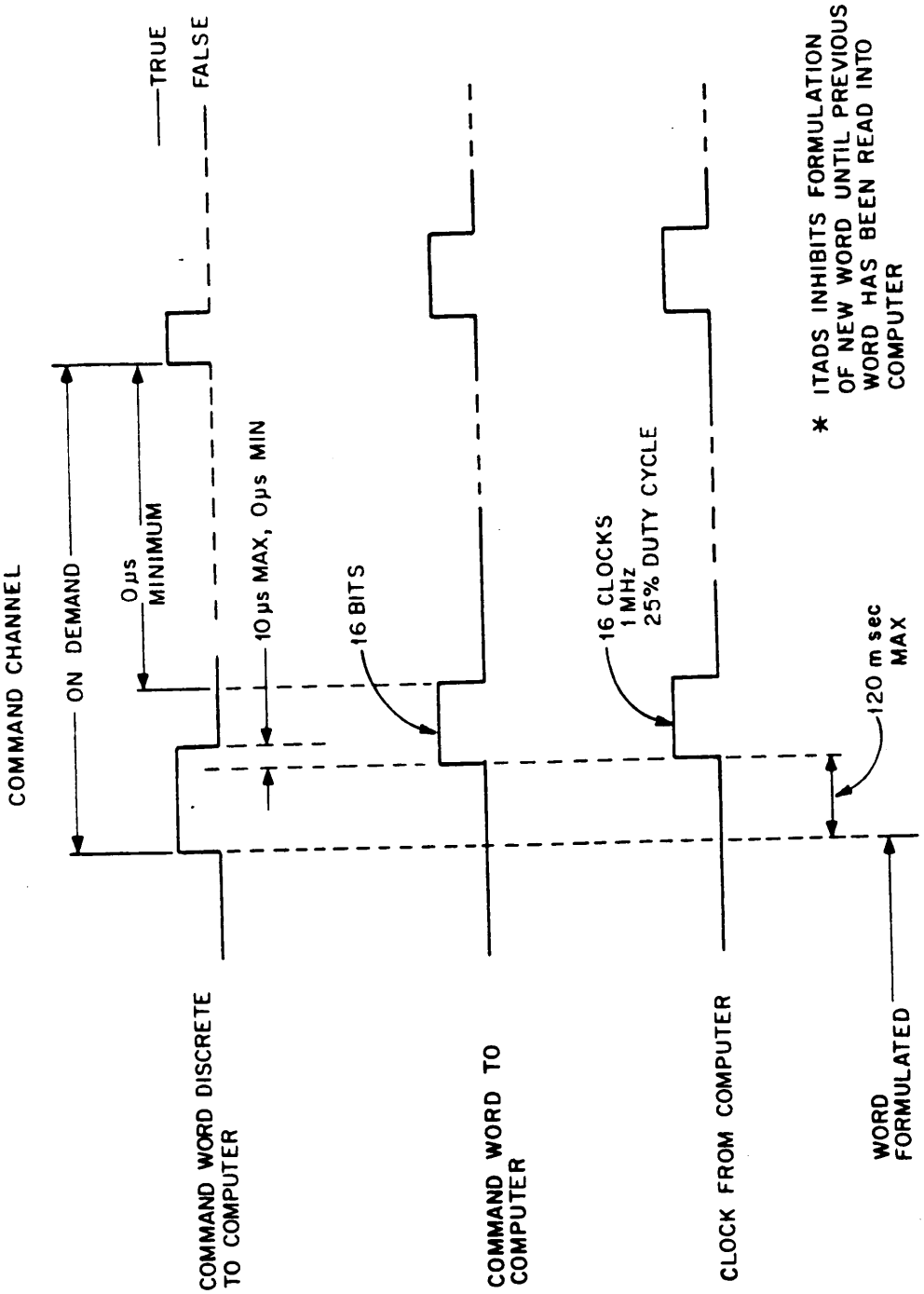


FIGURE 23. Command channel timing diagram.

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COMMAND WORD

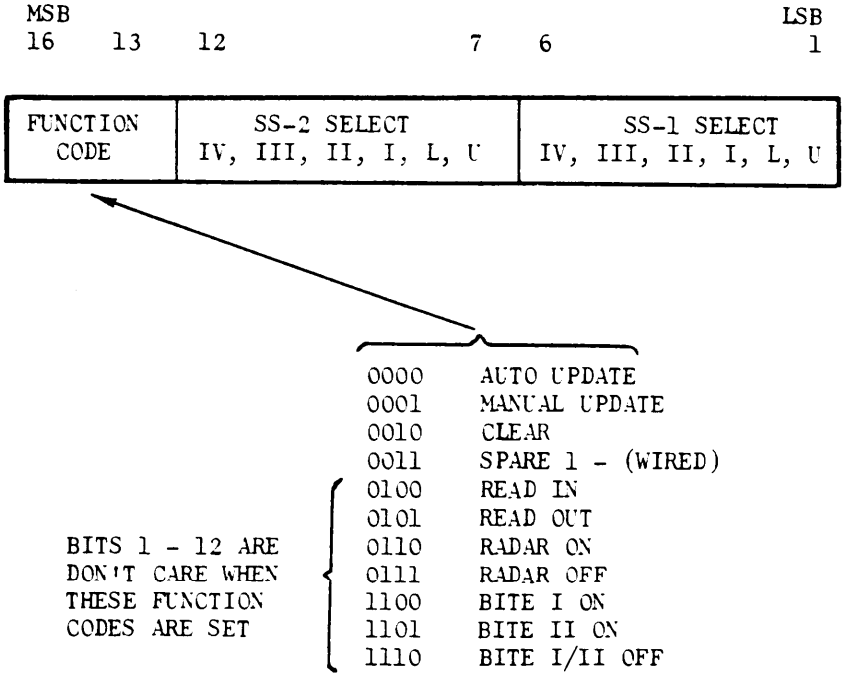


FIGURE 24. Command word format.

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*3.5.5.4.4.2 LED/MSD Channel - The LED/MSD Channel shall interface from the SDC to the Converter. The total channel shall consist of a data channel, a list discrete, a word discrete, and a clock. Timing shall be in accordance with Figure 25 and data word formats shall be in accordance with Figure 26. The list discrete shall be set true prior to clocking the Data to the Converter and will remain true until all words in the list have been transmitted. The word discrete shall be set true immediately before each word of the list is clocked to the Converter. It shall go false between words.

A list shall consist of 20 data words, 32 bits per word. The minimum time between lists shall be as shown in Figure 25. As shown in Figure 25, the SCALE/MESS/EDIT word will be the first word of the list and the SS-2 OMNI word the last. The 20 words will be in five different formats as shown in Figure 26.

The four MSD words shall control the 60 Marker Status Display Indicators with the first word being for marker indicators 01 through 16, the second 17 through 32, the third 33 through 48, the fourth 49 through 60.

*3.5.5.4.4.3 CRT Display Channel - The CRT Display Channel shall form an interface from the SDC to the Converter. It shall consist of a data channel, a list discrete, word discrete, and a clock. Figure 27 depicts the timing of the channel and Figure 28 illustrates data words format. Each CRT data word shall be either 64 or 96 bits in length. When conic information (BRG or LGTH/RAD) is associated with the X and Y position of that word, the 96 bits shall be transmitted. If conic information is not required, only 64 bits of data shall be transmitted to the Converter for example 33 - 96.

The word discrete shall be set true 2.25 µsec prior to the clock and will be set false 750 nsec after the clock, whether it be 64 or 96 bits in length. A list of data words shall be transmitted to the Converter. The word list shall be from 3 to 64 words in length. The minimum time between lists shall be as shown in Figure 27. The list discrete shall be set true 1 µsec before the word discrete and shall be set false 5 µsec after the word discrete falls. The “as required” words shall not be in any particular pattern within the list. The Converter shall display on the CRT only the data from the latest list update.

3.5.5.4.4.3.1 X and Y Position Data - The X and Y data shall be 12 bits each, consisting of 11 bits positional data (from CRT center) and 1 bit for sign: up, down, right, or left of CRT center. Bit and bit weight shall be as shown in Figure 28. Scale is defined as radius of nominal CRT display area in nautical miles. The LSB shall be transmitted first. Negative numbers shall be in “two’s” complement.

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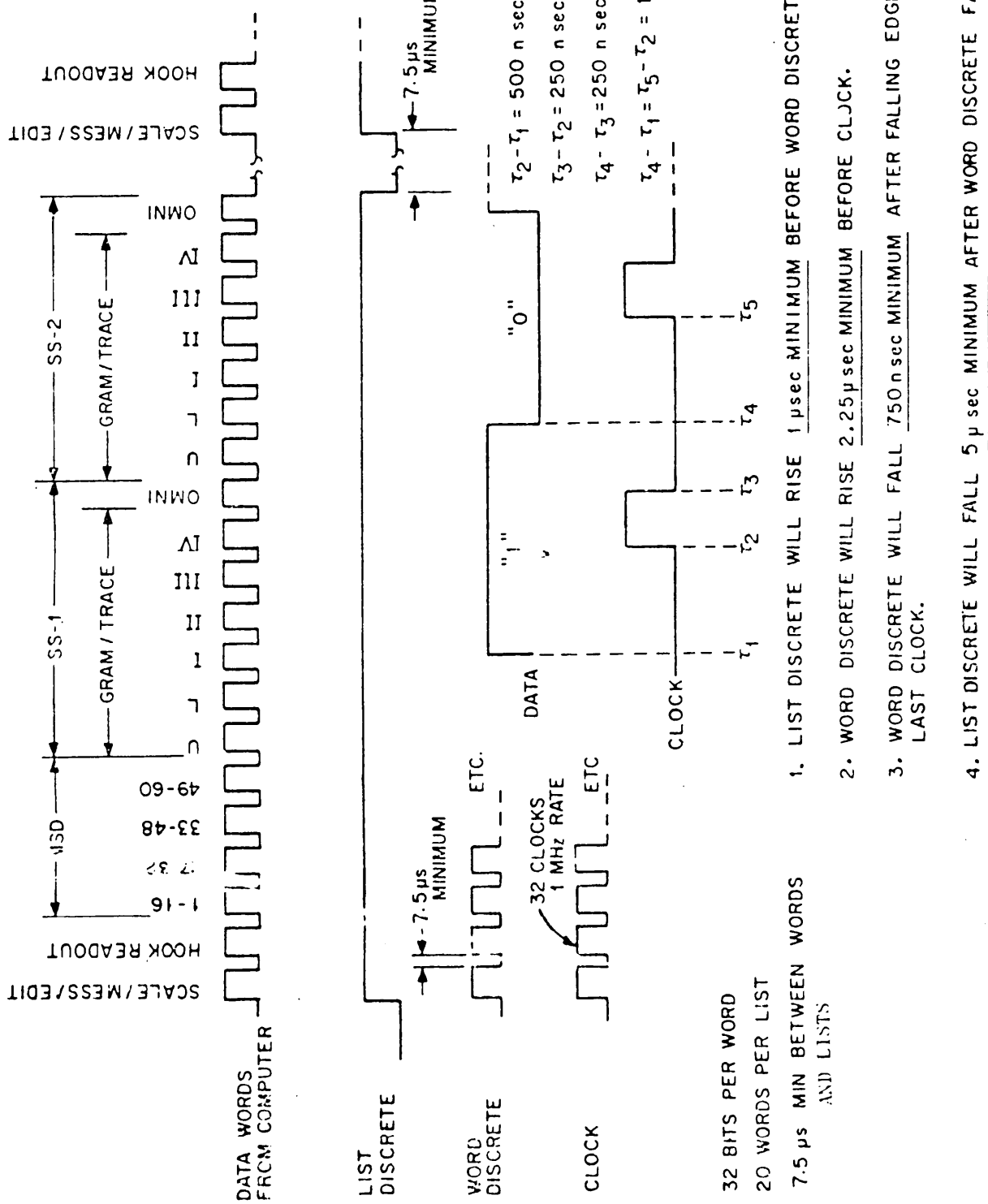


FIGURE 25. LED/MSD channel timing diagram.

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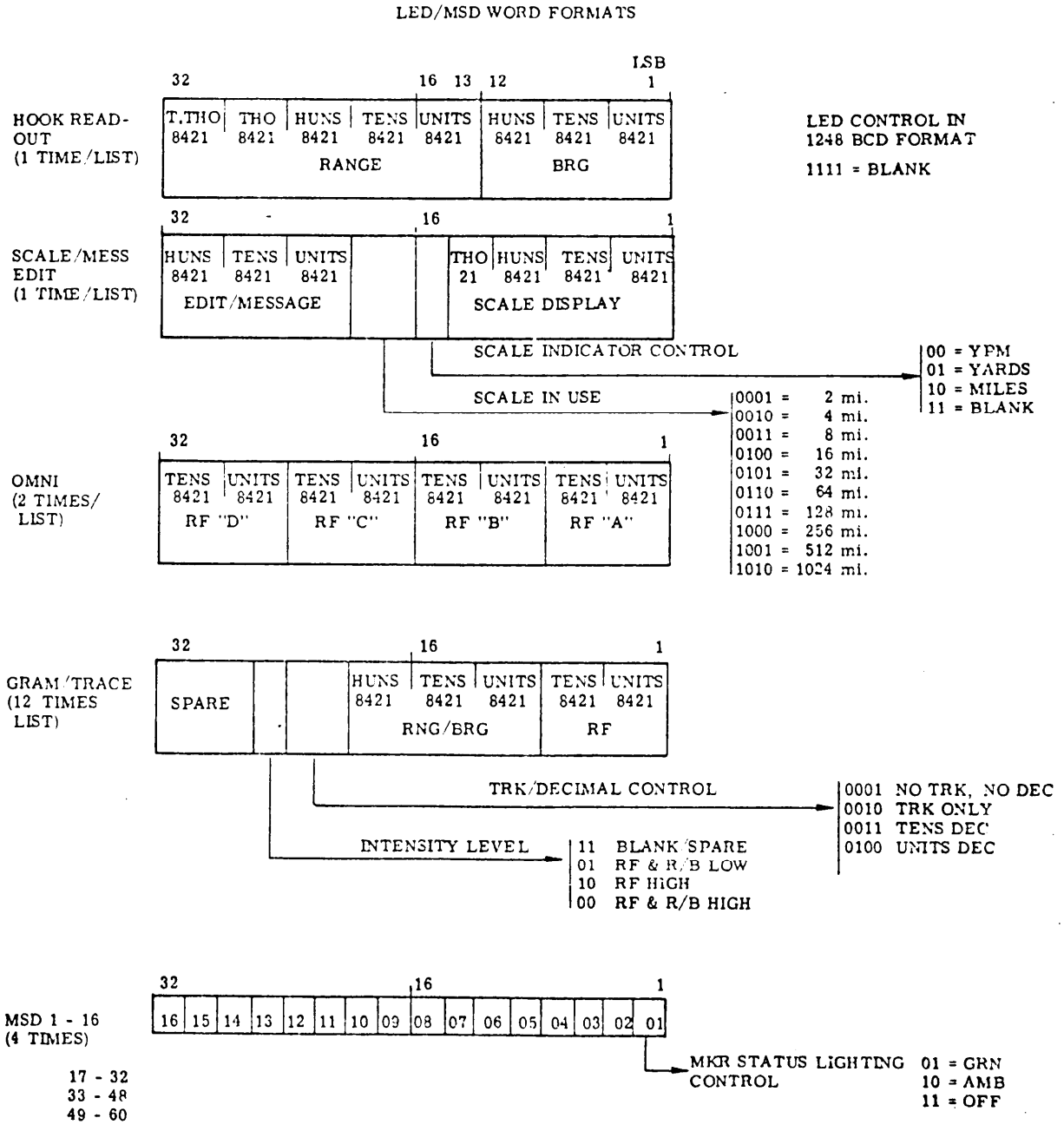
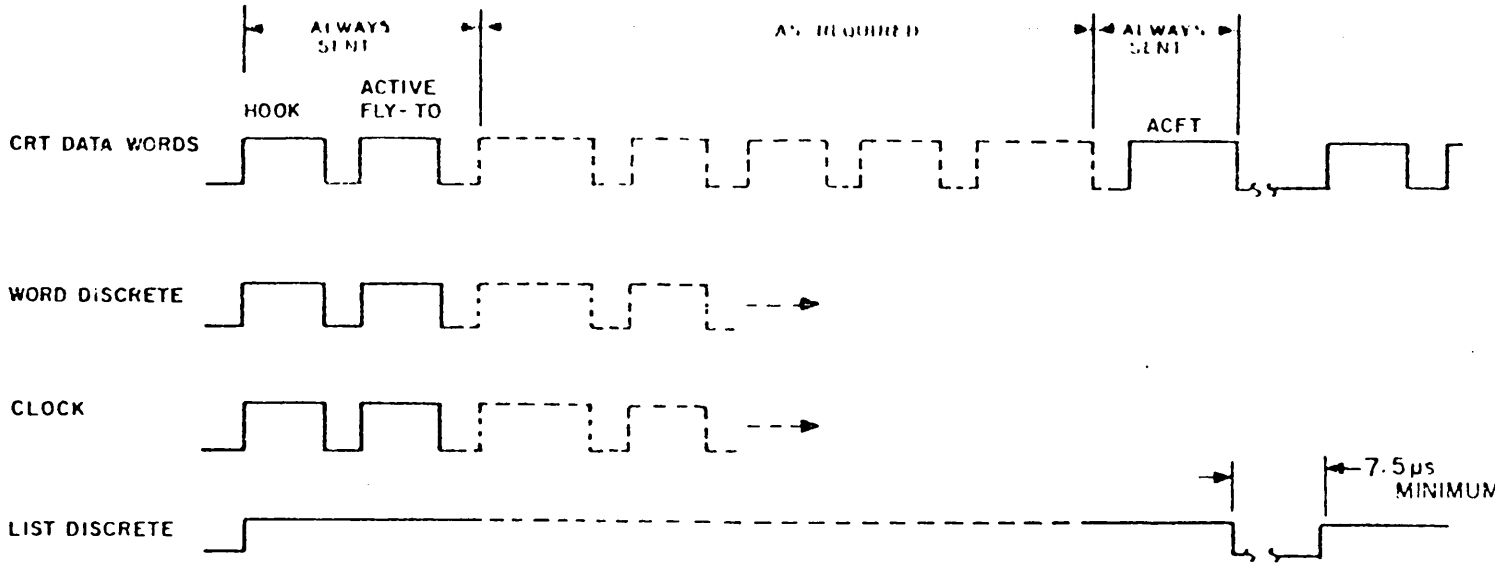


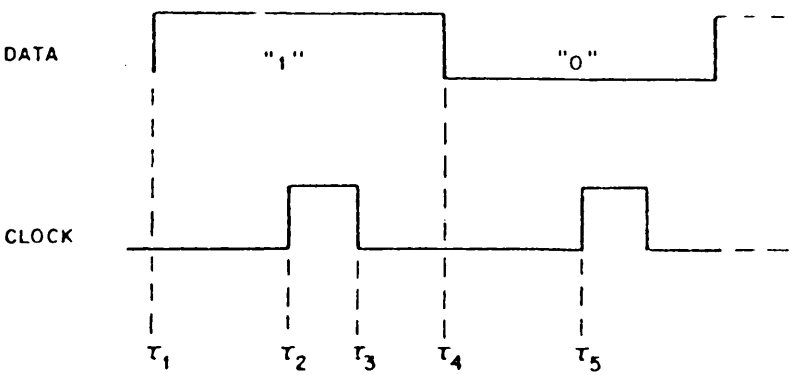
FIGURE 26. LED/MSD word formats.

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NOTES:

1. ⌐¬ = 64 BIT WORD INC POS & CHAR INFO ONLY.
2. ⌐‾¬ = 96 BIT WORD INC 1ST 64 + 32 BIT OF CONIC INFO.
3. MIN TIME BETWEEN CONSECUTIVE WORD DISCRETES = 2.5 μ sec.
4. LIST = 3 MINIMUM, 64 MAX
5. LIST DISCRETE WILL RISE 1 μ sec MINIMUM BEFORE WORD DISCRETE RISES.
6. WORD DISCRETE WILL RISE 2.25 μ sec MINIMUM BEFORE CLOCK.
7. WORD DISCRETE WILL FALL 750 n sec MINIMUM AFTER FALLING EDGE OF LAST CLOCK.
8. LIST DISCRETE WILL FALL 1 μ sec MINIMUM AFTER WORD DISCRETE FALLS.
9. ACFT WORD IS LAST WORD OF LIST.



$\tau_2 - \tau_1 = 500$ n sec

$\tau_3 - \tau_2 = 250$ n sec

$\tau_4 - \tau_3 = 250$ n sec

$\tau_4 - \tau_1 = \tau_5 - \tau_2 = 1$ μ sec

FIGURE 27. CRT Display Channel timing.

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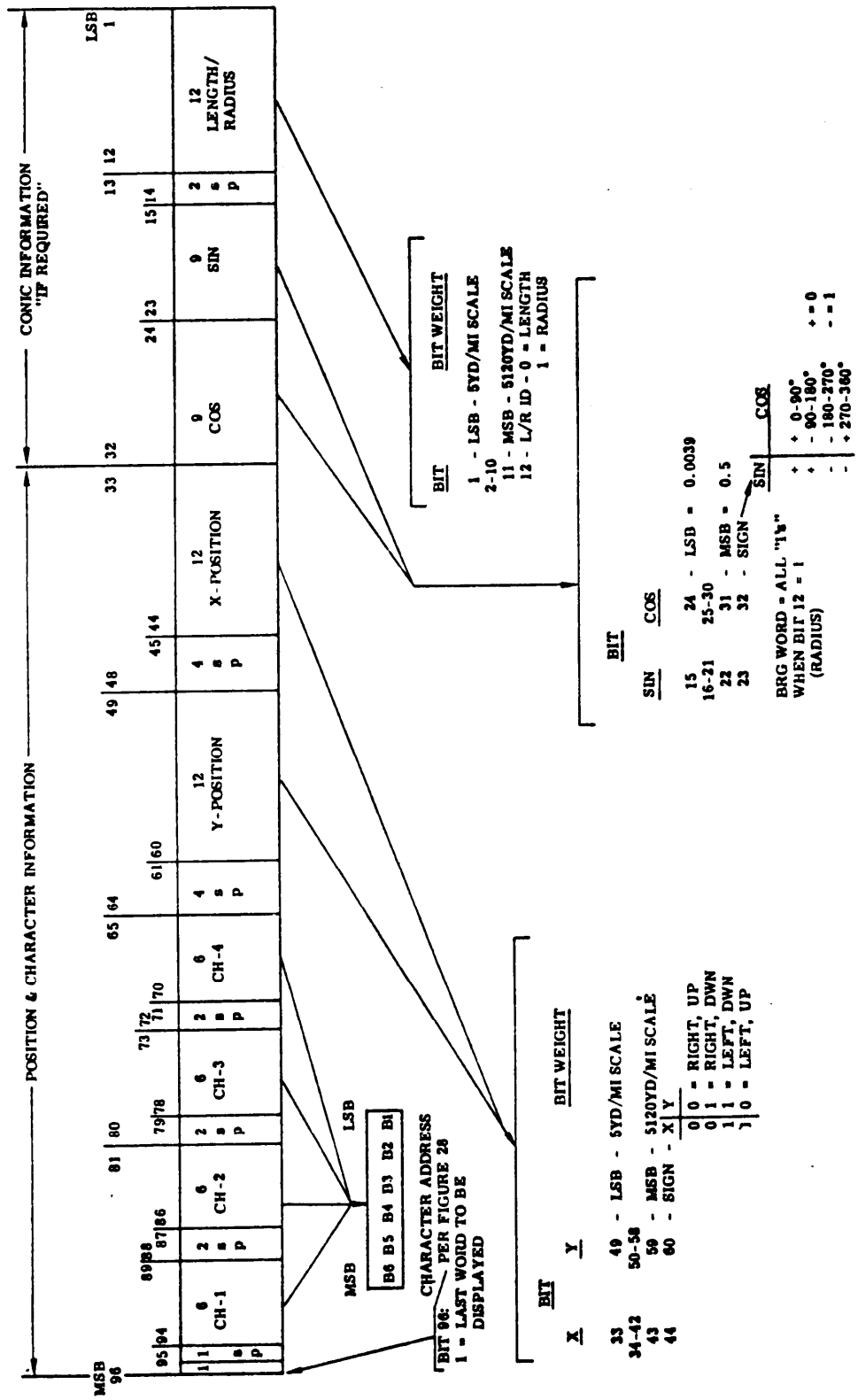


FIGURE 28. CRT Display word format.

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3.5.5.4.4.3.2 Character 1, 2, 3, and 4 Select - Characters 1, 2, 3, and 4 shall represent the character position within a group of four characters written from left to right, with Character 1 being the left, 4 the right, and the center of 2 being the marker position as defined by X and Y position. Each Character Select shall be capable of selecting any individual character in the character repertoire. The character repertoire shall consist of 48 characters similar to the 47 shown in the 5 X 7 character font of Figure 28. One spare of the 17 spares shown in Figure 28 shall be included to form a total of 48 characters (minimum).

*3.5.5.4.4.3.3 Length/Radius - The length of a bearing line, aircraft cursor, target predict cursor, or the radius of a circle to be displayed on the CRT shall be defined by bits 1 - 12. The conic origin shall be defined by the X, Y position with bit weight as shown in Figure 28.

3.5.5.4.4.3.4 Bearing Sine and Cosine - The bearing from the origin as defined by the X, Y position, in sine and cosine functions shall be 9 bits each consisting of 8 bits data and 1 bit for sign. Bits for sine and cosine, values of LSB and MSB, and sign of sine and cosine function in degrees shall be as shown in Figure 28.

3.5.5.4.4.4 CMPTR MALF Discrete - A CMPTR MALF discrete shall be provided from the SDC to the Converter. The Converter shall contain circuitry to provide continuous monitoring of the computer for anomalies and to enable malfunction display as described in Paragraph 3.5.5.4.6.

3.5.5.4.4.5 ITADS Status Discrete - The Converter shall set this discrete false during the "GO" condition and true during the "NO-GO" condition.

3.5.5.4.5 Symbol Generator - The Converter shall generate the following symbols for presentation on the CRT.

3.5.5.4.5.1 Symbology - The Converter shall be capable of displaying four characters on the CRT for each position word. The position data defines the center of the character matrix of the second character. Characters shall be as defined in Figure 29. Range circles and bearing lines shall be displayed on any marker. A Malfunction Symbol shall be provided by the hardware, as detailed in Paragraph 3.5.5.4.6. The Malfunction Symbol shall be a large "X" positioned in the center of the CRT. Display of the Malfunction Symbol shall result in removal of all other symbols.

3.5.5.4.5.2 Accuracy - The accuracy of the Converter is specified as the error between the digital position, range, or bearing at the input to the Converter and the analog voltage at the input to the Modified IP-886B/ASA-66.

(1) The buoy positions shall be within 20 yards on the 2 mile scale.
(2) The range circles shall be within 40 yards on the 2 mile scale.
(3) The bearing lines shall be within 1 degree.

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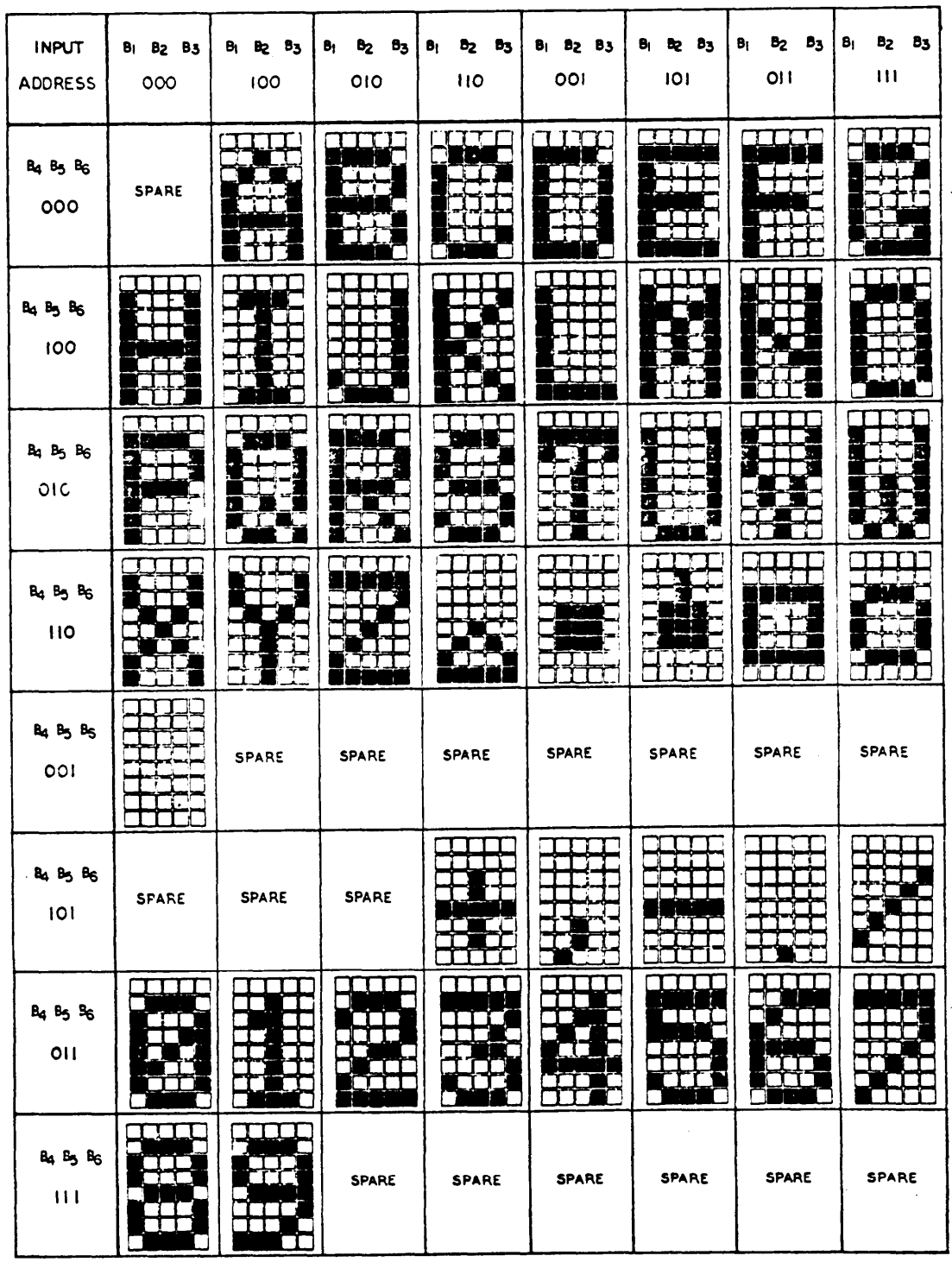


FIGURE 29. CRT Display characteristics.

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*3.5.5.4.5.3 Display Quality - The Converter shall be designed so that, in operation with the IP-886B/ASA-66 Tactical Data Display, the CRT shall be free of spurious and extraneous presentations and noise. Bearing lines and the HOOK cursor shall appear to originate on the position marker. Range circles shall be round and continuous. Modulation on the circles, lines, and characters shall be below usual threshold level. Update of associated range/bearing/position data for each marker shall be controlled by the Converter so that the associated data from the Computer is all of the same time frame. Cursor, circle, and marker brightness shall be balanced or a bench adjustment provided. The Converter shall not be required to compensate for variations in display quality introduced by the IP-886B\ASA-66.

3.5.5.4.6 NC Interface BITE - Circuits shall be provided within the Converter to provide continuous monitoring as follows:

(1) Detection of failure to receive input clock or data generated by the NC for a period exceeding approximately one second.

(2) CMPTR MALF Discrete. The CMPTR MALF discrete shall be interpreted to be active when either a normal logic "1" is received or when the differential signal level between the balanced transmission lines is held at less than 2.0 volts.

Detection of any one of the above abnormalities shall result in display of the Malfunction Symbol. (Reference Paragraph 3.5.5.4.5.1.)

3.5.5.5 Controls - The Converter shall contain the following controls depicted in Figure 21. Depression of any of the function PBI's shall set the associated function code and initiate a transmission of the command word to the computer. Depression of Gram/Trace Select PBI's shall set the associated bits in the command word. The function PBI's shall have opaque fields with illuminated legends. The Gram/Trace Select PBI's shall have illuminated fields with black legends.

3.5.5.5.1 ON/OFF - A guarded toggle switch shall be provided to turn the ITADS on in the ON position and off in the OFF position. Power for the IP-886B/ASA-66 shall not be a function of this switch.

3.5.5.5.2 GRAM TRACE/SELECT - This group shall contain 12 PBI's, 1 associated with each of the Gram/Trace readouts of both sensor station displays. These indicating pushbuttons shall be normally extinguished and shall illuminate amber when depressed and remain until either the CLEAR, MANUAL, or AUTO PBI's have been activated. These PBI's shall be labeled U, L, I, II, III, and IV corresponding to the associated Gram/Trace readout unit and located adjacent to the readout.

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3.5.5.5.3 RADAR - The Converter shall include an alternate action PBI providing radar mode control. The PBI shall be normally illuminated green, shall be amber when activated, and shall be non-illuminated and non-programmable when the APS-80 is not operating. When actuated green to amber, the PBI shall enable the radar sweeps and video and generate the RADAR ON function code in the command word. When depressed amber to green it shall disable the radar sweeps and video and generate the RADAR OFF function code in the command word. The radar mode shall be inhibited when on the 256, 512, and 1024 mile scales. When in the radar mode and scaling is changed to 256 miles or more, radar shall automatically be removed from the IP-886B/ASA-66 display, the RADAR OFF function code generated, and the PBI returned to green. Markers shall not be deleted as a function of the Radar Mode.

3.5.5.5.4 GEO/TAC - This shall be an alternate action PBI. The illumination shall change from green to amber for the function selected. The PBI shall control a discrete line pair at the computer interface, the discrete state corresponding to the illumination state of the PBI as follows:

PBI State	Illumination		Discrete
	GEO	TAC	
GEO	Amber	Green	True
TAC	Green	Amber	False

The switch illumination shall be set to GEO (amber) and TAC (green) when ITADS power is turned on and thereafter shall be an alternating function of switch action. The PBI shall have an opaque field with illuminated dual legend.

3.5.5.5.5 Radar Intensity - This control shall be a continuously variable knob which shall increase the brightness of the radar video when rotated in the clockwise direction. It shall not be possible to rotate this control in the counter-clockwise direction such that nominal radar video is not visible on the display.

3.5.5.5.6 Marker Intensity - This continuously variable control shall increase the intensity of the markers and associated lines and circles when rotated in the clockwise direction. With the control in the full counterclockwise position, the markers shall still be dimly visible.

3.5.5.5.7 LED Intensity - This continuously variable control shall increase the intensity of illumination of the LED displays on the Converter and the TDC when rotated in the clockwise direction. With the control in the full CCW position, all commanded segments of the nominally illuminated LED displays shall still be dimly visible.

3.5.5.5.8 RADAR CAL - This continuously variable control shall be a push-to-turn type and shall adjust the start time of the radar sweep for accurate calibration between the radar video and radar display deflection.

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3.5.5.5.9 BITE I/II - This PBI shall provide triple action control for Built In Test of the Converter. The operation and illumination of this PBI shall be as follows:

BITE PBI ACTION	ILLUMINATION			CODE TO COMPUTER	FUNCTION INITIATED
	BITE	I	II		
Normal	G	OUT	OUT	None	None
1st Depression	A	A	OUT	BITE-I	Built in Hardware Test (BITE - I)
2nd Depression	A	OUT	A	BITE-II	Software Controlled Test (BITE-II)
3rd Depression	G	OUT	OUT	BITE-OFF	Return to Normal

In the BITE-I mode the Converter shall perform a hardware selftest with performance indications being displayed on the LED's and ASA-66. Data from the Navigation Computer shall be ignored and the ITADS Status Discrete shall be held in the NO-GO condition while in BITE-I. When the RADAR mode is entered during BITE-I, a hardware self-test of the radar section contained in the Converter shall be performed. The RADAR self-test shall utilize the CRT. as an indicator of Radar performance.

In the BITE-II mode, all other controls and readouts on the Converter shall be allowed to function normally. When BITE is depressed, amber to green, all functions shall be restored to normal and the BITE-OFF function code shall be formulated in the Command Word.

3.5.5.5.10 LAMP TEST - This momentary action PBI, when activated, shall test the green and amber lamps of all PBI's, all segments of the LED displays, green and amber states of the Marker Status display indicators, and the illuminated state of the GRAM/TRACE select indicating pushbuttons.

3.5.5.5.11 SPARE 1 - This momentary action PBI shall be normally illuminated green, and when actuated shall generate the spare function code in the Command Word. The PBI shall not be labeled and shall be opaque.

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3.5.5.5.12 READ - This momentary action PBI shall be normally illuminated green. When depressed it shall generate the READ IN function code in the Command Word. When released it shall generate the READ OUT function code.

3.5.5.5.13 CLEAR - This momentary action PBI shall be normally illuminated green. When depressed, it shall generate the CLEAR function code in the Command Word and reset the selected Gram/Trace indicator(s).

3.5.5.5.14 MAN (Manual) - This momentary action PBI shall be normally illuminated green. When depressed, it shall generate the Auto Update Function Code in the Command Word, and reset the selected Gram/Trace indicator(s).

3.5.5.5.15 AUTO - This momentary action PBI shall be normally illuminated green. When actuated, it shall generate the Auto Update Function Code in the Command Word, and reset the selected Gram/Trace indicator(s).

3.5.5.5.16 PBI Illumination Intensity - All PBI illumination shall be of variable intensity under control of the external control for edge-lit panel lighting.

3.5.5.6 Readouts - The Converter shall contain the following readouts that shall be controlled by data received from the NC derived from LED/MSD words through the SDC interface.

3.5.5.6.1 GRAM/TRACE - Each of the 12 readouts shall contain:

A two digit, 7-segment LED readout for the RF number.

A three digit, 7-segment LED readout for DEG or KYDS display.

The eight Gram readouts (channels I, II, III, and IV of each sensor station) shall contain an indicating lamp for TRK display. This indicator shall be located between the RF and DEG/KYDS LEDS.

The readouts for the DEG/KYDS of the Upper and Lower Traces (U and L) of each sensor station shall contain decimals between the units and tens digits and between the tens and hundreds digits.

The RF and DEG/KYDS readouts shall have two levels of illumination, high and low, controlled by the computer with low being normal illumination.

The RF and DEG/KYDS value, track lamp, illumination, and decimal control shall be obtained from the GRAM/TRACE word as shown in Paragraph 3.5.5.4.4.2, and Figure 26.

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3.5.5.6.2 OMNI - A two digit, 7-segment LED readout shall be provided for each of the four OMNI SEARCH RF channels for each sensor station. The four readouts of each sensor station shall be labeled A, B, C, and D. The RF value to be displayed shall be obtained from the OMNI word as shown in Paragraph 3.5.5.4.4.2, and Figure 26.

3.5.5.6.3 MARKER STATUS - The Converter shall include a display group to indicate the status of markers for display on the CRT. The group shall consist of the numbers 01 through 60, each being capable of being illuminated green or amber, or being turned off. Illumination control shall be from the MSD words as shown in Paragraph 3.5.5.4.4.2, and Figure 26.

3.5.5.7 Panel Illumination - The Display shall contain an integrally illuminated information panel conforming to MIL-P-7788, Class 2-W.

3.5.5.8 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function	
1J1	DAGE 4679-1	N.S. Deflection	ASA-66
1J2	DAGE 4679-1	E.W. Deflection	
1J3	DAGE 4679-1	ITADS Video	
1J4	DAGE 4679-1	ITADS Unblank	
1J5	DAGE 9662-1	Trigger	RADAR
1J6	DAGE 9662-1	Pretrigger	
1J7	DAGE 9662-1	Video	
1J8	M81511/41FD02P1	Radar	
1J9	M81511/41FH02S1	Control Tray	
1J10	M81511/41FE01P4	Computer	
1J11	M81511/41FE03P1	Acft Power	
1J12	M81511/41FE01S2	AGE	

3.5.5.9 Input Power - The equipment shall utilize input power in accordance with Paragraph 3.3.12 as follows:

(1) aircraft power; 115 VAC, 400 Hz, 30, Category B, 455 VA maximum

(2) lighting power, 0 to 26 VAC, 1.0 Amp maximum

This input power shall be utilized to power both the Converter and the TDC (3.5.6).

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3.5.5.10 Mounting - The Converter shall contain mounting provisions designed to allow the unit to be slid forward out of the normally installed position for maintenance (Ref. Para. 3.5.14). The mount shall not be supplied as part of the equipment.

3.5.5*11 Color - The surface shall be painted lusterless gray, No. 36231, in accordance with FED-STD-595.

3.5.6 Tactical Display Control (TDC) C-9862/ASN-124 - The TDC shall meet the following requirements.

3.5.6.1 Function - The TDC shall provide function switches (PBI) associated with the control of data to be displayed on the CRT. It shall include readouts of scale in use on the CRT, HOOK cursor range and bearing, and Edit/Message. The panel shall conform to Figure 30.

3.5.6.2 Form Factor - The TDC including front panel protrusions, attachment points, and connector projections shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
5.000	20.500	11.500

Depth dimensions shall be modified in a mounted configuration to allow for an additional 2 inches permitting cover swing back.

3.5.6.3 Weight - The weight of the TDC shall not exceed 22 pounds.

3.5.6.4 Contents - The TDC shall contain circuitry necessary to set bits (LSB first in transmission) in a control word to the SDC and to send Track Ball pulse train to the SDC. All data words shall be transmitted serially.

3.5.6.4.1 TDC/SDC Interfaces - The TDC interface with the Computer Subsystem shall be as follows:

3.5.6.4.1.1 TMEC Channel - The TMEC channel shall interface from the TDC to the SDC. Figure 31 is a timing diagram of the TMEC channel and Figure 31 is the word format for the TMEC word. Each time a TMEC word is formulated, the TMEC word discrete shall be set true and held true until clocks are received from the computer. This discrete shall be set to false again within 10 µsec from the time the first clock is received from the computer.

The design shall be such that a new TMEC word cannot be formulated until the old (previous) TMEC word has been read into the Computer.

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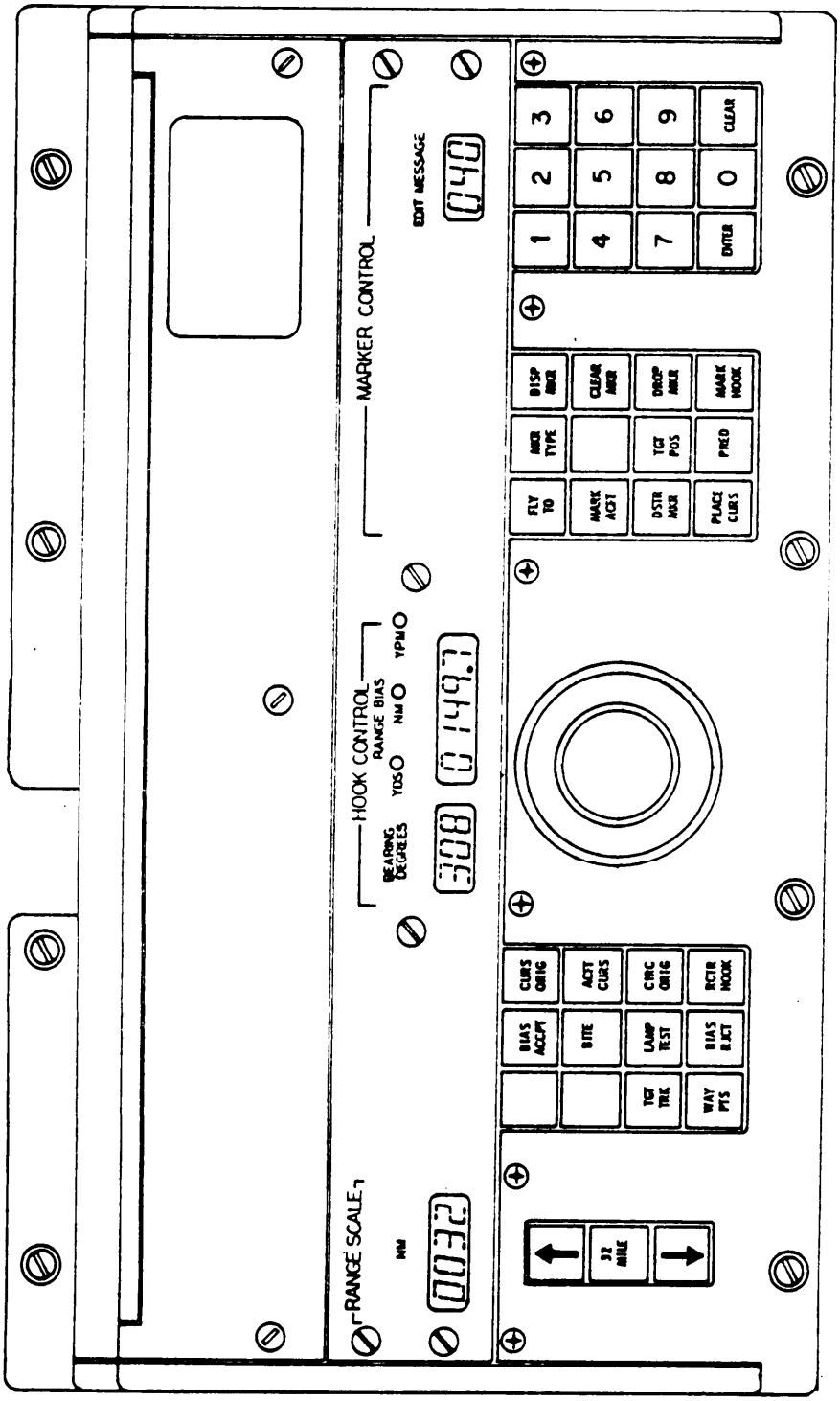


FIGURE 30. Tactical Display Control panel layout.

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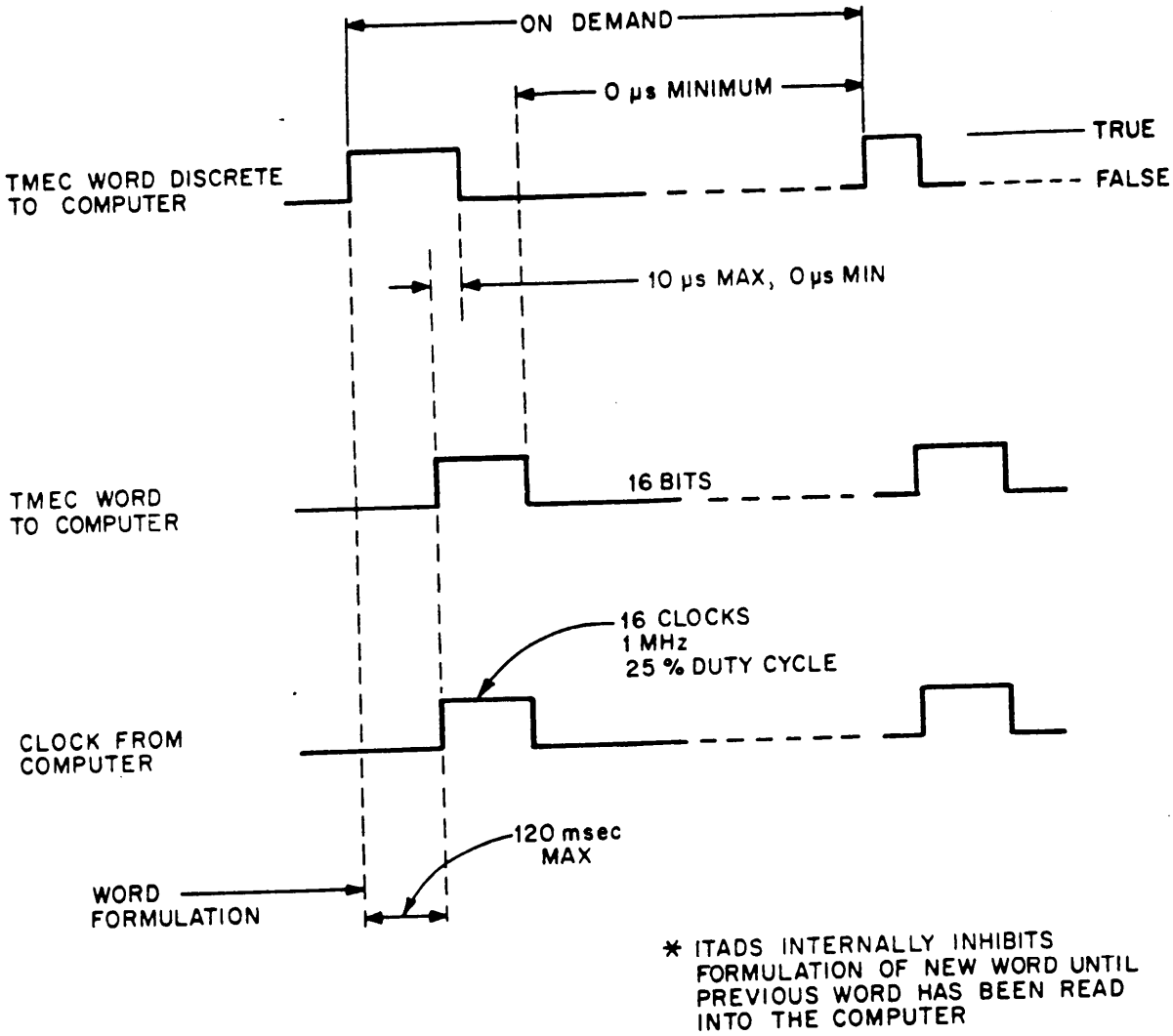
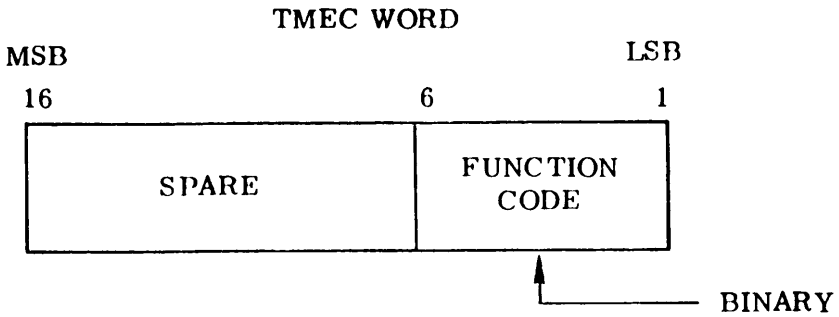


FIGURE 31. TMEC channel.

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FUNCTION CODE			FUNCTION CODE		FUNCTION CODE	
0	0	NUMERIC KEYS	16	BIAS ACCPT	32	DSTR MKR
1	1		17	BIAS RJCT	33	DROP MKR
2	2		18	WAYPOINTS IN	34	MKR TYPE
3	3		19	WAYPOINTS OUT	35	MARK HOOK
4	4		20	SPARE	36	CLEAR MKR
5	5		21	TGT TRKS IN	37	DISP MKR
6	6		22	TGT TRKS OUT	38	PLACE CURS
7	7		23	BITE 2 - ON	39	FLY-TO
8	8		24	BITE 2 - OFF	40	MARK ACFT
9	9		25	CIRC ORIG	41	PRED
10	SPARE 4 MOM PBI - WIRED		26	ACFT CURS	42	SPARE 3 MOM PBI - WIRED
11	CLEAR		27	CURS ORIG	43	TGT POS
12	RCTR HOOK		28	NO CURSOR	44	SPARE
13	SCALE UP		29	SPARE 2 IN (WIRED)	↓	
14	32 MILE		30	SPARE 2 OUT (WIRED)	62	
15	SCALE DOWN		31	SPARE	63	ENTER

FIGURE 32. TMEC word format.

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3.5.6.4.1.2 Track Ball Channel - The Track Ball pulse train shall be output to the SDC from the TDC. Four sets of buffered complementary pairs shall be provided, one each for the +X, -X, +Y, and -Y outputs. Pulse width shall be nominally 250 ns.

3.5.6.4.2 Converter Interface - The TDC shall interface directly with the Converter receiving logic power and LED illumination, LED display data, and 28 V power for panel lighting.

3.5.6.5 Controls - The TDC shall contain PBI's and a Track Ball. Depression of any of the PBI's except LAMP TEST shall cause the associated function code Figure 32 to be generated in the TMEC word that is then transmitted to the computer subsystem. The formulation and transmission of the TMEC word shall be such that a new TMEC word, resulting from rapid depression of a second PBI, is inhibited until the old TMEC word is input completely to the computer. Positional configuration and labeling of the controls shall be as depicted in Figure 30. The PBI's shall have opaque fields with illuminated legends.

LAMP Test actuation shall energize and illuminate all PBI's amber, all LED segments to "8", and all other indicators to an "on" state.

*3.5.6.5.1 Keysets - The TDC shall contain 39 PBI's as listed by label and type. PBI type 1 shall be momentary action, illuminated normally green, changing to amber when depressed. PBI type 2 shall be momentary action, illuminated green, changing and remaining amber when depressed, returning to green illumination when reset by depression or by ENTER PBI activation. PBI type 3 shall be alternate action, illuminated amber when activated "on" and green when deactivated "off". PBI type 4 shall be momentary action, illuminated green, when activated "on" flashes amber at 1 to 3 Hz rate under NC control. Spares shall not be labeled and shall be opaque.

Note: A permissible alternative shall be to have the 12 keys in the decimal keyset and the 3 scale keys normally illuminated green only.

TYPE 1

0 through 9	ENTER	CLEAR	SPARE 4
LAMP TEST	RCTR HOOK	↑ and ↓	32 MILE

TYPE 2

FLY-TO	MKR TYPE	DISP MKR	MARK ACFT
SPARE 3	CLEAR MKR	[3] DSTR MKR	PRED
DROP MKR	PLACE CURS	TGT POS	MARK HOOK

TYPE 3

[4] CURS ORIG	TGT TRK	SPARE 2	[5] CIRC ORIG
WAY PTS	BITE 2	[6] ACFT CURS	

TYPE 4

BIAS RJCT BIAS ACCPT

[3] Flashes amber at 1 to 3 Hz rate.
[4] Interlocked with CIRC ORIG and ACFT CURS.
[5] Interlocked with CURS ORIG and ACFT CURS.
[6] Interlocked with CURS ORIG and CIRC ORIG.

Interlocks shall preclude more than one function code to be active at any one time.

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3.5.6.5.2 Hook Control Trackball - The Trackball Control shall be similar to Orbit Instrument Corp. Part No. 55751. Direction sensing of the ball outputs shall be in the normal X-Y coordinate directions. One revolution of the ball in either direction of X or Y shall produce 480 ± 10% pulses. When no rotation of the ball occurs, regardless of the ball position, there shall be no pulses output in X or Y. When the ball is rotated in the + direction in X or Y, the + output pair of lines shall contain pulses while the - output pair shall not contain pulses, and conversely with opposite rotation.

3.5.6.6 Readouts - The TDC shall contain the following indicators.

3.5.6.6.1 Marker Control Indicator - The EDIT/MESSAGE indicator shall be a numeric readout consisting of three digits, each a seven segment LED display. Control of this readout shall be from the EDIT/MESSAGE portion of the SCALE/MESS/EDIT word from the computer during normal operation.

3.5.6.6.2 Hook Control Indicator - The BEARING DEGREES indicator shall be a numeric readout consisting of three digits, each a seven segment LED display. Display shall be normally controlled by the Bearing portion of the HOOK Readout word from the computer. It shall be located above the Trackball and to the left of the RANGE/BIAS readout.

The RANGE/BIAS numeric indicator shall consist of five digits, each a seven segment LED display with a decimal between the units and tens digit. The digit control shall be from the Range portion of the Hook Readout word from the computer. The decimal shall be illuminated when the scale indicator control portion of the SCALE/MESS/EDIT word from the computer is coded MILES. This readout shall be located above the Trackball and to the right of the BEARING DEGREES readout.

The YDS NM YPM indicators shall be a group of three lamps located above the FLANGE/BIAS readout that shall illuminate in accordance with the Scale Indicator control portion of the SCALE/MESS/EDIT word from the computer.

3.5.6.6.3 Range Scale Indicator - The NM readout shall be a four digit, seven segment LED display that shall be normally controlled by the Scale Display portion of the SCALE/MESS/EDIT word from the computer. It shall be located above the RANGE SCALE PBI's.

*3.5.6.7 Panel Illumination - An integrally illuminated information panel is not required. All LED readout indicators shall be capable of being dimmed from the LED-INTENSITY control on the Converter (Figure 21). The PBI's shall be capable of being dimmed by the external control for the edge-lit panel of the Converter.

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3.5.6.8 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function
2J1	M81511/41FH02P1	Converter-Interface
2J2	M81511/41FE01S3	AGE

3.5.6.9 System Power - The equipment shall utilize power from the Converter (Para. 3.5.5.9) as follows:

(1) 115 VAC, 400Hz, 30, Category B, 25 VA.

(2) 5 VDC, 56 watts.

3.5.6.10 Mounting - The TDC shall be designed for installation on the modified horizontal surface immediately below and in front of the ITAD and IP-886B/ASA-66 units.

3.5.6.11 Color - The surface shall be painted lusterless gray, No. 36231, in accordance with FED-STD-595.

3.5.6.12 Marker Placard - An adhesive placard shall be located on the TDC to identify normal marker usage, and shall be labeled as follows:

1 - 31	SONOBUOY	39	PRIORITY FLY-TO
32 - 33	SURFACE	40	TARGET PREDICT
34 - 35	RADAR	41 - 60	GENERAL PURPOSE
36 - 38	MAD		

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3.5.7 Navigation Computer (NC) CP-1224/ASN-124 - The NC shall meet the following requirements:

3.5.7.1 Function - The NC shall be a general purpose digital computer and shall be programmed for geographic and tactical navigation and to process data for the various tactical displays and controllers. The NC shall contain the program and switching output discretes to permit both ground and in flight operation and shall provide the control signals to maintain the CN-1231/ASN-84 Gyroscope Assembly as the stabilized inertial reference. The NC shall contain built-in-test circuitry and program subroutines which shall determine system Go or No-Go conditions.

3.5.7.2 Form Factor - The NC, exclusive of fan, handles, and connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
7.600	11.020	14.700

3.5.7.3 Weight - The weight of the NC shall not exceed 42 pounds.

3.5.7.4 Contents - The NC shall contain 32,768 words of memory. It shall be designed to accommodate expansion to 65,536 words of memory, with the growth in increments of 16 K. Memory growth shall require no physical size changes or electrical modifications to the NC. 7/ The memory harness and connectors necessary for memory growth to 65,536 words shall be present in the NC. Memory growth shall be accomplished by adding the desired number of memory planes. The NC shall contain the following major functions:

3.5.7.4.1 Processor - The processor shall be a parallel type with the following characteristics:

(1) Number System:	Two's complement, fixed point
(2) Word length:	16 bit instructions, 16/32 bit data
(3) Instruction Repertoire:	103 minimum instructions allowing single and double precision arithmetic, conditioned branch and logic operation, as defined in Figures 33 and 34.

7/ Modification to the memory discretes shall be accomplished on the memory harness card so that the discretes shall be appropriate to the particular memory configuration being installed.

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	Code	Time μs
Halt	000	N/A
Add Single Precision	001	1.0 [2]
Add Memory Single Precision	002	4.0
Add Interbank	003	1.0
Add Byte	004	4.0
Add Double Precision	101	1.0 [4]
Add Memory Double Precision	102	5.50
Subtract Single Precision	005	1.0 [2]
Subtract Memory Single Precision	006	4.0
Subtract Interbank	007	1.0
Subtract Byte	010	4.0
Subtract Double Precision	105	1.0 [4]
Subtract Memory Double Precision	106	5.5
Multiply Single Precision	011	3.5 [1]
Multiply Major/Minor Product	012	4.25 [1]
Multiply Memory Single Precision	013	5.75
Multiply Memory Major/Minor Product	014	6.5
Multiply Interbank	015	3.5
Multiply Interbank Major/Minor Product	016	4.25
Multiply Double Precision	111	5.50 [3]
Multiply Memory Double Precision	113	9.25
Divide Single Precision	017	10.75 [1]
Divide Memory Single Precision	020	13.0
Divide Interbank	021	10.75
Divide Double Precision	117	18.75 [3]
Divide Memory Double Precision	120	22.5
Transfer a Single Precision	022	1.0
Transfer 2's Complement	023	1.0
Transfer B Single Precision	024	1.0
Memory Transfer	025	1.75 + 1.5 (N) [6]
Memory to Memory Transfer	171	5.5 + 2.5 (M) [5]
Program to Memory Transfer	172	2.5 + 2.5 (M) [5]
Transfer a Double Precision	122	1.0
Transfer 2's Complement Double Precision	123	1.0
Transfer B Double Precision	124	1.0
Memory Transfer Double Precision	125	1.75 + 3.0 (N) [6]
Absolute Value Single Precision	026	1.0
Absolute Value Double Precision	126	1.0
Increment	027	1.0
Decrement	030	1.0
Increment Double Precision	127	1.0
Decrement Double Precision	130	1.0
Short Branch	031	1.75
Discrete Branch	032	2.5 [1]
Long Branch	033	1.75 [1]
Logical Branch A	034	3.25
Logical Branch B	035	3.25 [1]
Logical Branch C	036	3.25
Logical Branch D	037	3.25 [1]
Logical Branch E	040	2.50
Logical Branch M	134	6.25 [1]
Branch and Store	041	3.25 [1]
Branch Indirect	042	4.00
Logical Branch B Double Precision	135	3.25 [3]
Logical Branch D Double Precision	137	3.25 [3]
Logical Branch E Double Precision	140	2.50
Logical Branch M Double Precision	136	7.75 [3]

[1] Add 0.75 μs if one operand is a constant.
[2] Add 1.50 μs if one operand is a constant.
[3] Add 2.25 μs if one operand is a constant.
[4] Add 3.0 μs if one operand is a constant.
[5] M = Number of single precision words transferred (up to 32).
[6] N = Number of single or double precision words transferred (up to 8)

FIGURE 33. Instruction summary.

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	Code	Time µs
AND	043	1.0 [2]
Exclusive OR	044	1.0 [2]
Inclusive OR	045	1.0 [2]
NAND	046	1.0 [2]
AND Double Precision	143	1.0 [3]
Exclusive OR Double Precision	144	1.0 [3]
Inclusive OR Double Precision	145	1.0 [3]
NAND Double Precision	146	1.0 [3]
Shift Right Logical Single Precision	047	1.75 + 0.5 (M) [7]
Right Rotate Single Precision	050	1.75 + 0.5 (M) [7]
Shift Right Arithmetic Single Precision	051	1.75 + 0.5 (M) [7]
Shift Left Logical Single Precision	052	1.75 + 0.5 (M) [7]
Shift Right Logical Double Precision	147	1.75 + 0.5 (M) [7]
Right Rotate Double Precision	150	1.75 + 0.5 (M) [7]
Shift Right Arithmetic Double Precision	151	1.75 + 0.5 (M) [7]
Shift Left Logical Double Precision	152	1.75 + 0.5 (M) [7]
Input Single Precision	054	1.75 + 16.75 (N) [8]
Input Memory	055	4.0 + 17.0 (M) [7]
Output Single Precision	057	1.75 + 16.75 (N) [8]
Output Memory	060	3.25 + 17.5 (M) [7]
Read Memory to Bank Single Precision	061	2.5 + 1.5 (N) [8]
Read Byte	062	2.5 + 1.5 (N) [8]
Read Indirect Single Precision	063	4.0 + 1.5 (N) [8]
Read Memory to Bank Double Precision	161	2.5 + 3 (N) [8]
Read Indirect Double Precision	163	4.0 + 3 (N) [8]
Write Memory From Bank Single Precision	064	2.5 + 1.0 (N) [8]
Write Byte	065	2.5 + 1.0 (N) [8]
Write Indirector Single Precision	066	4.0 + 1.0 (N) [8]
Write Memory From Bank Double Precision	164	2.5 + 2.0 (N) [8]
Write Indirect Double Precision	166	4.0 + 2.0 (N) [8]
Read Indirect Register	162	1.75 + 1.5 (N) [8]
Write Indirect Register	165	1.75 + 1.0 (N) [8]
Interrupt	067	8.25
Interrupt Return	070	5.5
Execute Register	071	2.5 [6]
Execute Address	072	4.0 [1], [6]
Set Register	073	1.0
Set Bit	074	5.0 [5]
Set Bit Register	174	1.75
Sense Bit Register	175	3.25
Set Register Double Precision	173	1.0
Sense Bit	075	5.5 [4]
Set Discrete Sign	076	1.0
Set Discrete Field	077	1.0
Status to Working Register	167	1.75
Working Register to Status	170	1.0 [2]

[1] Add 0.75 µs if one operand is a constant.
[2] Add 1.5 µs if one operand is a constant.
[3] Add 3.0 µs if one operand is a constant.
[4] Subtract 2.25 µs if it is a status word bit.
[5] Subtract 3.25 µs if it is a status word bit.
[6] Add "executed operation" time.
[7] M = Number of single or double precision words transferred (up to 32).
[8] N = Number of single or double precision words transferred (up to 8).

FIGURE 34. Instruction summary. - continued

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(4) Execution Times:

	SINGLE PRECISION	DOUBLE PRECISION
Add/Subtract	1.0 µsec	1.0 µsec
Multiply	3.5 µsec	5.5 µsec
Divide	10.75 µsec	18.75 µsec

(5) Execution times include:

(a) Instruction look-ahead fetch
(b) Obtain two operands from file registers
(c) Complete operation
(d) Return result to file register

3.5.7.4.2 File Registers - The NC shall contain 16 file registers each 16 bits long. Access time shall be less than 15 nanoseconds.

3.5.7.4.3 Memory - The NC shall be capable of accepting up to 32K words of programmable memory. The words are 16 bits long. The memory shall operate at 1.0 microseconds maximum cycle time with 0.5 microseconds maximum access time.

3.5.7.4.3.1 Memory Protection - A memory protect capability shall be provided in the logic of the memory address circuits and shall operate as follows:

The region of lower memory shall be protected by establishing a hard wired address into the memory address logic. This address shall form a fixed boundary below which no write operation is allowed. If a write is attempted into an address below the protect boundary, the memory cycle shall be aborted and an interrupt is generated.

3.5.7.4.3.2 Addressing Modes - As a minimum the NC shall be capable of addressing the following modes:

A. Absolute
B. Indirect
C. Indexed
D. Immediate Operand
E. Relative addressing

3.5.7.4.3.3 Memory Loading - Control lines shall be incorporated at the external connectors to provide loading, modifying or extracting of contents of memory through the use of external equipment.

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3.5.7.4.4 Input/Output - As a minimum the NC shall contain 8 input data channels, 8 output data channels, 32 input discretes, and 32 output discretes. The NC shall have the expansion capability of 16 output channels, 16 input channels, 64 input discretes and 64 output discretes. All input lines will be 2 wire differential line receivers (Fairchild µA9615 or equivalent) terminated with 130 ohms dc. All output lines will be 2 wire differential line drivers (Fairchild µA9614 or equivalent). In addition to the differential input discretes there shall be three high-level, single-ended logic level discretes.

3.5.7.4.4.1 Data Channels - All data channels shall transfer 16 bit binary words serially, least significant bit first, at a rate of 1.0 MHz. The I/O clock shall supply all serial data transfer rates. For data output from the NC, the data line shall be stable for 250 nanoseconds after the trailing edge of the clock pulse.

3.5.7.4.4.2 Differential Discretes - Discretes shall be differential signals in accordance with Figure 18. Input discretes are sensed as steady logic “1” or “0” on any of 32 lines. Output discretes shall be of two types:

(1) Set discrete line to logic “1” or logic “O”.

(2) Supply 250 nanosecond logic “1” pulse.

3.5.7.4.4.2.1 Interrupts - The NC shall accept a minimum of eight levels of interrupts - designated level 1 through level 8. Interrupt priority shall be from level 1 through level 8 in decreasing priority - where priority of an interrupt level identifies which interrupt will be serviced first when more than one interrupting event occurs concurrently.

3.5.7.4.4.3 Clock - The NC shall have a 16 MHz clock whose accuracy shall be greater than 50 parts per million.

3.5.7.4.4.4 DMA Channel - The NC shall provide a 16-bit parallel one half duplex, differential DMA channel which shall be capable of 8-bit data byte addressing. The channel operates autonomously on a “cycle stealing” basis. The maximum data rate is 570,000 16-bit words per second.

3.5.7.4.4.5 Built-In Test - BIT features shall be incorporated into the NC to provide GO/NO-GO indications of the NC readiness automatically on the NC.

3.5.7.4.4.6 Software Diagnostic Capability - The NC shall be logically and electrically designed to facilitate the isolation of a malfunctioning of three or less modules through diagnostic programming.

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3.5.7.4.4.7 High Level Discretes - The SC shall accept three, high level, single-ended logic level discretes defined as follows:

0 to +4.5 VDC = Logic Zero

Open, or + 5.5 to + 15 VDC = Logic One

3.5.7.5 Input Power

3.5.7.5.1 Aircraft Power - The equipment shall utilize Aircraft Power in accordance with Paragraph 3.3.12 as follows:

115 VAC, 400 Hz, 3 0, Category B, 430 VA

3.5.7.5.2 System Power The equipment shall utilize System Power as follows for a period of 10 seconds minimum (Paragraph 3.5.10) should power interruption occur.

72 VDC to 64 VDC, 340 watts, regulated according to power requirements, from the Computer Battery Unit.

3.5.7.6 Mounting - The NC shall be mounted on a shockmount. The shockmount shall not be provided as part of the equipment.

*3.5.7.7 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function
J1	MS27478Y14D18PA	Primary Power
J2	SJTP02RE24-35S(023)	Test
J3	SJTP02RE18-35P(023)	SDC
J4	SJTP02RE18-35PB(023)	SDC
J5	SJTP02RE18-35SA(023)	Keyer Control
J6	SJTP02RE18-35SC(023)	SDC
J7	SJTP02RE20-35SA(023)	Tape Loader
J8	SJTP02RE20-35P(023)	SDC
J9	MS27508E8F35P	Fan Power

3.5.7.8 Color - The NC case shall be lusterless gray, No. 36231, as specified in FED-STD-595.

3.5.8 Signal Data Converter Unit (SDC) CV-3289/ASN-124 - The SDC shall meet the following requirements:

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3.5.8.1 Function - The SDC receives input analog signals from external equipments and converts these signals into digital data for processing by the NC. It provides conversion of digital data from the NC to synchro and DC outputs. It provides level shifting of pulse trains and discretes for NC interface with external equipments. It provides lighting and logic power to external equipments. It provides conversion of input signals from the OMEGA Receiver Converter OR-90/ARN-99(V)1 from parallel to serial digital data. It provides resolver to synchro buffer to interface with the CN-1231/ASN-84 Gyroscope Assembly azimuth output to the MX-6985/ASN-50 Compass Adapter Compensator.

3.5.8.2 Form Factor - The Signal Data Converter, exclusive of fan, connector projections and handle protrusion, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
9.000	15.875	22.000

3.5.8.3 Weight - The weight shall not exceed 54 pounds.

3.5.8.4 Contents - The SDC shall accommodate system input/output functions wherever necessary to reduce or eliminate the addition of functions to the NC. Spare card capacity of 4 card slots shall be provided. The power supply shall be sufficient to supply the SDC when all card slots are filled and all other system growth requirements are fulfilled assuming average SDC board power requirements and existing voltages. Connector provision for additional input/output capability shall be made. Connectors shall be functionally separated to the greatest extent possible to facilitate aircraft installation.

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The SDC shall contain the following major functions:

3.5.8.4.1 Analog Inputs - The following external analog inputs shall be converted for digital transmission to NC:

INPUT	TYPE	SOURCE
True Air Speed (8)	3 wire synchro, 11.8 V line to line, 400 Hz, Phase A	CPK-28/A24G-9 True Air Speed
Radar Bearing (8)	3 wire synchro, 11.8 V line to line, 400 Hz, Phase A	IP-414A/APA-125 Radar Indicator
Drift Angle	3 wire synchro, 11.8 V line to line, 400 Hz Phase A	C-4418/APN-153(V) Doppler Control Panel
Azimuth Resolver (8) Single Speed	4 wire resolver, 5 V line to line, 400 Hz 25° lead from Phase A	CN-1231/ASN-84 Gyroscope Assembly
Azimuth Resolver (8) 8 Speed	4 wire resolver, 5 V line to line 400 Hz 65° lead from Phase A	CX-1231/ASN-84 Gyroscope Assembly
Magnetic Heading (8)	3 wire synchro, 11.8 V line to line, 400 Hz Phase A	MX-6985/ASN-50 Compass Adapter Compensator
Roll Gimbal Temperature	2 wire DC voltage, +2 VDC to +5 VDC range	CN-1231/ASN-84 Gyroscope Assembly

(8) Reference is directed to Paragraph 3.5.8.4.1.2.

3.5.8.4.1.1 Input Impedance - All synchro and resolver inputs shall have an input impedance of not less than 5,000 ohms. Roll gimbal temperature input shall have an input impedance of not less than 10,000 ohms.

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3.5.8.4.1.2 Analog to Digital (A/D) Conversion Accuracy - The A/D conversion shall be made to 12 bits of resolution. the conversion accuracy over the temperature range -55°C to +55°C shall be:

(1) Within ±12 arc minutes statically and dynamically up to, but not greater than, an angle rate of change of 15 degrees per second for all synchro and resolver inputs, except conversion accuracy for asterisk (*) inputs can be ± 18 arc minutes.

(2) Within ± 10% of full scale value for roll gimbal temperature.

3.5.8.4.2 Analog Outputs - The SDC shall provide the following analog outputs:

OUTPUT	TYPE	MAXIMUM LOAD
Course Command	3 wire synchro, 11.8 V line to line, 400 Hz, Phase A	AF/A24J-1 HSI (2 units)
100's Distance	3 wire synchro, 11.8 V line to line, 400 Hz Phase A	AF/A24J-1 HSI (2 units)
10's Distance	3 wire synchro, 11.8 V line to line, 400 Hz Phase A	AF/A24J-1 HSI (2 units)
1's Distance	3 wire synchro, 11.8 V line to line, 400 Hz Phase A	AF/A24J-1 HSI (2 units)
Drift Angle	3 wire synchro, 11.8 V line to line, 400 Hz Phase A	AF/A24J-1 HSI (2 units)
N-S Distance	3 wire synchro, 11.8 V line to line, 400 Hz Phase A	ID-499/ASA-13 Position Indicator and PT-396/AS Plotting Board
E-W Distance	3 wire synchro, 11.8 V line to line, 400 Hz, Phase A	ID-499/ASA-13 Position Indicator and PT-396/AS Plotting Board

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OUTPUT	TYPE	MAXIMUM LOAD
True Heading	3 wire synchro, 11.8 V line to line, 400 Hz, Phase A	AF/A24J-1 HSI (2 units) ID-663/U BDHI (1 unit) MW-2 Compass PT-396/AS Plotting Board MX-2230(XN-1)ASQ Monitor APS-80 Search Radar R-976B/ALD-2 ECM Receiver Indicator
Autopilot Steering	2 wire AC, 0 to 4.5 V, 400 Hz, Phase A, 0.390 VRMS/DEGREES	PB20-N Autopilot
Course Deviation Bar	2 wire meter movement drive, 0 to ± 485 µA into 330 ohm load	AF/A24J-1 HSI (2 units)

3.5.8.4.2.1 Output Accuracy - The digital to synchro conversion shall be within ± 45 minutes. Autopilot steering and course deviation bar conversion shall be within ± 10% of full scale.

3.5.8.4.3 Doppler Ground Speed Interface - The doppler ground speed shall be input as a pulse train from Pin Y of 2J1 on RT-680/APN-153(V) Doppler Receiver-Transmitter. The input pulse train is a negative going 4 volt pulse with respect to the Doppler Ground Speed Reference (+25 VDC). This signal shall be converted into a digital data word proportioned to the frequency of the pulse train and conditioned for output to the NC in accordance with 3.5.8.4.8.

3.5.8.4.4 Radar Range Interface - Radar range shall be converted from pulse width to digital data word. Radar range is input from Radar Buffer Converter and output to NC.

3.5.8.4.4.1 Inputs - The following signals shall be put on 2 wire differential line receivers (Fairchild µA9615 or equivalent) terminated with 130 ohms dc.

Input Signal	Description
Range Scale Factor Gate	Low range is 990 msec gate or less, High range is 2200 msec gate or greater.
Range Gate	Gate width is proportional to target range. Scaling is low range: 12.36 µsec/nm high range: 6.18 µsec/nm

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3.5.8 .4.4.2 Outputs - The radar range signal shall be transferred on the data channel of para. 3.5.8.4.8.

3.5.8.4.5 Discrete Signal Conditioning - The SDC shall format discretes for interface with NC. All input discretes shall be converted to 2 wire differential line driver (Fairchild µA9614 or equivalent). All output discretes shall be received from the NC as 2 wire differential line receivers (Fairchild µA9615 or equivalent) terminated with 130 ohms dc.

3.5.8.4.5.1 Discrete Inputs - The following discrete inputs shall be received and converted to NC signal level.

Input	Signal Level
Doppler Memory	28 VDC
Doppler Test	28 VDC
Doppler Normal	28 VDC
Search Stores Release	SPDT isolated contacts, 1 amp max
Kill Stores Release	SPDT isolated contacts, 1 amp max
Mark On Top	SPDT isolated contacts, 1 amp max
AHRS No-Go	SPDT isolated contacts, 1 amp max
ITAD No-Go	2-wire differential line driver
Battery Bite GO (IMS)	SPDT isolated contacts

3.5.8.4.5.2 Discrete Outputs - The following discretes shall be received from the NC and output to external equipments. These discretes outputs shall be relay contacts that are open or connected to aircraft ground.

Output	Signal Level
Distance Shutter	200 ma DC
1000's Shutter	200 ma DC
IMS No-Go	200 ma DC
True Heading Valid	200 ma DC

3.5.8.4.5.3 Battery Test Command - The SDC shall route the Battery Test Command from the NC to the IMS Power Supply. When the NC is off, the Battery Test Command shall be false.

3.5.8.4.6 Omega Interface - The SDC shall interface the computer subsystem with the AN/ARN-99(V)1 Omega Navigational Set in accordance with MIL-N-81675(AS). Parallel data from the Omega Receiver Converter shall be converted to serial data to the NC.

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3.5.8.4.6.1 Digital Inputs - All digital inputs shall be 2 wire differential line receivers (Fairchild µA9615 or equivalent).

Input	Description
Omega Data 2^0 through 2^{29}	30 parallel data lines from Omega Receiver to Omega data register
Omega In Clock	Clock used to remove data serially from Omega data register
Omega Out Clock	Clock used to enter Omega command word serially into Omega command word register.
Receiver Interrupt	Interrupt line from Omega Receiver indicating data is available.
Omega Data Strobe	Single pulse used to parallel enter Omega data into Omega data register and then change this register from parallel enter to serial shift mode.
Omega Input Acknowledge	Single pulse used to change Omega data register from serial shift to parallel enter mode.
NC Master Reset	Perform the same function as Omega Input Acknowledge during initialization.
Omega Interrupt Enable	Level discrete that sets interrupt enable.
Omega External Function Request	Input from Omega Receiver buffered to NC.

3.5.8.4.6.2 Digital Outputs - All digital outputs shall be 2-wire differential line drivers (Fairchild µA9614 or equivalent).

output	Description
Omega Command 2^0 through 2^{23}	24 parallel lines to the Omega receiver from Omega command word register
Omega Interrupt	Discrete derived from Receiver interrupt.
Interrupt Enable	Discrete to Omega Receiver
Omega In	Data output to remove data serially from Omega data register
Omega External Function	Discrete from NC direct to Omega Receiver. Noted for reference only.
Input Data Request	Discrete to NC buffered from Omega Receiver.

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3.5.8.4.7 Inertial Platform Interface - The SDC shall provide interface with the CN-1231/ASN-84 Gyroscope Assembly. This interface function shall utilize protected power from the PP-4964/ASN-84 Power supply.

3.5.8.4.7.1 Accelerometer Sensing Signals - Accelerometer inputs of plus and minus pulses for both X and Y accelerometers shall be provided. The Plus and minus pulses shall be accumulated in up-down counters for each acceleraometer. Accumulator output shall be available upon request by the NC.

3.5.8.4.7.2 Gyro Torquing Signals - Gyro torquing commands from the NC shall be converted into gyro pulse torquing commands for X, Y, and Z gyro torquing channels.

3.5.8.4.7.3 Resolver to Synchro Conversion - The SDC shall also provide conversion of the azimuth resolver single speed analog input as noted in 3.5.8.4.1 to 3 wire synchro, 11.8 V line to line, 400 Hz, Phase A signal.

3.5.8.4.8 DPU Interface - Transmission of data between the SDC and NC shall be on two wire differential line drivers (Fairchild µA9614 or equivalent) and receivers (Fairchild µA9615 or equivalent).

3.5.8.4.8.1 Data Channels - Serial binary data channels shall be used for transferring 16-bit words from NC to the SDC and from SDC to the NC.

3.5.8.4.8.2 SDC Interrupt - A 0.5 µsec pulse to the NC shall signal that the SDC has finished the last task assigned by the NC.

3.5.8.4.8.3 Command Clocks - I/O clocks, consisting of sixteen (16) 250 µsec pulses at 1.0 MHz, shall be used to transfer data from the NC to the SDC, and from SDC to the NC.

*3.5.8.4.8.4 Command Reset - A 250 nanosecond pulse from the NC to the SDC shall signal that the next data transfer will be a new command word from the NC. The command word specifies the type of data transfer to follow and the device with which the transfer will be made.

3.5.8.4.8.5 4 MHZ Clock - A continuously running, precision clock at 4 MHz shall be provided for use by SDC conversion circuits.

3.5.8.4.9 AQA-7 Interface - The SDC shall obtain range, bearing and program data from the SS-1 and SS-2 AQA-7 (DIFAR) memory units (Paragraph 6.8) and provide these data to the NC on the data channel of 3.5.8.4.8. The SDC interface shall be compatible with AQA-7's in accordance with MIL-C-81486 and AS-1283 when installed in any combination in SS-1 and SS-2.

3.5.8.4.10 ARR-72 Interface - The SDC shall interface with the eight Dual Channel Control Indicators, four each at Sensor Station 1 and Sensor Station 2. (Refer to Paragraph 6.8.) Each DCCI contains two channels of RF intonation. Each channel provides a two-digit RF number from 00 through 31 in BCD format of six bits. The SDC shall interleave the RF numbers obtained from the ARR-72 on the data channel of 3.5,8.4.8 to the NC.

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3.5.8.4.11 ITADS interface - The SDC shall provide the computer subsystem interface with the ITADS as defined by Paragraphs 3.5.5 and 3.5.6.

3.5.8.4.12 Altimeter Interface - The SDC shall interface with an AAU-21A or AAU-32A Digital Encoder Altimeter in accordance with MIL-A-81403 or MIL-A-81852.

3.5.8.4.13 BITE - The SDC shall include a latching GO/NO-GO indicator for continuous monitoring of SDC status. Manual reset of the indicator shall be provided adjacent to the indicator. Hardware shall be incorporated to monitor the status of the SDC power supply. As a minimum, provision shall be made for software testing of critical SDC hardware functions. Provisions shall also be made for reporting these status indications to the NC for display on the Digital Display Indicator.

3.5.8.5. Power

3.5.3.8.5.1 Aircraft Power - The equipment shall utilize aircraft power in accordance with paragraph 3.3.12 as follows:

(1) 115 VAC, 400 Hz, 3 0, Category B, 320 VA
(2) 0-26 VAC, 400 HZ, 1 0, 1.0 Amp.
(3) 28 VDC, 1.4 watts
(4) 26 VAC, 400 Hz, 1 0, 6 VA

3.5.9.5.2 System Power - The equipment shall utilize power from PP-4964/ASN-84 Power Supply as follows:

(1) +5 VDC, 4 watts
(2) +15 VDC, 0.5 watts
(3) -15 VDC, 0.3 watts
(4) +26 VDC, 1 watt
(5) -26 VDC, 1 watt

*3.5.8.5.3 Power Outputs - The SDC shall be capable of supplying the following power to equipment external to the SDC:

(1) 115 V, 400 Hz, 3 0, 56 watts
(2) 0-26 VAC, 400 Hz, 1 0, 0.8 Amp
(3) +13.5 VDC ± 5% at 1 Amp
(4) -13.5 VDC ± 5% at 1 Amp
(5) +5 VDC ± 5% at 8 amp., with respect to chassis ground, less than 100 mV RMS ripple content, short circuit protected.
(6) Analog ground shall be isolated from chassis ground.

3.5.8.6 Mounting - The SDC shall be mounted on shockmount. The shockmount shall not be supplied as part of the equipment.

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*3.5.8.7 Electrical Connections - Connections to external circuits shall be provided as listed:

Reference Designation	Receptacle Type	Function
J1	MS27478Y12D98P	Primary Power
J2	M81511/51FF03S1	Keyer Control Power
J3	M81511/51FE03S1	MPC Power
J4	SJTP02RE16-35P(023)	NFIB
J5	MS27505E15F35P	Analog outputs
J6		Growth
J7	MS27508E8F35P	Fan Power
J8	MS27505E21F35P	Omega
J9	MS27505E21F35PB	NC
J10	MS27505E19F35PB	ASN-84
J11	MS27505E21F35PA	Omega
J12	MS27505E19F35P	ARR-72
J13	MS27505E19F35PA	ARR-72
J14	MS27505E11F35P	Altimeter
J15	SJTP02RE14-35SA(023)	MPC/MEC
J16	SJTP02RE16-35SB(023)	ITADS/Radar Buffer
J17		Growth
J18	SJTP02RE20-35S(023)	NC
J19	SJTP02RE14-35S(023)	AQA-7
J20	SJTP02RE16-35S(023)	BU

3.5.8.8 Color - The SDC case shall be lusterless gray, No. 36231 as specified in FED-STD-595.

3.5.9 Radar Buffer Converter CV-3290/ASN-124 - The Radar Buffer Converter shall meet the following requirements:

3.5.9.1 Function - The Radar Buffer Converter processes outputs from the IP-414A/APA-125 Radar Indicator for transmission to the SDC.

3.5.9.2 Form Factor - The Radar Buffer Converter, exclusive of connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
2.18	4.40	4.70

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3.5.9.3 Weight - The weight of the Radar Buffer Converter shall not exceed 1.5 pounds.

3.5.9.4 Contents - The Radar Buffer Converter shall contain the circuits necessary to develop Range Gate Bar and Scale Factor Gate Bar outputs for the SDC from inputs of Strobe and Scale Factor Gate from the IP-414A/APA-125 Radar Indicator in accordance with MIL-I-19165. The Buffer outputs shall not be adversely affected when cursor 3 is active.

3.5.9.5 Mounting - The Radar Buffer Converter shall be designed to mount in any attitude. Mounting dimensions shall be per Figure 35.

3.5.9.6 Color - The case of the Radar Buffer Converter shall be lusterless gray, No. 36231 per FED-STD-595.

*3.5.9.7 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function
J1	M39012/24-0001	Range Strobe Input
J2	SJTP02RE8-35S(023)	Range S.F. Gate Input
J3	SJTP02RE10-35P(023)	Range Output and Power Input

3.5.9.8 Aircraft Power - The equipment shall utilize aircraft power in accordance with Paragraph 3.3.12 as follows:

115 V, 400 Hz, 1 0, 0.1 Amp

3.5.10 Battery Unit BA-597/ASN-124 - The Battery Unit shall meet the following requirements:

3.5.10.1 Function - The Battery Unit shall provide battery power to the NC during power interruptions and shall provide sequencing and interlock circuits for control of the Inertial Measurement System AN/ASN-84.

3.5.10.2 Form Factor - The Battery Unit, including connectors, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
8.500	6.500	13.000

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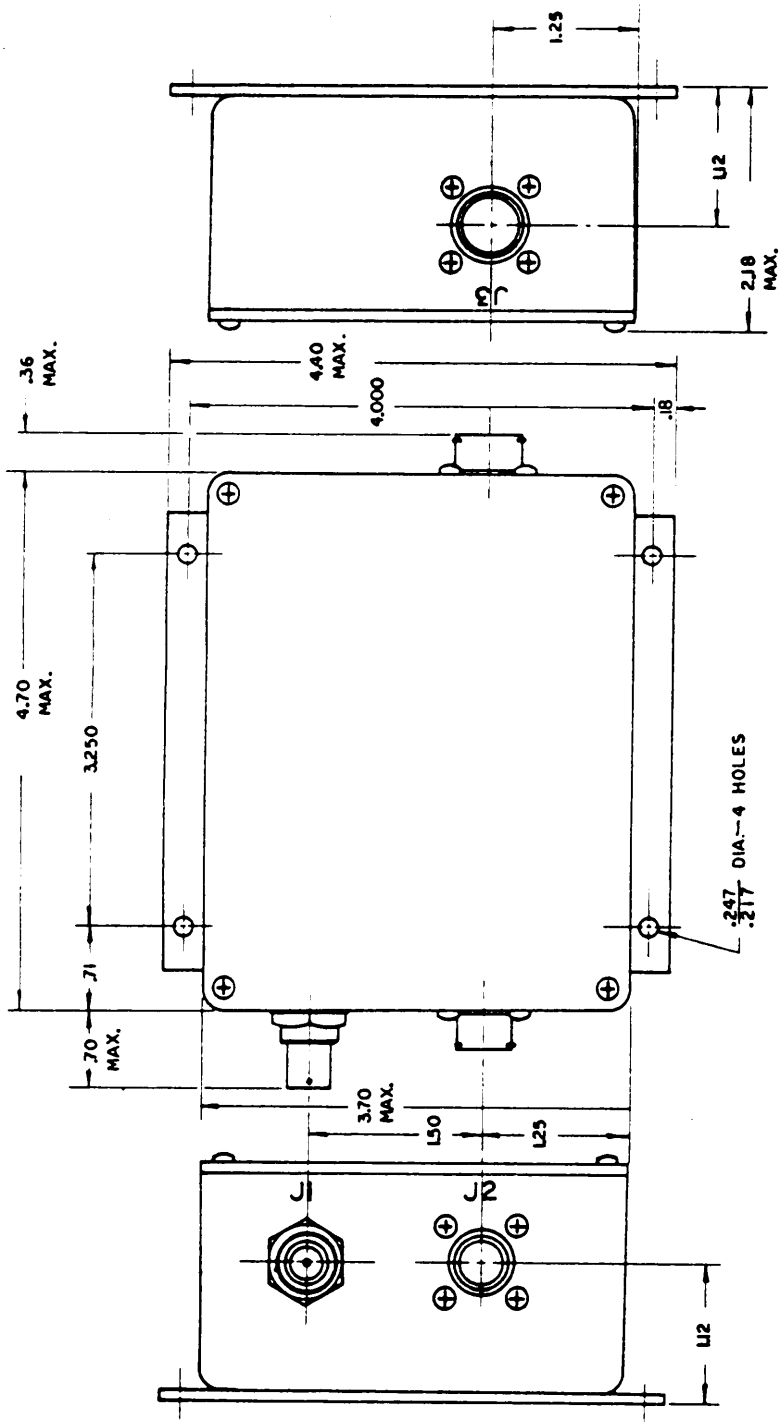


FIGURE 35. Radar Buffer Converter mounting dimensions.

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3.5.10.3 Weight - The weight of the Battery Unit shall not exceed 20 pounds.

3.5.10.4 Contents - The Battery Unit shall consist of the following major functions:

3.5.10.4.1 Battery - The battery shall provide power for DPU operation during momentary loss of aircraft power. The battery shall be capable of providing 340 watts for at least 10 seconds with a forty-five minute period of normal operation prior to the next discharge, and a maximum of twelve such cycles in twelve hours. Voltage of the battery at no load shall be 72 V minimum and at rated load may reduce to 64 V.

3.5.10.4.2 Battery Charger - The unit shall contain a charger to maintain the battery charge when power is ON. The charger shall be designed in accordance with requirements of the battery selected. The battery charger shall contain environmental controls to prevent charging at temperatures under 0°C or over 50°C, and other safety features as required by the battery.

3.5.10.4.3 System Power Controls - The Battery Unit shall accept mode select signals from the Mode Controller and provide power to the other units of the system in accordance with the following:

3.5.10.4.3.1 Power Off - The Battery Unit shall provide no power. When the selector switch is positioned to the OFF position from any other position, an interlocking circuit shall be operated such that the 2 sustaining battery supplies (the Battery Unit and the PP-4964/ASN-84 Power Supply) will not attempt to maintain system operation.

3.5.10.4.3.2 Computer - Three phase 115V 400 Hz aircraft power shall be applied to the NC and the SDC by external relays under Battery Unit control. The Battery Unit shall provide +28 VDC, 127 milliamps per relay for these turn-on relays.

3.5.10.4.3.3 Align - When the Mode Selector Switch is turned to ALIGN, the Battery Unit shall apply three phase 115V 400 Hz aircraft power to the PP-4964/ASN-84 Power Supply by an external relay and the Battery Unit shall force discrete outputs D1, D2, D3 and D5 in the logic “1” state for 10 seconds.

Between the time 10 seconds and 16 seconds after selection of ALIGN, the Battery Unit shall check input discretes D1, D2, D3 and D5; if during this time interval any of these do not maintain logic “1” state, the Battery Unit shall automatically shut the NC, SDC, and PP-4964 Power Supply OFF.

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3.5.10.4.3.4 NAV - Power applied during ALIGN shall remain during NAV.

3.5.10 .4.3.5 SLAVE - Power applied during ALIGN shall remain during SLAVE.

3.5.10.4.3.6 ATTD - When the Mode Selector Switch is turned to ATTD, the Battery Unit shall force the Back Up Level discrete (D6) to the CN-1231 Gyroscope Assembly (GA) and shall also turn off power to the NC and to SDC.

3.5.10.4.3.7 Reversing Selector Switch - Reversal of the Mode Selector Switch from ATTD to SLAVE, SLAVE to NAV, NAV to ALIGN or from ALIGN to CMPTR shall cause no reversion in system operation. Reversal of the Mode Selector Switch to OFF shall be recognized and shall cause the system to shut off; a new start sequence shall be required to turn on again.

3.5.10.4.4 Power Failure and Power Shutdown - Power failure and shutdown shall require the following:

3.5.10.4.4.1 Shutdown - Normal operating procedure shall require the pilot to turn the Mode Selector Switch to OFF prior to shutting off aircraft power, in order to avoid unnecessary Battery Unit and PP-4964 Power Supply use at each shutdown.

3.5.10.4.4.2 Low Battery - The Battery Unit shall indicate the existence of low battery voltage or open battery circuit by issuing a high level single ended logic discrete to the NC.

3.5.10.4.4.3 Power Failure Operation

(1) Less than 10 seconds: Continue the IMS, the NC and the Battery Unit on battery. Return to normal when power reappears in less than 10 seconds.

(2) Greater than 10 seconds nominal: Turn off the IMS, SDC, and NC power.

3.5.10.4.4.4 Selector OFF - After the automatic shutdown of Paragraph 3.5.10.4.4.3(2), the Battery Unit shall not allow the system to turn ON unless the Mode Selector Switch has been reset to OFF position, and returned to one of the ON positions.

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3.5.10.4.5 IMS Sequencing and Interlocks - IMS sequence and interlock controls shall be provided as follows:

(1) During the first ten seconds of ALIGN Mode, supply discretes D1, D2, D3 and D5 to IMS.

(2) Prevent turn ON of the gyro unless logic ones for D1, D2, D3 and D5 are available from the NC.

(3) Upon receipt of a logic discrete Operate Mode Interlock from the computer, switch discrete output lines D1, D2, D3, D5, D9, and D10 to the logic zero state and maintain this state until the system is switched to OFF.

(4) During SLAVE and during ATTD switch discrete output line D6 (back up level) to the logic one state and as long as no erection cutout signal is present maintain this state until the system is switched to OFF.

(5) Upon receipt of an Erection Cutout Signal during SLAVE or ATTD, switch discrete output line D6 to the logic zero state and maintain this state as long as the cutout signal is present.

(6) Route discrete D11 from the NC to the GA.

(7) Upon receipt of input discrete D6 from the NC, send output discrete D6 (or'd also with ATTD and SLAVE Mode Selector Switch positions) to the GA.

(8) Upon receipt of a NC Fail signal (Fault is high), the Battery Unit shall close isolated switch contacts to indicate attitude mode to the Mode Controller.

3.5.10.5 Input Power

*3.5.10.5.1 Aircraft Power - The equipment shall utilize Aircraft Power in accordance with Paragraph 3.3.12 as follows:

115 VAC, 400 Hz, 3 0, Category B, 90 VA

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3.5.10.5.2 System Power - The equipment shall utilize system power as follows:

5 VDC, 400 mAmp, regulated, from the PP-4964/ASN-84 Power Supply (not available during CMPTR)

*3.5.10.6 Mounting - The Battery Unit shall be mounted on a shockmount (Ref. Para. 3.5.15). The shockmount shall not be supplied as part of the equipment.

*3.5.10.7 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function
J1	SJTP02RE16-35SA(023)	SDC Interface
J2	SJTP02RE14-35SB(023)	Node Controller
J3	SJTP02RE12-35S (023)	Gyro Assembly
J4	SJTP02RE16-26P(023)	Interconnecting Box

3.5.10.8 Color - The Battery unit case shall be lusterless gray, No. 36231, as specified in FED-STD-595.

3.5.11 Mode Controller C-9856/ASN-124 - The Controller shall meet the following requirements:

3.5.11.1 Function - The Controller shall provide the pilot with control and monitoring of the Inertial Measurement Set (IMS) modes and equipment status. The configuration of the front panel shall conform to Figure 36.

3.5.11.2 Form Factor - The Mode Controller, exclusive of front panel protrusions, attachment points, but including connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
2.630	5.750	5.000 (behind mounting surface)

3.5.11.3 Weight - The weight of the Controller shall not exceed 2 pounds.

3.5.11.4 Contents - The Controller shall contain logic circuitry and power supplies for control and display of inertial navigation system modes and status.

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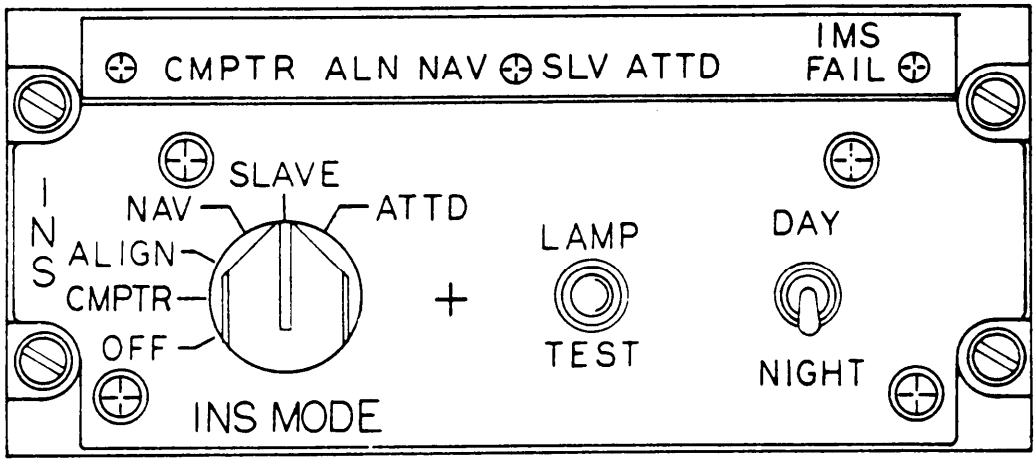


FIGURE 36. Mode Controller.

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3.5.11.5 Controls - The Controller shall provide the following controls:

3.5.11.5.1 INS Mode Switch - A six-position three deck rotary switch shall control the state of output discretes to the system for mode control. It shall be necessary to lift the switch in order to continue rotation when moving between the following positions:

OFF & CMPTR or (CMPTR & OFF)
CMPTR & ALIGN or (ALIGN & CMPTR)
NAV & SLAVE or (SLAVE & NAV)
SLAVE & ATTD or (ATTD & SLAVE)

3.5.11.5.1.1 OFF - In this position of the mode switch, the ON discrete to the Battery Unit shall be open.

3.5.11.5.1.2 CMPTR - This position of the mode switch shall provide continuity of the CMPTR discrete to the Battery Unit.

3.5.11.5.1.3 ALIGN - This position of the mode switch shall provide continuity of the Align discrete to the Battery Unit and shall provide the Align code to the NC.

3.5.11.5.1.4 NAV (Navigate) - This position of the mode switch shall provide continuity of the Navigate discrete to the Battery Unit and shall provide the Navigate code to the NC.

3.5.11.5.1.5 SLAVE - This position of the mode switch shall provide continuity of the Slave discrete to the Battery Unit and shall provide the Slave code to the NC.

3.5.11.5.1.6 ATTD (Attitude) - This position of the mode switch shall provide continuity of the Attitude discrete to the Battery Unit.

3.5.11.5.2 LAMP TEST Switch - A pushbutton switch shall be provided for the push to test function. Depressing the switch shall cause all the annunciators on the panel to illuminate while the switch is held down.

3.5.11.5.3 DAY NIGHT Switch - A two-position toggle switch shall be provided to control annunciator lighting intensity. In the DAY position, the switch shall cause full 5V lighting power to be available for lighting the annunciators. In the NIGHT position, the switch shall select a lower voltage to be available for lighting the annunciators, to be determined by the transformer tap utilized.

3.5.11.6 Displays - The Controller shall provide six annunciators of mode and status. Lamp supply shall be developed from aircraft 26 V, 400 Hz, routed through the DAY NIGHT control function. The mode annunciator legends shall be green when illuminated. The IMS FAIL annunciator legend shall be amber when illuminated. The legends shall not be legible in bright sunlight when not illuminated.

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3.5.11.6.1 Power Sources - Logic and signal power sources shall be as follows:

ANNUNCIATOR	LOGIC POWER SOURCE	SIGNAL SOURCE
ALN (Align) NAV (Navigate)	SDC SDC	NC (LD/R) NC (LD/R)
SLV (Slave) ATTD (Attitude)	SDC N/A	NC (LD/R) Battery Unit (SW)
IMS FAIL	LMS PS	IMS PS (SEL) IMS GA (SEL)
CMPTR	SDC	NC (LD/R)
LD/R = Line Driver/Receiver. SEL = Single Ended Logic SW = Switch		

3.5.11.6.2 Logic Signal Inputs, Differential Type (LD/R) - All differential logic input circuitry shall be similar to a differential line receiver type µA9615 and shall be compatible with the signal supplied by a differential line driver similar to µA9614. Input impedance shall be nominal 130 ohms dc, balanced line load. Logic levels shall be as defined in Figure 4.

3.5.11.6.3 Logic Signals, Single Ended (SEL) - Single ended logic signals shall be unbalanced steady state signals as follows:

Logical one “1” = 2.4 to 5.5 VDC
Logical zero “0” = 0 to 0.4 VDC

3.5.11.6.4 Logic Requirements - Logic circuits shall be incorporated to control specific annunciators as follows:

The IMS FAIL annunciator shall illuminate when either of the single ended logic (SEL) discretes, PS or GA, go to Zero during the ALIGN, NAV, SLAVE and ATTD modes. The IMS FAIL annunciator shall illuminate whenever logic power is lost during the ALIGN, NAV, SLV, and ATTD modes. The ATTD annunciator shall illuminate upon a switch closure signal input. This signal shall be inoperative during the OFF, CMPTR, and ATTD modes.

The ALN, NAV, SLV, and CMPTR annunciators shall illuminate whenever the logic signal input is “1” and shall be inoperative whenever the logic power is lost.

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3.5.11.7 Input Power

3.5.7.11.7.1 Aircraft Power - The equipment shall utilize aircraft power in accordance with Paragraph 3.3.12 as follows:

(1) 26 VAC, 400 Hz, 1 0, 0.8 Amp
(2) 0-26 VAC, 400 Hz, 1 0, 0.4 Amp

3.5.11.7.2 System Power - The equipment shall utilize system power as follows:

(1) 5 VDC, 0.4 Amp, regulated, from the SDC
(2) 5 VDC, 0.1 Amp, regulated, from the PP-4964/ASN-84 Inertial Power Supply.

3.5.11.8 Panel Illumination - The Controller shall incorporate an integrally illuminated information panel per MIL-P-7788, Class 1-R, Type V. The panel shall utilize 26 V, 400 Hz aircraft dimmable power.

*3.5.11.9 Electrical Connections - Electrical connections to external circuits shall be as listed:

Reference Designation	Receptacle Type	Function
J1	SJTP02RE14-35PB(023)	Logic Power and Discretes
J2	SJTP02RE10-98P(023)	Lighting Power

3.5.11.10 Mounting - The Controller shall be a console mounted unit in accordance with MIL-C-6781, Type 1.

3.5.11.11 Color - The Controller shall be painted lusterless black, No. 37038, in accordance with FED-STD-595.

3.5.12 Control-Indicator C-9863/ASN-124 - The Control-Indicator shall meet the following requirements:

3.5.12.1 Function - The Control-Indicator shall provide selection of radio navigation aids or the Navigation-Display Set as the source of data for the HSI. It shall also provide an annunciator to indicate when True Heading data is being displayed. The front panel shall conform to Figure 37.

3.5.12.2 Form Factor - The Control-Indicator, excluding front panel protrusions, but including the rear connector, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
1.500	2.500	2.200 (behind mounting surface)

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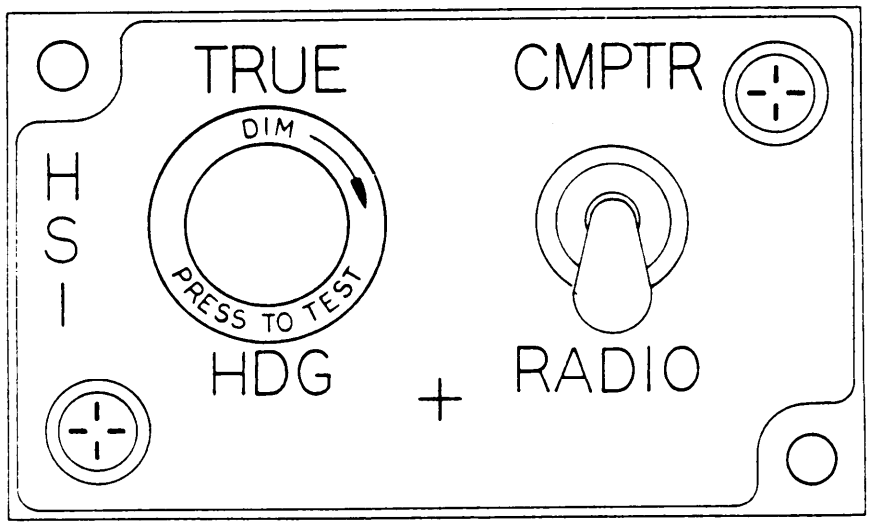


FIGURE 37. Control Indicator.

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3.5.12.3 Weight - The weight of the Control-Indicator shall not exceed 1 pound.

3.5.12.4 Controls - The Control-Indicator shall provide a two-pole double throw toggle switch for selecting CMPTR or RADIO. The switch contacts shall provide continuity between a separate pair of lines in each position.

3.5.12.5 Displays - The Control-Indicator shall provide an indicator lamp to signal when True Heading data is being displayed on the HSI. The lamp shall be illuminated green when the True Heading Lamp power is supplied to the Control-Indicator. The lamp shall provide integral push-to-test function and adjustable brightness control.

3.5.12.6 Aircraft Power - The equipment shall utilize aircraft lighting power as follows:

(1) 26 VAC, 400 Hz, 1 0, 0.05 Amp.

(2) 0-26 VAC, 400 Hz, 1 0, 0.20 Amp.

3.5.12.7 Panel Illumination - The Control-Indicator shall incorporate an integrally illuminated information panel per MIL-P-7788, Class 1-R, Type V. The panel shall utilize 26 V, 400 Hz, aircraft dimmable power.

*3.5.12.8 Electrical Connections - Connections to external circuits shall be as listed.

Reference Designation	Receptacle Type	Function
J1	SJTP02RE10-35PB(023)	To Aircraft and system

3.5.12.9 Mounting - The Control-Indicator shall meet the requirements of MIL-C-6781 for Type II control panels.

3.5.12.10 Color - The Control-Indicator shall be painted lusterless black, No. 37038, in accordance with FED-STD-595.

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3.5.13 Interconnecting Box J-3349/ASN-124 - The Interconnecting Box shall meet the following requirements:

3.5.13.1 Function - The Interconnecting Box shall provide signal and power distribution as necessary to simplify the system interconnect and preserve electromagnetic compatibility of the system when interconnected and installed in the aircraft. The Interconnecting Box shall normally be considered part of the aircraft installation and procured as such. (Refer to Paragraph 6.2.)

3.5.13.2 Form Factor - The Interconnecting Box, exclusive of atttachment points and connector projections, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
6.0	10.0	6.0

3.5.13.3 Weight - The weight of the Interconnecting Box shall not exceed 5 pounds.

3.5.13.4 Contents - The Interconnecting Box shall provide terminal boards and external connector receptacles for the routing of interconnecting wiring in such a way as to simplify aircraft cable design EMI filters shall be provided as necessary. Relays shall be provided for control of primary power to the NC, the SDC, and the PP-4964/ASN-84 Inertial Power Supply.

3.5.13.5 Mounting - The Interconnecting Box shall be designed to mount in any attitude. Mounting dimensions shall be in accordance with Figure 38.

3.5.13.6 System Power - The equipment shall utilize system power as follows:

(1) 28 VDC, 12 watts, relay power from the Battery Unit.

*3.5.13.7 Electrical Connections - Connections to external circuits shall be provided as listed.

Reference Designation	Receptacle Type	Function
J1	SJTP02RE12-35P(023)	IMS Signals
J2	SJTP02RE12-35SA(023)	Aircraft Signals
J3	SJTP02RE14-18S(023)	IMS Power
J4	SJTP02RE16-26S(023)	Battery Unit Power
J5	SJTP02RE22-21P(023)	Aircraft Power
J6	SJTP02RE18-32S(023)	Switched Power

3.5.13.8 Color - The case of the Interconnecting Box shall be lusterless gray, No. 36231, per FED-STD-595.

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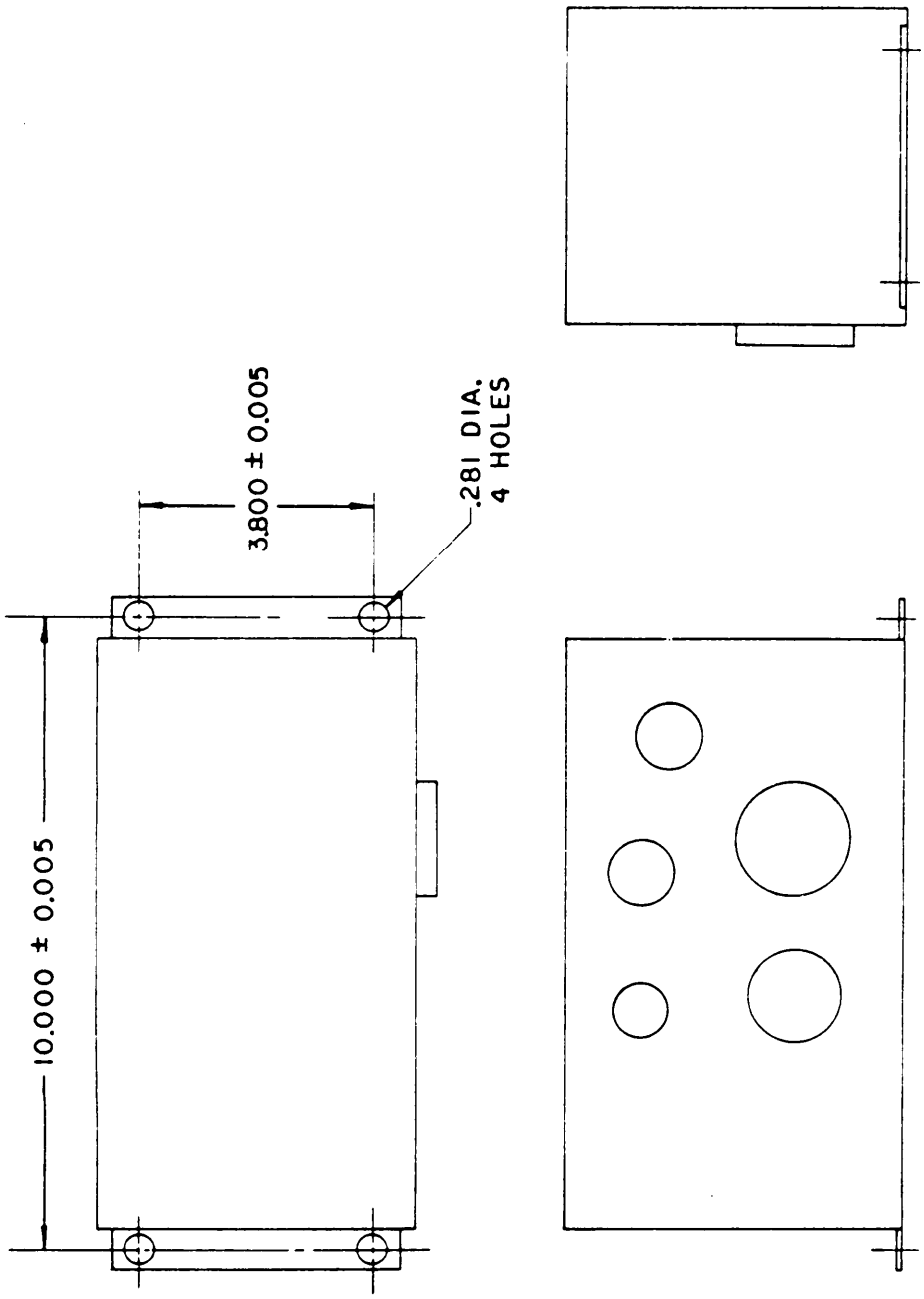


FIGURE 38. J-3349/ASN-124 Interconnecting Box mounting dimensions.

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3.5.14 Mounting Base, MT-4950/ASN-124 - The Base shall adapt the mounting provisions of the AN/ASA-66 Tactical Display Group and the CV-3332/ASN-124 Converter-Interface to the TACCO'S station of the P-3B aircraft. Slide shall be provided to support the Converter and the IP-886B/ASA-66 Display in a withdrawn position for maintenance. The base shall provide vibration and shock isolation.

3.5.14.1 Form Factor - The Base shall not exceed the following dimensions (inches)

Height	Width	Depth
23.0	23.0	27.0

3.5.14.2 Weight - The weight of the base shall not exceed 20 pounds.

3.5.14.3 Mounting - Mounting provisions shall be as shown in EI-632-14.

3.5.14.4 Isolation - The resonant frequency of the loaded Base shall be 10 Hz or less and transmissiblity at resonance shall not exceed 5.

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3.5.16 Mounting Base, MT-4953/ASN-124 - The Base shall adapt to the mounting provisions of the BA-597/ASN-124 Battery Unit and shall provide vibration and shock isolation.

3.5.16.1 Form Factor - The Base shall not exceed the following dimensions (inches):

Height	Width	Depth
3.0	7.5	15.0

3.5.16.2 Weight - The weight of the Base shall not exceed 3 pounds.

3.5.16.3 Mounting - Mounting provisions shall be as shown in EI-632-11.

3.5.16.4 Isolation - The resonant frequency of the loaded Base shall be 10 Hz or less and transmissibility at resonance shall not exceed 5.

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4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests
(2) Initial production Tests
(3) Acceptance Tests
(4) Integration Testing
(5) Flight Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.7. The Government inspector and the procuring activity shall be advised 30 days prior to when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - When required by contract, one of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor, when required by contract, shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper correction action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - When required by contract, the equipment shall be reworked by the Contractor by replacing all wear or damaged items on completion of the initial production tests. After reworking, the contractor shall resubmit the equipment for acceptance.

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4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall maintain test reports showing results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test

4.4.1.1 Examination of Product - Each equipment shall be examined to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

*4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

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An Initial Production Sample Test may be conducted on one of the first ten equipments offered for acceptance at the direction of the procuring activity. If an initial production sample test is conducted as above, no Sample Test need be conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

*4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

4.4.3 Reliability Assurance Tests

4.4.3.1 Reliability Development Test

4.4.3.1.1 Test Details - Reliability Assurance Tests shall consist of a Reliability Development Test in accordance with AR-104. Four (4) systems shall be tested. Accumulated operating time on each test sample shall be a minimum of 500 hours. The total cumulative operating time shall be 3000 hours.

The environmental conditions shall comply with the requirements of MIL-STD-781B test level E except that the temperature extremes shall be -30°C and +55°C.

The duty cycle (power on time) shall be in accordance with MIL-STD-781B except that for temperature cycling there shall be time to stabilize at low temperature plus time to stabilize at high temperature, plus sufficient time at high temperature to allow for a total cycle time of 12 hours. Equipment performance shall be measured and recorded at high temperature following a period of stabilization as indicated by prior thermal survey.

4.4.3.1.2 Test Failures and Corrective Action - Failures shall be classified in accordance with AR-104 except that reclassification shall be permitted when approved by the procuring activity. Failures of illuminated indicators (where readily replaceable by the operator) and time totalizing meters shall not be used in calculating system reliability.

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Corrective action/recurrent control shall be applied in accordance with 4.4.5.

Corrective action shall only be incorporated in the test samples with procuring activity approval. Continued testing need not await formal approval by the procuring activity of the contractor proposed corrective action.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level E shall be used except the temperature extremes shall be -30°C and +55°C and the vibration shall be 2 G ± 10% peak acceleration value at any non-resonant frequency between 20 and 60 Hz measured at the mounting points on the equipment.

* 4.4.3.2.1 All Equipment Test- - Each system produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used, the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in-operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month, the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than reject decision. Each WRA produced for the purpose of spares shall be tested for 50 hours with the final 25 hours failure free. Prior to the 50 hour test a burn-in period equivalent to the burn-in used on production systems produced for aircraft installs is required.

4.4.4 Special Tests - Special tests shall be conducted when contracted for on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

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4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests. The contractor may opt, at his own risk, to incorporate proposed corrective actions in the equipment prior to receipt of formal approval from the procuring activity.

(4) Where practical, include a test in the individual tests to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

Should a failure occur during parts testing, assembly tests, and field testing, a detailed analysis shall be performed with procuring activity concurrence that a pattern failure is present. When corrective action has been established, the above steps (3) and (4) shall be taken.

Failure data shall be recorded and categorized for failures occurring within the contractor's facility. Recording, reporting, and categorizing of failures occurring in the field shall be the responsibility of the procuring activity.

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4.5 Integration Tests - Integration Tests shall be conducted in accordance with system test procedures as approved by the procuring activity. To validate the system electrical and programmed performance, these tests shall be conducted on one system prior to flight test validation.

4.6 Flight Tests - A flight test validation program shall be conducted by the procuring activity under the conditions stated in Paragraph 3.4.2 to ensure compliance with the accuracies required by Paragraph 3.4.2.2.

4.7 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require additional applicable tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance in accordance with Paragraph 4.4.1. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

4.10 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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5.0 PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6.0 NOTES

6.1 Intended Use - The System, when interfaced with the associated equipment of Paragraph 6.8 shall provide the P-3B aircraft and crew tactical plot stabilization of selected markers, geographic navigation, and display of tactical and sensor data.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours. The requirements of Paragraph 3.2 shall apply to the non-repairable assembly and not the discrete parts contained therein.

6.6 Type Designations - The parentheses () when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the procuring activity assumes no liability whatsoever for any inaccuracies in these notations. Users are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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6.8 Associated Equipment

AAU-21A or AAU-32A Barometric Altimeter per MIL-A-81403(AS) and MIL-A-81852(AS)

AN/ASN-50 Attitude and Heading Reference System per MIL-A-23670A

AF/A24J-1 Horizontal Situation Indicator Group per MIL-H-27269

PB20N Autopilot as described in NAVAIR 01-75PAA-2-11; 05-45ADA-1, -2

ID-663/U Bearing, Distance, Heading Indicator per MIL-I-22075A

AN/APS-80 Radar Set per MIL-R-21113

AN/APA-125 Indicator Group per MIL-I-19165

ID-499/ASA-13 Position Indicator per MIL-C-18191 (WEP)

PT-396/AS Tactical Display Plotting Board per MIL-P-18987

AN/APN-153(V) Radar Navigation Set per MIL-N-23407

CPK-28A24G-9 True Airspeed Computer per NAVWEPS 01-75PAA-2-7.5, ref. AF/A24G-9 True Airspeed System

AN/ARN-99(V)1 Omega Navigation Set per MIL-N-81675

AN/AQA-7(V) Sonar Computer-Recorder Group per MIL-C-81486

C-7617/ARR-72 Dual Channel Control Indicator per MIL-R-81859

Part of the Inertial Navigation Set AN/ASN-84 (Ref. MIL-N-81497), consisting of:

- CN-1231/ASN-84 Gyroscope Assembly
- PP-4964/ASN-84 Power Supply
- MT-3947/ASN-84 Electrical Equipment Rack

Tactical Data Display Group AN/ASN-66 per MIL-D-81358(AS)

Tactical Data Display Group AN/ASA-66B per MIL-D-85009(AS)

ECM Receiver Indicator R-976B/ALD-2 per MIL-T-19877

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Rotating Dial Compass Repeater Indicator Type MW-2 per MIL-I-21685 (WEP)

Error Voltage Monitor MX-223/ASQ per MIL-M-22747

Model A206 Navigation Forward Interconnect Box per NAVAIR 01-75PAA-2-7

6.9 WRA MTBF - The MTBF of paragraph 3.3.2.3 is allocated for each WRA as follows:

Controller, Marker Entry	95,238 hours
Control, Keyer	1,536 hours
Indicator, Digital Display	2,100 hours
Controller, Marker Plotter	2,375 hours
Converter-Interface	325 hours
Control, Tactical Display	1,333 hours
Computer, Navigation	450 hours
Converter, Signal Data	475 hours
Converter. Radar Buffer	37,715 hours
Battery Unit	7,553 hours
Controller, Mode	17,153 hours
Control-Indicator (2)	263,158 hours
Interconnecting Box	164,000 hours

6.10 Metric Conversion Table - The following table provides the metric equivalent for all measurements contained in this specification:

0.001 in - 0.003 cm	3.265 in - 8.293 cm
0.0065 in - 0.017 cm	3.500 in - 8.89 cm
0.0075 in - 0.019 cm	3.84 in - 9.754 cm
0.012 in - 0.031 cm	7.40 in - 11.18 cm
0.034 in - 0.086 cm	4.500 in - 11.43 cm
0.060 in - 0.152 cm	4.70 in - 11.94 cm
0.125 in - 0.318 cm	5.000 in - 12.7 cm
0.25 in - 0.635 cm	5.250 in - 13.335 cm
0.35 in - 0.889 cm	5.75 in - 14.61 cm
0.50 in - 1.27 cm	5.750 in - 14.605 cm
1.500 in - 3.81 cm	6.500 in - 16.51 cm
1.875 in - 4.763 cm	7.5 in - 19.05 cm
2.0 in - 5.08 cm	7.600 in - 19.30 cm
2.18 in - 5.54 cm	7.880 in - 20.015 cm
2.200 in - 5.588 cm	8.500 in - 21.59 cm
2.500 in - 6.35 cm	9.000 in - 22.86 cm
2.630 in - 6.680 cm	11.020 in - 27.99 cm
3.0 in - 7.62 cm	11.500 in - 29.21 cm
3.000 in - 7.620 cm	13.000 in - 33.02 cm

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14.700 in	37.34 cm
15.0 in	38.1 cm
15.875 in	40.32 cm
16.4 in	41.66 cm
18.000 in	45.72 cm
20.500 in	52.07 cm
22.000 in	55.88 cm
23.00 in	58.42 cm
27.00 in	68.58 cm
12 ft	365.7 cm
30 ft	914.40 cm
100 ft.	3048.01 cm
250 ft	7620.02 cm
500 ft	15240.03 cm
9 yd	822.96 cm
20 yd	1828.80 cm
25 yd	2286.00 cm
35 yd	3200.41 cm
40 yd	3657.61 cm
2 mi	3.22 km
256 mi	412.00 km
512 mi	823.99 km
1024 mi	1647.97 km
0.5 nmi	0.927 km
1.2 nmi	2.224 km
1.5 nmi	2.78 km
2.4 nmi	4.448 km
2.5 nmi	4.63 km
3.5 nmi	6.486 km
1 lb	453.59 gm
1.5 lb	680.39 gm
2 lb	907.18 gm
3 lb	1360.78 gm
8 lb	3628.74 gm
20 lb	9071.84 gm
22 lb	9979.02 gm
42 lb	1.9050.86 gm
54 lb	24493.97 gm
64 lb	29029.89 gm
225 lb	102.06 gm
0.1 knot	0.515 cm/sec
3 knot	154.44 cm/sec
5 knot	257.4 cm/sec
20 knot	1029.6 cm/sec
200 knot	10296.0 cm/sec
1092.2 knot	56226.46 cm/sec
2184.5 knot	112458.06 cm/sec
3276.7 knot	168684.52 cm/sec
30 MIL	762.00 micron
1 oz/Sq ft	.0305 gm/cm^2

*6.11 This specification is under the cognizance of AIR-53351C.

Preparing Activity - NAVY (AS)

Project NO. 5826 - N148

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APPENDIX
AN/ASN-124 INTERFACE WITH ARINC-561 INS

10. SCOPE

10.1 Scope - The equipment covered by this appendix shall enable the Navigation Display Set to be interconnected with an ARINC-561 INS in lieu of ASN-84 components.

10.2 Classification - The equipment covered by this Appendix shall consist of the following items:

Assigned AN Nomenclature	Type Designation	Applicable Paragraph
Controller, Marker Entry	C-9850/ASN-124	3.5.1
Control, Keyer	C-9857/ASN-124	3.5.2
Indicator, Digital Display	ID-2035/ASN-124	3.5.3
Controller, Marker Plotter	C-9855/ASN-124	3.5.4
Converter-Interface	CV-3332/ASN-124	3.5.5
Control, Tactical Display	C-9862/ASN-124	3.5.6
Computer, Navigation	CP-1224/ASN-124	3.5.7
Converter, Signal Data	CV-3502/ASN-124	30.5.1
Converter, Radar Buffer	CV-3290/ASN-124	3.5.9
Battery Unit	BA-597/ASN-124	3.5.10
Controller, Mode	C-10422/ASN-124	30.5.2
Control-Indicator (2)	C-10421/ASN-124	30.5.3
Interconnecting Box	J-3349/ASN-124	3.5.13

10.3 Associated Equipment - The equipment shall operate to the requirements of this specification with the associated equipment listed in Paragraph 6.8, as modified by paragraphs 30.2 and 40.1.

20. APPLICABLE DOCUMENTS

20.1 General - The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for previous procurement. However, in lieu of the issue number listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design, performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

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Standards

MS 33638C(ASG)	Cases, Instrument, Flange-Mounted, Aircraft

Commercial Specifications

ARINC Characteristics No. 561-8	Air Transport Inertial Navigation System - INS

*20.2 Availability of Documents

20.2.1 Military Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105 Tabor Avenue, Philadelphia, Pennsylvania 19120.

20.2.2 Commercial Specification - When requesting the ARINC Characteristic No. 561-8 refer to both title and number. Copies of this characteristic may be obtained for a fee upon request to Aeronautical Radio, Incorporated, Documents Section, 2551 Riva Road, Annapolis, Maryland 21401.

30. REQUIREMENTS

The provisions of paragraph 3.0 apply except as modified herein.

30.1 Input Electrical Power - No power shall be required from the INS.

30.2 System Definition - The set shall consist of those items of paragraph 10.2. The system shall operate in conjunction with the associated P-3B aircraft equipment listed in paragraph 6.8, except the ASN-84 components, and shall operate in conjunction with the ARINC 561 INS, reference paragraph 40.1. The functional block diagram for this system configuration is shown in Figure 39. The analog output signal distribution diagram for this system configuration is shown in Figure 40.

30.2.1 Hardware - The CV-3502 Signal Data Converter (SDC) shall interface with the ARINC 561 INS instead of the ASN-84. The resolver to synchro buffer for platform azimuth shall not be required. The CV-3502 shall be designed so that conversion to the CV-3289, and vice versa, shall be possible by changing only these inertial interface printed circuit cards.

The C-10422 Mode Controller (MC) shall provide ON-OFF power control for the Navigation Display Set (except ITADS). It shall replace the C-9856 MC for this application. It shall also apply power to the OR-90/ARN-99(V)1 and provide an annunciator for “Pre-Select Heading”.

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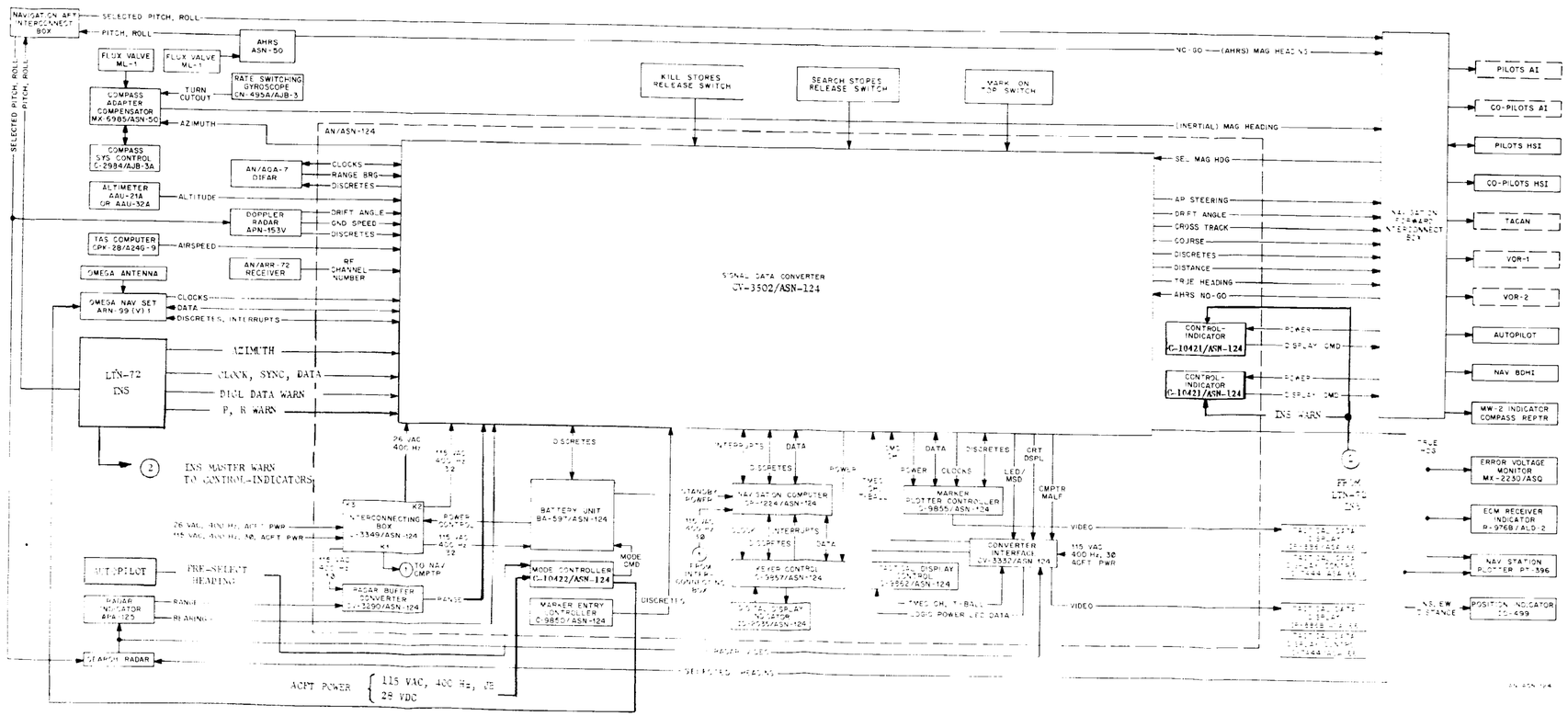


FIGURE 39. Functional block diagram, Navigation-Display Set, AN/ASN-124, ARINC-561 INS interconnect.

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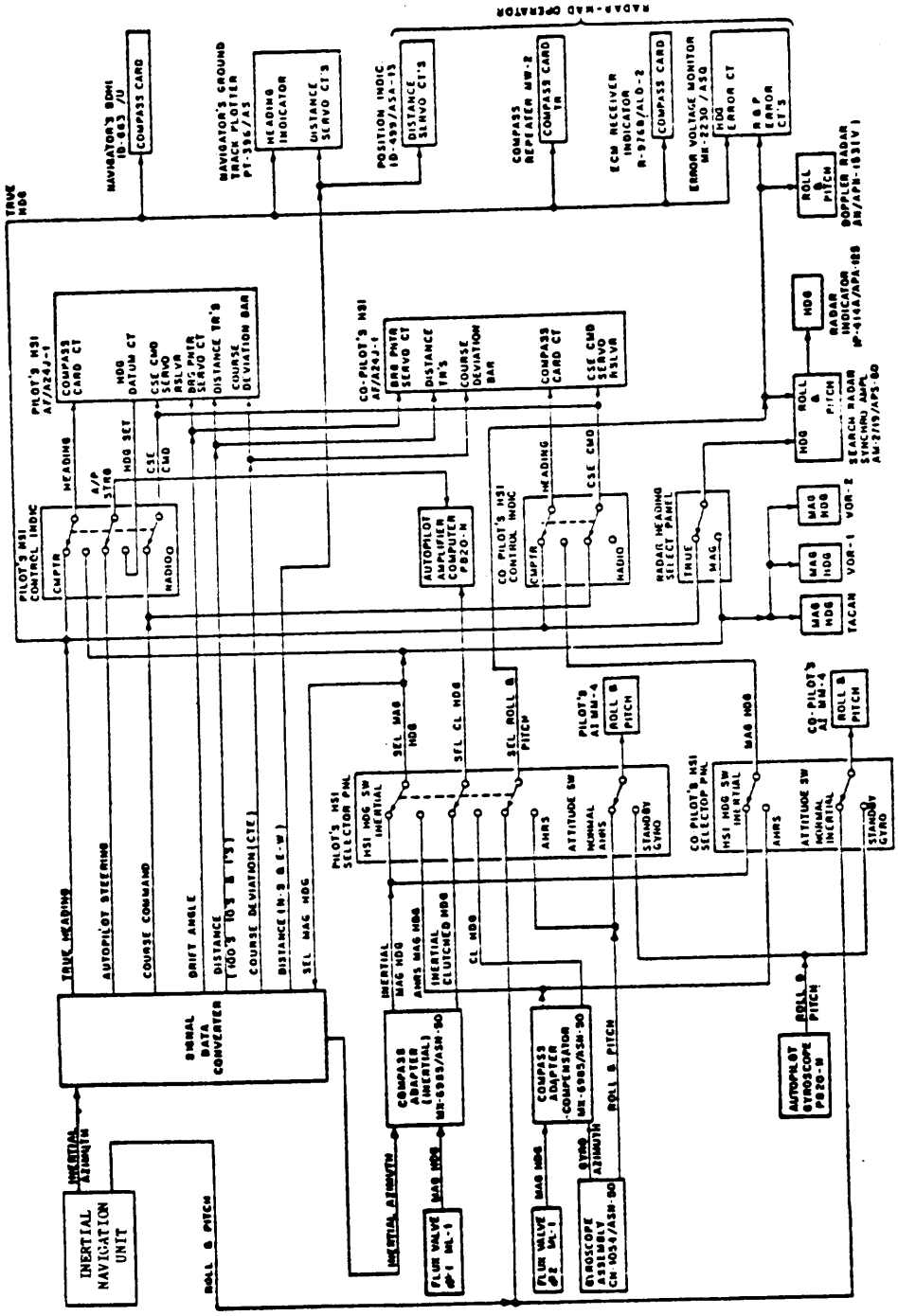


FIGURE 40. Analog Output Signal Distribution Diagram, ARINC-561 INS interconnect.

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The C-10421 Control-Indicator (C-I) shall be used in place of the C-9863 C-I for this application. An amber indicator light is provided for INS WARNING in the cockpit. Other functions remain the same.

30.2.2 Software - The system shall be programmed to perform the following basic functions:

(1) Navigation
(2) Control Panel Interface
(3) Display Interface
(4) System Self Test and Fault Isolation

30.3 Operating Modes- The pure inertial (IN) solution shall be deleted from the ASN-124 and shall be provided by the ARINC 561 INS. The auxiliary inertial (XN) solution shall also be provided by the ARINC 561 INS. The ASN-124 system shall utilize velocity and position data derived from the XN solution and supplied by the ARINC 561 INS. XN shall be selectable in the ASN-124 for geographic or tactical navigation. It shall not be possible to position update XN in the ASN-124. In dead-reckoning (DR) modes, inertial heading shall be True Heading supplied by the ARINC 561 INS.

30.4 Accuracy

30.4.1 Navigation-Display Set Accuracy - There shall be no requirement for IN/XN navigation accuracy for the ASN-124 in GEONAV or TACNAV.

30.5 Detail Requirements

30.5,1 Converter Signal Data, CV-3502/ASN-124 - The SDC shall meet the requirements of 3.5.8 except as modified herein.

30.5.1.1 Function - With the IMU Interface SRA and the IMU Azimuth Buffer SRA removed, the CV-3502 shall be interchangeable with the CT-3289. Installation of the ARINC 561 INS Interface SRA'S shall be the only change required for use with the ARINC 561 INS.

30.5.1.2 Inertial System Interface - The SDC shall receive serial binary digital data from the INS. The data channel shall be in accordance with ARINC 561. The following parameters shall be received from the INS:

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Parameter	Octal Code Label	Range	Units	Resolution	Significant Bits [10]
True Heading-TH	014	0 to ± π	Radians	$2\ \pi/2^{13}$	12
N-S Velocity-V	066	±3276.7	Knots	0.1 knot	15
E-W Velocity-V_E	067	±3276.7	Knots	0.1 knot	15
Latitude - LAT [9]	010	0 to ± π/2	Radians	$2\ \pi/2^{20}$	19
Longitude - LON [9]	011	0 to ± π	Radians	$2\ \pi/2^{20}$	19
Waypoint 1 LAT	040	0 to ± π/2	Radians	$2\ \pi/2^{18}$	17
Waypoint 1 LON	041	0 to ± π	Radians	$2\ \pi/2^{18}$	17
Waypoint 2 LAT	042	0 to ± π/2	Radians	$2\ \pi/2^{18}$	17
Waypoint 2 LON etc. . . through	043	0 to ± π	Radians	$2\ \pi/2^{18}$	17
Waypoint 9 LAT	060	0 to ± π/2	Radians	$2\ \pi/2^{18}$	17
Waypoint 9 LON	061	0 to ± π	Radians	$2\text{-}1/2^{18}$	17

[9] These present position data shall include the effect of position updates.

[10] The number of significant bits does not include sign bit.

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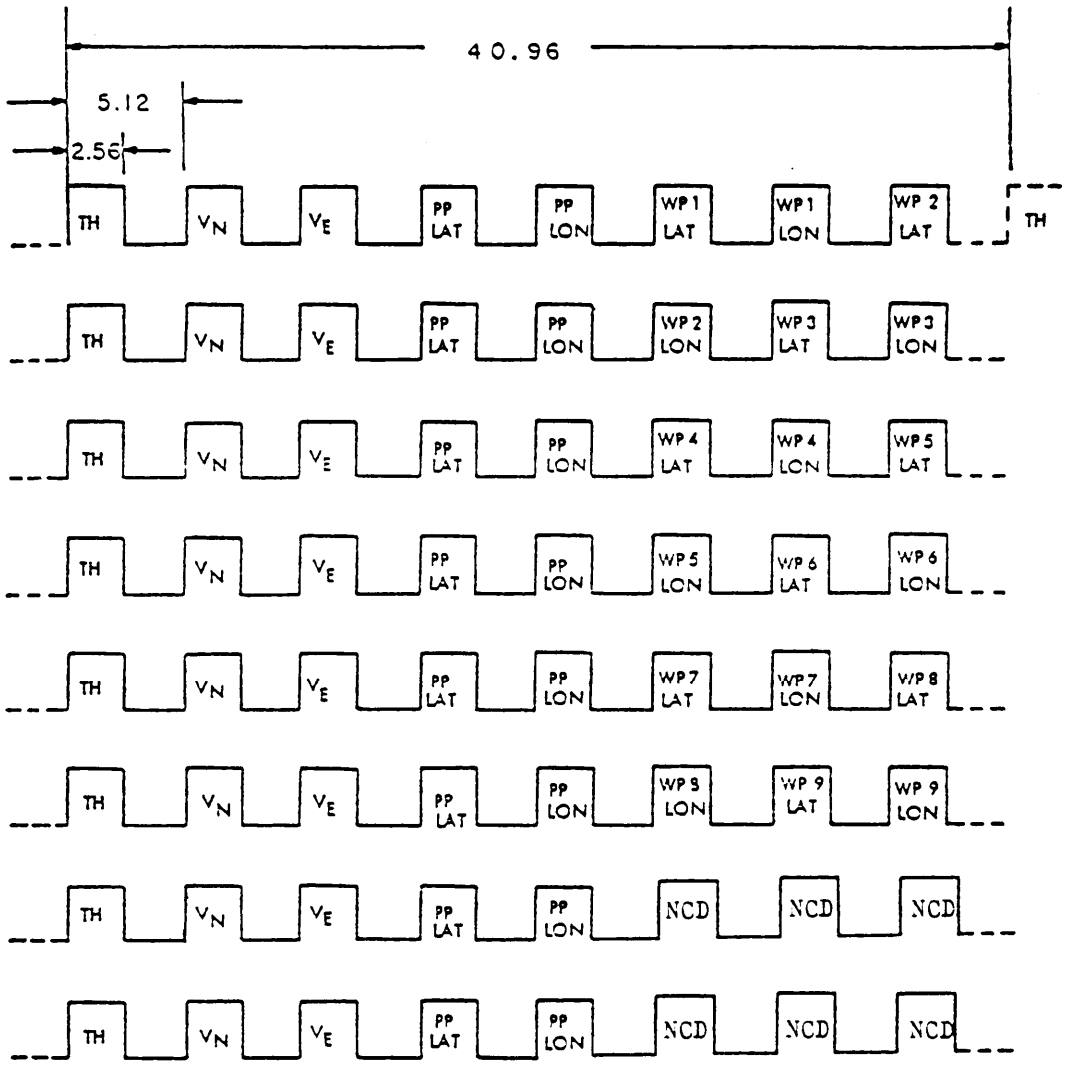
The data shall be output in eight frames of eight words each, comprising a complete data cycle of 64 words. (See Figure 41 attached.) The complete data cycle shall then be recycled continuously without interruption. The first five words of each eight-word frame shall be the five parameters TH, VN, VE, LAT, LON as shown above. VN, VE data shall be fresh each frame and coordinated. TH and LAT/LON shall be fresh data at least every other frame. If "no computed data" is indicated for a velocity or position parameter, the other velocity or position parameter shall also be coded "no computed data". The remaining three word times of each frame of eight words shall be used to transmit waypoint data, beginning with Waypoint 1 LAT/LON and Waypoint 2 LAT in frame 1 and ending with Waypoint 8 LON and Waypoint 9 LAT/LON in frame 6. The last three words of 7 and 8 shall be coded "no computed data". Waypoint words shall normally be coded "no computed data" unless (1) the operator selects MT on the CDU or (2) the operator inserts Waypoint Latitude or Longitude. When the operator selects RMT, all eighteen waypoint LAT/LON words shall be output with valid data, in one 64 word cycle. The output shall then return to transmission of no computed data in the waypoint words. When the operator inserts a Waypoint Latitude or Longitude, the INS shall utilize the next available 64 word cycle to transmit the inserted valid data in the appropriately numbered waypoint words. The output data shall then return to transmission of no computed data.

When the operator selects CDU Test in Standby or Align, the INS shall output test values as follows:

PARAMETER	VALUE	
True Heading	119.9707°	(Odd bits 19-29)
N/S Velocity	1092.2 Kts	(Even bits 16-28)
E.W Velocity	2184.5 Kts	(Odd bits 15-29)
Latitude	59.99977°N	(Even bits 12-28)
Longitude	119.99989°E	(Odd bits 11-29)

During test mode the status/sign matrix of the data words shall indicate "functional test" (bit 31 = 0, bit 30 = 1) in accordance with ARINC 561. The test values of latitude and longitude shall also be transmitted in the word locations of waypoint 1-9 as long as the CDU is in Test.

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Times in milliseconds
One bit time = 80 micro-seconds
NCD = no computed data

FIGURE 41. 64 word complete data cycle.

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30.5.1.3 Input Power - No input power shall be required from the INS.

30.5.2 Controller, Mode C-10422/ASN-124 - The Controller shall meet the following requirements:

30.5.2.1 Function - The Controller shall provide ON/OFF power control of the NC, SDC, and the OR-90/ARN-99(V)1 Receiver-Converter. The configuration of the front panel shall conform to Figure 42.

30.5.2.2 Form Factor - The Controller, exclusive of front panel protrusions and attachment points, but including connector projections, shall be formed within the following maximum dimensions (inches):

Height	Width	Depth
2.630	5.750	4.500 (behind mounting surface)

30.5.2.3 Weight - The weight of the Controller shall not exceed 2 pounds.

30.5.2.4 Controls - The Controller shall provide the following control:

30.5.2.4.1 Power - A two-position rotary switch shall control the state of output discretes to the ASN-124, and shall switch power to the OR-90/ARN-99. It shall be necessary to lift the switch in order to change position.

30.5.2.5 Displays - The Controller shall provide the fallowing display:

30.5.2.5.1 Auto-Pilot Engage - The Controller shall provide an indicator lamp to signal when the auto-pilot is engaged. The lamp shall be illuminated amber when the +28 VDC lamp power is supplied to the Controller.

30.5.2.6 Lighting Power - The equipment shall utilize aircraft lighting power in accordance with paragraph 3.3.12 as follows:

0-26 VAC, 400 Hz, 1 0, 0.4 Amp.

30.5.2.7 Panel Illumination The Controller shall incorporate an integrally illuminated information panel per MIL-P7788, Class 2W, Type V. The panel shall utilize 26V, 400 Hz, aircraft dimmable power.

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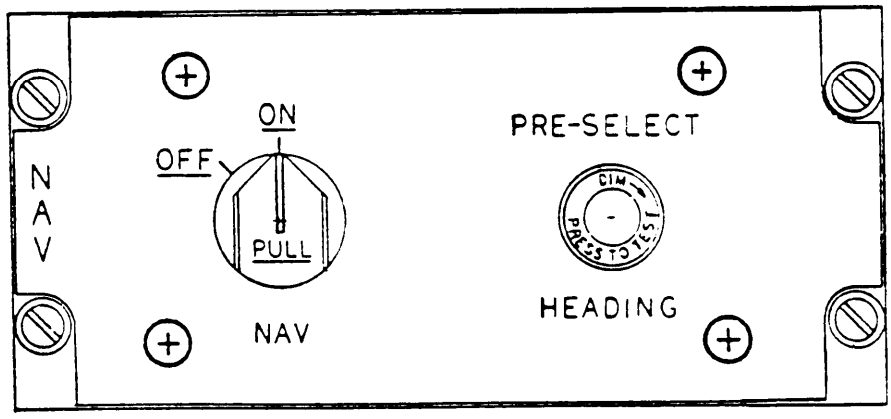


FIGURE 42. Mode Controller front panel.

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30.5.2.8 Electrical Connections - Electrical Connections to external circuits shall be as follows:

Reference Designation	Receptacle Type	Function
J1	SJTP02RE14-35PB(023)	Power and discretes
J2	SJTP02RE14-18P(023)	Power

30.5.2.9 Mounting - The Controller shall be a console mounted unit in accordance with MIL-C-6781 Type 1.

30.5.2.10 Color - The controller shall be painted lusterless gray, No. 36231, in accordance with FED-STD-595.

30.5.3 Control-Indicator, C-10421/ASN-124 - The Control-Indicator shall meet the following requirements.

30.5.3.1 Function - The Control-Indicator shall provide selection of radio navigation aids or the Navigation-Display Set as the source of data for the HSI. It shall provide an annunciator to indicate when True Heading data is being displayed, and an annunciator to indicate INS Warning. The front panel shall conform to Figure 43.

30.5.3.2 Form Factor - The Control-Indicator, excluding front panel protrusions, but including the rear connector, shall be formed within the following maximum dimensions (inches).

Height	Width	Depth
3.265	3.265	2.200 (behind mounting surface)

30.5.3.3 Weight - The weight of the Control-Indicator shall not exceed 2 pounds.

30.5.3.4 Controls - The Control-Indicator shall provide a two-pole double throw toggle switch for selecting CMPTR or RADIO. The switch contacts shall provide continuity between a separate pair of lines in each position.

30.5.3.5 Displays

30.5.3.5.1 True Heading - The Control-Indicator shall provide an indicator lamp to signal when True Heading data is being displayed on the HSI. The lamp shall be illuminated green when the True Heading Lamp power is supplied to the Control-Indicator. The lamp shall provide integral push-to-test function and adjustable brightness control.

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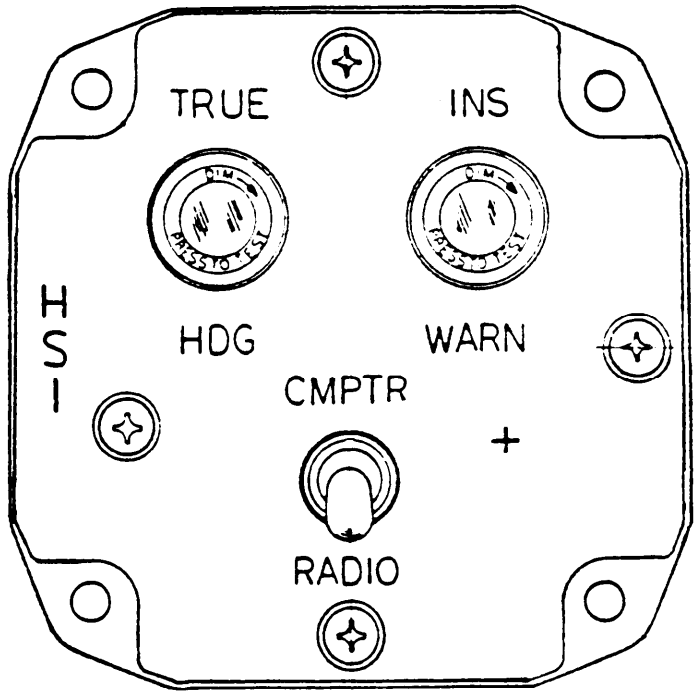


FIGURE 43. Control-Indicator front panel.

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30.5.3.5.2 INS Warn - The Control-Indicator shall provide an indicator lamp to signal when the INS annunciates INS Master Warning on the INS Control Display Unit. The Control-Indicator shall include a lamp driver circuit when shall present a load of 20,000 or more ohms referenced to aircraft ground and shall operate according to the following logic:

Greater than 17 VDC = WARN

Less than 3.5 VDC = NOT WARN

The lamp shall be illuminated amber when the INS signals WARN. The lamp shall provide integral push-to-test function and adjustable brightness control.

30.5.3.6 Aircraft Power - The equipment shall utilize aircraft power as follows:

(1) 0 - 26 VAC, 400 Hz, 1 0, 0.6 amp.
(2) 28 VDC, 4 watts

30.5.3.7 Panel Illumination - The Control-Indicator shall incorporate an integrally illuminated information panel per MIL-P-7788, Class 1R, type V. The panel shall utilize 26V, 400 Hz, aircraft dimmable power.

30.5.3.8 Electrical Connections - Connections to external circuits shall be as listed.

Reference Designation	Receptacle Type	Function
J1	SJTP02RE10-35PB(023)	To aircraft and system

30.5.3.9 Mounting - The Control-Indicator shall meet the requirements of MIL-C-6781 for Type II control panels, and shall utilize the mounting dimensions of MS33638.

30.5.3.10 Color - The Control-Indicator shall be painted lusterless black, No. 37038, in accordance with FED-STD-595.

40. NOTES

40.1 Associated Equipment - Inertial Navigation Set per ARINC 561-8.

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MIL-N-85005A(AS) Navigation Display Set, AN/ASN-124

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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